Lecture Notes on Mathematical Analysis

Single and Multivariable Calculus

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Contents

Preface

These notes are for students who know the main ideas of single-variable calculus and want to understand why its central statements are true. They develop rigorous real analysis before rebuilding calculus from limits, completeness, continuity, differentiation, and Riemann integration. The intended audience is an advanced undergraduate beginning proof-based mathematics. Familiar algebra and ordinary calculus are assumed, but the set-theoretic, logical, topological, and finite-dimensional linear-algebraic tools needed later are developed in the text.

Part I (Chapters 1–9) constructs the real number system and develops one-variable analysis through Riemann integration and the Fundamental Theorem of Calculus. Part II (Chapters 10–13) develops the finite-dimensional topology and linear algebra needed for multivariable differentiation and Riemann integration, and ends with differential forms and the general Stokes theorem. Measure theory, Lebesgue integration, complex and functional analysis, partial differential equations, and deeper differential geometry are deliberately outside the scope; the final chapter points toward them as natural continuations.

AI-assisted preparation. OpenAI Codex was used extensively in drafting, revising, proofreading, and preparing the LaTeX source for these notes. The author determined the scope and mathematical organization, made the final decisions about retained content, and is responsible for approving and verifying the correctness of the published version.

How to Read These Notes

These notes start from familiar calculus questions but use a different standard of explanation: every formal assertion has hypotheses, and every main claim is justified from material already established or from an explicitly identified external theorem. A definition fixes meaning; a theorem requires proof unless it is marked as external. Examples provide intuition, but not proof of a general assertion. Occasionally, an explicitly identified informal preview uses familiar notation to motivate a later formal development; such previews are not used as formal justification.

The chapters are designed to be read in order. Exercises are part of the course: they supply routine practice, short proofs, counterexamples, and connections among the main results. A reader should attempt them after the relevant section, consulting the stated hypotheses and earlier results rather than relying on calculus folklore. The bibliography distinguishes the one substantial imported topological result from further-reading references.

Part I

Single-Variable Calculus and Analysis

Chapter 1

Motivation and the Transition from Calculus to Analysis

1.1 Why Calculus Needs Foundations

Calculus describes change and accumulation. Its central calculations often begin with an informal phrase: "make the increment arbitrarily small," "add infinitely many pieces," or "approach a point." These phrases are useful, but they conceal questions that computation alone cannot answer. What does it mean to approach? Why is a limiting value unique? Why may a limiting process be interchanged with addition, differentiation, or integration?

Mathematical analysis supplies precise meanings and proofs. The precision is not a bureaucratic replacement for intuition. It identifies the hypotheses that make a calculation reliable and exposes plausible-looking arguments that fail.

Example 1.1 (A number suggested by geometry)**.** The diagonal of a square of side length 1 has a length customarily denoted by $\sqrt{2}$. If this length were a rational number p/q in lowest terms, then $p^2 = 2q^2$. Thus p^2 is even. An odd integer has the form $2k+1$, whose square is $4k(k+1)+1$ and is odd; hence p is even. Write $p = 2r$. Then $q^2 = 2r^2$, so the same argument makes q even. This contradicts that p/q was in lowest terms. Hence no rational number has square 2.

The example does not yet construct $\sqrt{2}$; rather, it points out why rational numbers are not sufficient for familiar geometry. Chapter 3 will construct a number system containing such limits and isolate the property that makes it adequate: completeness.

1.2 A First Look at Limits

Consider the values $1, 1/2, 1/3, \ldots$. Their terms become as small as we wish. A rigorous definition should express exactly that statement without appealing to a last term, because there is no last positive integer.

Informal preview of convergence. A sequence $(a_n)_{n\in\mathbb{N}}$ of real numbers **converges to** a real number L when, for every positive tolerance ε, all sufficiently late terms a_n lie within distance ε of L. In symbols, the intended statement is

$$\text{for every } \varepsilon > 0 \text{ there is } N \in \mathbb{N} \text{ such that } n \geq N \Longrightarrow |a_n - L| < \varepsilon.$$

This is not yet a formal definition: the quantifiers, real numbers, and natural numbers will be developed formally in Chapters 2–4.

For instance, $1/n$ converges to 0: given $\varepsilon > 0$, choose a natural number $N > 1/\varepsilon$. Then $n \geq N$ implies $0 < 1/n \leq 1/N < \varepsilon$. The existence of such an N is an Archimedean property, which will be proved from the construction of the real numbers rather than silently assumed.

1.3 The Structural Ideas of Analysis

Limits
: A limit records stable behavior at arbitrarily fine scales. The order of the quantifiers matters: a proposed tolerance is given first, and the required stage may depend on it.

Completeness
: A number system is complete when a process that should determine a number does not leave a gap. The least-upper-bound principle and the assertion that Cauchy sequences converge are two important formulations.

Continuity
: A function is continuous at a point when small changes in input force small changes in output. The exact definition uses the same quantifier pattern as a limit.

Differentiation
: The derivative is the best linear approximation to a function at a point. In one variable it is a number; in several variables it is a linear map.

Integration
: Integration formalizes the limiting process of approximating accumulated quantities by finite sums. The existence of an integral, and its relationship to derivatives, are the substance of the Fundamental Theorem of Calculus.

1.4 Warnings That Motivate Hypotheses

Example 1.2 (Pointwise convergence need not preserve continuity)**.** For $n \in \mathbb{N}$, define $f_n : [0,1] \to \mathbb{R}$ by $f_n(x) = x^n$. Every f_n is continuous. For $0 \leq x < 1$, the values x^n tend to 0, whereas $f_n(1) = 1$ for every n. Thus the pointwise limiting function is

$$f(x) = \begin{cases} 0, & 0 \leq x < 1, \\ 1, & x = 1, \end{cases}$$

which is not continuous at 1. Chapter 6 will prove that **uniform** convergence supplies the missing hypothesis.

Example 1.3 (A bounded set may lack a rational supremum)**.** Let $S = \{q \in \mathbb{Q} : q > 0 \text{ and } q^2 < 2\}$. The set is nonempty (for example $1 \in S$) and bounded above in $\mathbb{Q}$ (for example 2 is an upper bound). It has no least rational upper bound. Informally, any such bound would have to be $\sqrt{2}$, which is not rational. A complete proof, including the needed rational approximations, will be given in Chapter 3.

1.5 Road Map

Chapter 2 establishes the language of proof. Chapters 3–4 build $\mathbb{R}$ and its sequence theory. Chapters 5–9 reconstruct one-variable calculus from limits. The second part then enlarges the setting to finite-dimensional spaces, develops multivariable differentiation and integration, and explains the classical integral theorems as cases of the general Stokes theorem.

Exercises

Exercise 1.1. Let (a_n) be a sequence of real numbers that converges to both L and M. Using the informal definition above, formulate a strategy for proving $L = M$. The complete proof appears after the necessary order facts have been proved.

Chapter 2

Basic Set Theory, Logic, Functions, and Relations

2.1 Sets

Mathematical analysis constantly describes collections: the possible inputs of a function, the points of an interval, or the terms of a sequence. The language of sets lets us describe such collections precisely and compare them without relying on a picture or an informal list.

We take
$$\mathbb{N} = \{1, 2, 3, \ldots\}$$
as the primitive discrete number system. Its usual arithmetic means that addition and multiplication have their familiar laws and keep natural numbers natural. Its induction property says that a statement true for (1), and inherited from each number to its successor, is true for every natural number. Its well-ordering property says that every nonempty collection of natural numbers has a smallest member. These principles will be used repeatedly, and induction is stated formally in Theorem 2.4. We write
$$\mathbb{N}_0 = \mathbb{N} \cup \{0\}.$$

The integers and rational numbers will be constructed in this chapter, and the real numbers in Chapter 3. A **set** is a collection of objects, called its elements. We write $x \in A$ when x is an element of A.

Definition 2.1 (Subsets and set equality)**.** For sets A, B, write $A \subseteq B$ if every element of A is an element of B. Define $A = B$ to mean both $A \subseteq B$ and $B \subseteq A$.

The empty set $\varnothing$ has no elements, and is a subset of every set. Here "and" requires both stated conditions, "or" allows either condition (including both), and $x \notin B$ means that x is not an element of B. For any sets A, B, define

$$A \cup B = \{x : x \in A \text{ or } x \in B\}, \qquad A \cap B = \{x : x \in A \text{ and } x \in B\},$$
$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}.$$

If $A \subseteq X$, its complement is $A^c = X \setminus A$. The power set is $\mathcal{P}(A) = \{E : E \subseteq A\}$.

Proposition 2.2 (De Morgan's laws)**.** If $A, B \subseteq X$, then $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$.

Proof. For $x \in X$, membership in $(A \cup B)^c$ means that it is false that $x \in A$ or $x \in B$. This is equivalent to $x \notin A$ and $x \notin B$, namely $x \in A^c \cap B^c$. The second identity follows identically with "and" and "or" exchanged. ☐

Definition 2.3 (Cartesian product)**.** For sets A, B, define $A \times B = \{(a, b) : a \in A,\ b \in B\}$, where $(a, b) = (a', b')$ means $a = a'$ and $b = b'$.

2.2 Logic and Proofs

Definitions specify the objects of mathematics, while proofs explain why their asserted properties follow. This section introduces the small amount of logical language needed to state definitions and organize proofs clearly.

A **proposition** is a statement that is either true or false. A **predicate** is a statement with a variable whose truth depends on the value of that variable. For example, let $P(n)$ mean "n is even." Then $P(4)$ is a true proposition, whereas $P(n)$ is not a complete proposition until n is assigned a value or is quantified.

If P and Q are propositions, then P and Q means that both are true, and P or Q means that at least one is true. This is the inclusive mathematical meaning of "or." For instance,

$$n > 0 \text{ and } n < 5$$

requires both inequalities, while "n is even or divisible by 3" permits either property or both. The **negation** of P, written

$$\neg P,$$

means that P is false. Thus the negation of $x > 3$ is $x \leq 3$. Negating a mathematical statement means negating its full meaning, not merely placing the word "not" at an arbitrary point.

The implication

$$P \Rightarrow Q$$

means "if P, then Q"; P is its **hypothesis** (or antecedent), and Q is its **conclusion** (or consequent). For example,

$$n \text{ is divisible by } 4 \quad \Longrightarrow \quad n \text{ is even.}$$

Its **contrapositive** is

$$\neg Q \Rightarrow \neg P,$$

and an implication is logically equivalent to its contrapositive. The **converse** is $Q \Rightarrow P$; it is a different assertion and does not follow automatically. The notation

$$P \Longleftrightarrow Q$$

means "P if and only if Q," often abbreviated "iff": both $P \Rightarrow Q$ and $Q \Rightarrow P$ hold. A proof of an iff statement normally proves the two directions separately.

Quantifiers turn predicates into propositions. The statement

$$\forall x \in A\ P(x)$$

means that $P(x)$ holds for every $x \in A$, while

$$\exists x \in A\ P(x)$$

means that there is at least one $x \in A$ for which $P(x)$ holds. For example,

$$\forall n \in \mathbb{N} \quad n+1 > n,$$

whereas

$$\exists n \in \mathbb{N} \quad n^2 = 4.$$

The first says something about every natural number; the second asserts the existence of a witness, here $n = 2$. After predicates, quantifiers, and negation have been introduced, their basic interaction can be written as

$$\neg(\forall x \in A\ P(x)) \iff \exists x \in A\ \neg P(x),$$

$$\neg(\exists x \in A\ P(x)) \iff \forall x \in A\ \neg P(x).$$

In words, "every student passed" is false exactly when there is a student who did not pass; "there is a student who passed" is false exactly when every student failed.

2.2.1 Writing Proofs

Logic becomes useful here as a guide to proof writing. To prove a universal implication

$$\forall x \in A \quad P(x) \Rightarrow Q(x),$$

begin by letting $x \in A$ be arbitrary, suppose $P(x)$, and show $Q(x)$. The word "arbitrary" matters: the argument must work for every allowed x, not only for a convenient example. To prove an existential statement, give a **witness** and verify that it has the required property. To prove that an object exists uniquely, first prove existence and then show that any two objects with the property are equal. This pattern will recur for limits, least upper bounds, inverse functions, and implicit functions.

Set equality is proved by showing both inclusions

$$A \subseteq B \qquad \text{and} \qquad B \subseteq A.$$

Similarly, two functions with the same domain and codomain are equal when their values agree at every point of the domain. A direct proof derives the conclusion from the hypotheses. A contrapositive proof establishes $\neg Q \Rightarrow \neg P$ in order to prove $P \Rightarrow Q$. A proof by contradiction assumes the hypotheses and the negation of the desired conclusion, then derives an impossibility.

A proof should indicate why each nontrivial step follows: from a hypothesis, a definition, an earlier theorem, or an algebraic calculation. Calculations are part of a proof only when their purpose is clear, so short explanatory sentences around a nontrivial chain of formulas are often helpful. The aim is clarity and visible logical dependence, not rigid prose templates. The next theorem records the induction method: prove a base case, assume the statement at an arbitrary stage as the induction hypothesis, and then prove the induction step.

Theorem 2.4 (Induction)**.** Let $P(n)$ be a proposition for each $n \in \mathbb{N}$. If $P(1)$ is true and $P(n) \Rightarrow P(n+1)$ for every $n \in \mathbb{N}$, then $P(n)$ is true for all $n \in \mathbb{N}$.

Proof. This is the induction axiom for $\mathbb{N}$. Equivalently, it follows from the well-ordering principle that each nonempty subset of $\mathbb{N}$ has a least element: a least n for which $P(n)$ failed could be neither 1 nor a successor of a number for which P holds. $\square$

2.3 Relations and Functions

Many mathematical statements compare two objects: two numbers may be equal, one number may divide another, or one set may be contained in another. A relation records exactly which ordered pairs of objects are being regarded as related.

Definition 2.5 (Binary relation)**.** Let A and B be sets. A **binary relation** R from A to B is a subset

$$R \subseteq A \times B.$$

If $A = B$, then R is a **binary relation on** A. When

$$(a, b) \in R,$$

we commonly write

$$a\ R\ b.$$

Example 2.6 (Familiar relations)**.** Equality, the usual numerical order, divisibility on $\mathbb{N}$, and inclusion between subsets of a fixed set are all binary relations. For example, on

$$A = \{1, 2, 3, 4\},$$

the divisibility relation is the subset

$$R = \{(a, b) \in A \times A : a \text{ divides } b\}.$$

It contains, among others,

$$(1, 2), \quad (1, 4), \quad (2, 4), \quad (3, 3),$$

but

$$(3, 4) \notin R.$$

Thus the statement that a relation is a set of ordered pairs has concrete content. Not every relation is an equivalence relation or an order: for example, on $\mathbb{N}$, the relation

$$m\ R\ n \quad \Longleftrightarrow \quad n = m + 1$$

is neither, since it is not reflexive.

Definition 2.7 (Equivalence relation)**.** An **equivalence relation** is a binary relation $\sim$ on A such that, for all $a, b, c \in A$, it is **reflexive**, **symmetric**, and **transitive**:

$$a \sim a, \qquad a \sim b \Longrightarrow b \sim a, \qquad a \sim b \text{ and } b \sim c \Longrightarrow a \sim c.$$

Its equivalence class at a is

$$[a] = \{b \in A : b \sim a\}.$$

Equivalence relations classify elements by grouping those regarded as the same for a specified purpose. Equality is the simplest example. Congruence modulo a natural number, used after the construction of $\mathbb{Z}$ below, is a further familiar example.

Proposition 2.8. If $\sim$ is an equivalence relation, then A is the union of its equivalence classes, and two classes are either equal or disjoint.

Proof. Reflexivity gives $a \in [a]$ for every a. If $x \in [a] \cap [b]$, symmetry and transitivity imply $a \sim b$. Then $y \sim a$ implies $y \sim b$, and conversely, so $[a] = [b]$. $\square$

Definition 2.9 (Partial order)**.** An **order relation** or **partial order** on a set X is a binary relation $\leq$ on X that is reflexive, antisymmetric, and transitive: for all $x, y, z \in X$,

$$x \leq x,$$

$$x \leq y \text{ and } y \leq x \Longrightarrow x = y,$$

and

$$x \leq y \text{ and } y \leq z \Longrightarrow x \leq z.$$

A set equipped with a partial order is a **partially ordered set**, or **poset**. Antisymmetric does not mean "not symmetric"; it is the specific implication displayed above.

Example 2.10 (Partial orders and incomparability)**.** The power set $\mathcal{P}(X)$, ordered by inclusion, is a poset. If X contains 1 and 2, then

$$\{1\} \not\subseteq \{2\}, \qquad \{2\} \not\subseteq \{1\},$$

so these two sets are not comparable. Divisibility is another partial order: on positive integers, define

$$m \leq_d n \quad \Longleftrightarrow \quad m \text{ divides } n.$$

Definition 2.11 (Total order and strict order)**.** A **total order**, or **linear order**, is a partial order $\leq$ on X such that every two elements are comparable:

$$x \leq y \quad \text{or} \quad y \leq x \qquad (x, y \in X).$$

For a total order, define its associated strict order by

$$x < y \quad \Longleftrightarrow \quad x \leq y \text{ and } x \neq y.$$

We take $\leq$ as the primitive relation. The usual order on $\mathbb{N}$ is a total order, whereas inclusion on $\mathcal{P}(X)$ is generally only a partial order.

Thus binary relations have two importantly different specializations: equivalence relations classify or group elements, whereas partial orders compare elements; total orders are the partial orders with no incomparability.

Definition 2.12 (Function and inverse images)**.** A function $f : A \to B$ assigns exactly one $f(a) \in B$ to every $a \in A$. Its graph

$$\{(a, f(a)) : a \in A\} \subseteq A \times B$$

is therefore a relation for which every $a \in A$ occurs in exactly one ordered pair. For $E \subseteq A$ and $F \subseteq B$, set $f(E) = \{f(x) : x \in E\}$ and $f^{-1}(F) = \{x \in A : f(x) \in F\}$. It is **injective** if $f(a) = f(a')$ implies $a = a'$, **surjective** if $f(A) = B$, and **bijective** if both.

Definition 2.13 (Composition and identity functions)**.** Let $f : A \to B$ and $g : B \to C$. Their **composition** is the function $g \circ f : A \to C$ defined by

$$(g \circ f)(a) = g(f(a)) \qquad (a \in A).$$

For every set A, the **identity function** $\mathrm{id}_A : A \to A$ is defined by $\mathrm{id}_A(a) = a$.

Proposition 2.14 (Basic laws of composition)**.** Let $f : A \to B$, $g : B \to C$, and $h : C \to D$ be functions.

(a) $h \circ (g \circ f) = (h \circ g) \circ f$.

(b) $f \circ \mathrm{id}_A = f$ and $\mathrm{id}_B \circ f = f$.

(c) If f and g are injective, then $g \circ f$ is injective; if f and g are surjective, then $g \circ f$ is surjective. In particular, a composition of bijections is a bijection.

(d) If f and g are bijections, then $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Proof. For $a \in A$, both functions in the first assertion have value $h(g(f(a)))$, proving equality. The identity assertions follow from $\mathrm{id}_A(a) = a$ and $\mathrm{id}_B(f(a)) = f(a)$. If $(g \circ f)(a) = (g \circ f)(a')$, injectivity of g gives $f(a) = f(a')$, and injectivity of f gives $a = a'$. If $c \in C$, surjectivity of g gives $b \in B$ with $g(b) = c$, and surjectivity of f gives $a \in A$ with $f(a) = b$; then $(g \circ f)(a) = c$. Finally,

$$(f^{-1} \circ g^{-1}) \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ \mathrm{id}_B \circ f = \mathrm{id}_A,$$

and the analogous calculation in the other order gives id_C. A two-sided inverse is unique: if u and v are both two-sided inverses of a function w, then

$$u = u \circ (w \circ v) = (u \circ w) \circ v = v.$$

This proves the last assertion without using a later result. □

Theorem 2.15 (Inverse function)**.** If $f : A \to B$ is bijective, there is a unique function $f^{-1} : B \to A$ satisfying $f^{-1}(f(a)) = a$ and $f(f^{-1}(b)) = b$.

Proof. For each $b \in B$, surjectivity gives an a with $f(a) = b$, and injectivity makes it unique. Define $f^{-1}(b)$ to be that element. The identities and uniqueness follow immediately. □

Proposition 2.16 (Inverse-image rules)**.** For $f : A \to B$ and $F, G \subseteq B$, one has $f^{-1}(F \cup G) = f^{-1}(F) \cup f^{-1}(G)$, $f^{-1}(F \cap G) = f^{-1}(F) \cap f^{-1}(G)$, and $f^{-1}(B \setminus F) = A \setminus f^{-1}(F)$.

Proof. For example, $x \in f^{-1}(F \cap G)$ if and only if $f(x) \in F$ and $f(x) \in G$, precisely when $x \in f^{-1}(F) \cap f^{-1}(G)$. The other two identities use the same membership calculation. □

Proposition 2.17 (Image and inverse-image rules)**.** Let $f : A \to B$, let $E, H \subseteq A$, and let $F, G \subseteq B$. Then

$$f(E \cup H) = f(E) \cup f(H), \qquad f(E \cap H) \subseteq f(E) \cap f(H),$$

and equality holds in the second formula when f is injective. Moreover,

$$E \subseteq f^{-1}(f(E)), \qquad f(f^{-1}(F)) \subseteq F,$$

and equality holds in the first inclusion when f is injective and in the second when f is surjective.

Proof. Each assertion follows by unpacking membership. For example, if

$$y \in f(E \cap H),$$

then $y = f(x)$ for some $x \in E \cap H$, so y belongs to both $f(E)$ and $f(H)$. Conversely, if f is injective and

$$y = f(e) = f(h)$$

with $e \in E$ and $h \in H$, then $e = h \in E \cap H$. The remaining statements follow similarly; the equality cases use respectively injectivity to recover a unique preimage and surjectivity to supply one. □

2.4 The Natural Numbers, Integers, and Rational Numbers

The natural numbers are the starting point for the arithmetic used throughout the notes. A fully set-theoretic construction of $\mathbb{N}$, for example by finite ordinals, belongs to foundations rather than analysis. We therefore take $\mathbb{N}$ and $\mathbb{N}_0$ with their usual arithmetic, induction, and well-ordering structure as primitive. The next two quotient constructions explain the algebraic chain

$$\mathbb{N} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q}$$

before Chapter 3 treats the analytic extension

$$\mathbb{Q} \longrightarrow \mathbb{R}.$$

Passing from $\mathbb{N}_0$ to $\mathbb{Z}$ permits subtraction, and passing from $\mathbb{Z}$ to $\mathbb{Q}$ permits division by nonzero elements.

Definition 2.18 (The integers)**.** On $\mathbb{N}_0 \times \mathbb{N}_0$, define

$$(a, b) \sim_{\mathbb{Z}} (c, d)$$

if and only if

$$a + d = b + c.$$

The set of equivalence classes is denoted by

$$\mathbb{Z} = (\mathbb{N}_0 \times \mathbb{N}_0)/ \sim_{\mathbb{Z}} .$$

Write $[a, b]_{\mathbb{Z}}$ for the class of (a, b), interpreted as the formal difference $a - b$. Define

$$[a, b]_{\mathbb{Z}} + [c, d]_{\mathbb{Z}} = [a + c, b + d]_{\mathbb{Z}},$$

$$[a, b]_{\mathbb{Z}}[c, d]_{\mathbb{Z}} = [ac + bd, ad + bc]_{\mathbb{Z}},$$

and

$$-[a, b]_{\mathbb{Z}} = [b, a]_{\mathbb{Z}}.$$

Set

$$0_{\mathbb{Z}} = [0, 0]_{\mathbb{Z}}, \qquad 1_{\mathbb{Z}} = [1, 0]_{\mathbb{Z}},$$

and define

$$\iota_{\mathbb{Z}} : \mathbb{N}_0 \longrightarrow \mathbb{Z}, \qquad \iota_{\mathbb{Z}}(n) = [n, 0]_{\mathbb{Z}}.$$

Finally, define the order by

$$[a, b]_{\mathbb{Z}} < [c, d]_{\mathbb{Z}} \quad \Longleftrightarrow \quad a + d < b + c.$$

Proposition 2.19 (Integer arithmetic)**.** The relation $\sim_{\mathbb{Z}}$ is an equivalence relation, and the operations and order in the preceding definition are independent of representatives. The map $\iota_{\mathbb{Z}}$ is injective and preserves addition, multiplication, and order. With these operations, $\mathbb{Z}$ has commutative, associative, distributive addition and multiplication, identities $0_{\mathbb{Z}}, 1_{\mathbb{Z}}$, additive inverses, and no zero divisors. Its displayed order is total and is compatible with addition and multiplication by positive elements.

Proof. Reflexivity and symmetry of $\sim_{\mathbb{Z}}$ are immediate. If

$$a + d = b + c, \qquad c + f = d + e,$$

then

$$a + d + f = b + c + f = b + d + e.$$

Cancellation in $\mathbb{N}_0$ gives $a + f = b + e$, proving transitivity.

An operation written with representatives gives an operation on equivalence classes only when changing representatives does not change the resulting class. This is what **well defined** means here. For the displayed sum, suppose

$$(a, b) \sim_{\mathbb{Z}} (a', b'), \qquad (c, d) \sim_{\mathbb{Z}} (c', d').$$

Then

$$\begin{aligned}(a + c) + (b' + d') &= (a + b') + (c + d')\\ &= (b + a') + (d + c')\\ &= (b + d) + (a' + c').\end{aligned}$$

Thus

$$(a + c, b + d) \sim_{\mathbb{Z}} (a' + c', b' + d'),$$

so addition is well defined. Expanding the products and repeatedly using the defining equalities proves the same for the displayed product; this is the formal identity

$$(a - b)(c - d) = ac + bd - (ad + bc).$$

The comparison defining the order is likewise unchanged after adding the equalities associated with equivalent representatives. These checks show that all formulas descend to classes.

The associative, commutative, and distributive laws now follow from the corresponding arithmetic identities in $\mathbb{N}_0$. The formulas for $0_{\mathbb{Z}}$, $1_{\mathbb{Z}}$, and $-[a, b]_{\mathbb{Z}}$ give the required identities and additive inverses. The order says exactly that a formal difference is positive when its first entry exceeds its second; the order laws follow by adding inequalities and multiplying positive differences in $\mathbb{N}_0$. Consequently, a product can vanish only when one factor does. Finally,

$$[m, 0]_{\mathbb{Z}} + [n, 0]_{\mathbb{Z}} = [m + n, 0]_{\mathbb{Z}},$$

$$[m, 0]_{\mathbb{Z}}[n, 0]_{\mathbb{Z}} = [mn, 0]_{\mathbb{Z}},$$

and

$$[m, 0]_{\mathbb{Z}} < [n, 0]_{\mathbb{Z}} \iff m < n,$$

which proves the assertions about $\iota_{\mathbb{Z}}$. □

Definition 2.20 (The rational numbers)**.** On

$$\mathbb{Z} \times (\mathbb{Z} \setminus \{0_{\mathbb{Z}}\}),$$

define

$$(a, b) \sim_{\mathbb{Q}} (c, d)$$

if and only if

$$ad = bc.$$

The set of equivalence classes is denoted by

$$\mathbb{Q} = (\mathbb{Z} \times (\mathbb{Z} \setminus \{0_\mathbb{Z}\}))/ \sim_\mathbb{Q} .$$

Write $[a, b]_\mathbb{Q}$, or simply a/b, for the class of (a, b). Define

$$[a, b]_\mathbb{Q} + [c, d]_\mathbb{Q} = [ad + bc, bd]_\mathbb{Q},$$
$$[a, b]_\mathbb{Q}[c, d]_\mathbb{Q} = [ac, bd]_\mathbb{Q},$$

and

$$\iota_\mathbb{Q} : \mathbb{Z} \longrightarrow \mathbb{Q}, \qquad \iota_\mathbb{Q}(a) = [a, 1_\mathbb{Z}]_\mathbb{Q}.$$

The additive inverse of $[a, b]_\mathbb{Q}$ is $[-a, b]_\mathbb{Q}$; if $a \neq 0_\mathbb{Z}$, its multiplicative inverse is $[b, a]_\mathbb{Q}$. Define positivity by

$$[a, b]_\mathbb{Q} > 0 \iff ab > 0_\mathbb{Z}.$$

As in every ordered arithmetic system, define

$$r < s \iff s - r > 0.$$

Proposition 2.21 (Rational arithmetic)**.** The relation $\sim_\mathbb{Q}$, the operations, and the positivity relation in the preceding definition are well defined. The map $\iota_\mathbb{Q}$ is an injective order-preserving arithmetic embedding. The resulting system $\mathbb{Q}$ satisfies the ordered-field axioms stated in Chapter 3.

Proof. The only new point needed for the equivalence relation is cancellation of a nonzero integer. If

$$ad = bc, \qquad cf = de,$$

then

$$adf = bcf = bde.$$

Cancelling the nonzero factor d gives $af = be$, proving transitivity. Again, well definedness asks whether a formula has the same output class after representatives are changed. For addition, suppose

$$ab' = a'b, \qquad cd' = c'd.$$

Then

$$\begin{aligned}(ad + bc)b'd' &= ab'dd' + bb'cd' \\ &= a'bdd' + bb'c'd \\ &= (a'd' + b'c')bd.\end{aligned}$$

Hence

$$(ad + bc, bd) \sim_\mathbb{Q} (a'd' + b'c', b'd'),$$

so rational addition is well defined. The same cross-multiplication proves representative independence for multiplication.

The displayed formulas for additive and multiplicative inverses verify the inverse laws directly. If

$$[a, b]_\mathbb{Q} = [c, d]_\mathbb{Q},$$

then multiplying $ad = bc$ by bd shows that ab and cd have the same sign, because nonzero squares are positive in $\mathbb{Z}$. Thus positivity is well defined. It is closed under addition and multiplication, and every nonzero rational is either positive or has positive negative; these facts follow by choosing the signs of numerator and denominator. Hence the order is compatible with the operations. The remaining field identities descend from the integer identities through the displayed fraction formulas. Finally, the definition of $\iota_\mathbb{Q}$ immediately gives injectivity and preservation of arithmetic and order. □

2.5 Countability

Definition 2.22 (Countability)**.** Sets A, B have the same cardinality if a bijection $A \to B$ exists. A set is **finite** if it is empty or bijective with $\{1, \ldots, n\}$ for some $n \in \mathbb{N}$; it is **countably infinite** if it is bijective with $\mathbb{N}$; and **countable** if finite or countably infinite.

Proposition 2.23 (Elementary finite-set facts)**.** Every subset of a finite set is finite. If A and B are finite, then

$$A \cup B, \qquad A \times B$$

are finite. If A has m elements and B has n elements, then

$$A \times B$$

has mn elements.

Proof. Transport the question along bijections with initial segments of $\mathbb{N}$. A subset of

$$\{1, \ldots, m\}$$

can be listed in increasing order and therefore has at most m elements. The union is listed by first listing A and then retaining the members of B not already listed. Finally, the ordered pairs

$$(i, j), \qquad 1 \leq i \leq m, \quad 1 \leq j \leq n,$$

are enumerated by

$$(i, j) \longmapsto n(i - 1) + j,$$

which is a bijection onto $\{1, \ldots, mn\}$. □

Theorem 2.24 (Diagonal enumeration)**.** The set $\mathbb{N} \times \mathbb{N}$ is countable. If, for every $n \in \mathbb{N}$, a listing of a set A_n is given, then

$$\bigcup_{n=1}^{\infty} A_n$$

is countable.

Proof. For $s \geq 2$, the sth diagonal is the finite list of pairs (n, k) with $n + k = s$, ordered by increasing n. Every $(n, k) \in \mathbb{N} \times \mathbb{N}$ belongs to the unique diagonal $n + k$, so concatenating these finite lists gives a listing of $\mathbb{N} \times \mathbb{N}$ without omissions or repetitions.

Now let $a_{n,1}, a_{n,2}, \ldots$ be the given listing of A_n. Replace the pair (n, k) in the preceding diagonal list by $a_{n,k}$. If x belongs to the displayed union, then $x \in A_n$ for some n and hence $x = a_{n,k}$ for some k; therefore this new sequence contains x. It can contain repetitions, either because a listing repeats an element or because two sets overlap. Read the sequence from left to right and retain an entry exactly when it has not occurred earlier. If infinitely many entries remain, their positions give a bijection from $\mathbb{N}$ onto the union. If only finitely many remain, they give a finite listing. Thus the union is countable. □

Corollary 2.25. The rational numbers are countable.

Proof. The explicitly constructed integer classes are listed exactly once by

$$0, 1, -1, 2, -2, \dots,$$

so $\mathbb{Z}$ is countable. Hence $\mathbb{Z} \times \mathbb{N}$ is countable: compose its first-coordinate listing with the diagonal enumeration from Theorem 2.24. The map

$$\mathbb{Z} \times \mathbb{N} \longrightarrow \mathbb{Q}, \qquad (p, q) \longmapsto p/q,$$

is surjective because every rational class has a numerator in $\mathbb{Z}$ and, after changing both signs if necessary, a positive denominator in $\mathbb{N}$. Applying this map to a listing of $\mathbb{Z} \times \mathbb{N}$ gives a listing of $\mathbb{Q}$; retaining the first occurrence of each value makes it a finite listing or a bijection with $\mathbb{N}$. □

Theorem 2.26 (Cantor's diagonal theorem)**.** The interval $(0, 1)$ is uncountable.

Proof. This familiar decimal argument is an informal preview: Chapter 3 constructs the real numbers, and Chapter 4 later interprets decimal notation through limits of finite truncations. It is included here to display the diagonal idea before that analytic machinery is available.

Suppose, for contradiction, that $x_1, x_2, \dots$ listed $(0, 1)$. Give each x_n the decimal expansion that does not end in infinitely many 9s, and let d_n be its nth digit. Define $y = 0.e_1e_2\dots$, where $e_n = 1$ if $d_n \neq 1$, and $e_n = 2$ if $d_n = 1$. This decimal uses only 1 and 2, so it represents a number strictly between 0 and 1 and does not have an alternative expansion ending in infinitely many 9s. Its nth digit differs from d_n, so $y \neq x_n$ for every n. This contradicts the assumed listing. □

2.6 Choice

Definition 2.27 (Axiom of choice)**.** The **axiom of choice** asserts that for every indexed family $(A_i)_{i \in I}$ of nonempty sets, there is a function

$$c : I \to \bigcup_{i \in I} A_i$$

such that $c(i) \in A_i$ for every $i \in I$. Such a function is a **choice function**.

The assertion is invisible for a finite family when one can explicitly name elements. Its content is the right to make infinitely many selections even when no rule distinguishes a preferred member of each set. A useful mental image is an unlabelled collection of nonempty boxes: a choice function selects one item from every box at once, without requiring an algorithm for the selections.

Why should analysis care? Choice packages many local existence statements into a global object. For example, it yields the familiar unrestricted form of the countable-union theorem: for a sequence of nonempty countable sets, it permits one to select a listing for each set, after which the proof of Theorem 2.24 applies. It also supports the standard existence of vector-space bases and, in more advanced analysis, the extension principles underlying Hahn–Banach and the existence of ultrafilters. Those later results are not prerequisites for this text.

There is a trade-off. The advantage is powerful general existence theorems and concise structural arguments. The disadvantage is that a choice argument often supplies no explicit object and no procedure for finding one. In some foundational settings, very strong forms of choice also yield counterintuitive decompositions of sets. Accordingly, we distinguish an explicit construction from an existence proof using choice, and state the latter use when it occurs. The standard equivalents of the full axiom are Zorn's lemma and the well-ordering theorem; their proofs are outside the scope of these notes.

For the core development through Chapters 3–9, the proofs will be arranged to use explicit sequences and finite choices whenever possible. The construction of $\mathbb{R}$ uses a specified rational representative at a time, and the Bolzano–Weierstrass proof in Chapter 4 selects indices recursively from explicitly nonempty sets. If a later chapter invokes a general basis, a nonconstructive global selection, or a result equivalent to choice, the dependence will be identified at that point.

Exercises

Exercise 2.1. Prove that the set of all finite strings on a fixed finite alphabet is countable.

Exercise 2.2. Let $f : A \to B$ and let $E \subseteq A$.

(a) Prove that
$$f(f^{-1}(f(E))) = f(E).$$

(b) Prove that
$$E \subseteq f^{-1}(f(E)).$$

(c) Prove that
$$f^{-1}(f(E)) = E$$
whenever f is injective.

(d) Give an explicit example showing that the equality in the preceding part can fail when f is not injective.

Exercise 2.3. Suppose that
$$(a, b) \sim_{\mathbb{Z}} (a', b') \qquad \text{and} \qquad (c, d) \sim_{\mathbb{Z}} (c', d').$$
Verify directly that
$$(a + c, b + d) \sim_{\mathbb{Z}} (a' + c', b' + d').$$
Thus the addition of integer classes is well defined.

Exercise 2.4. Prove directly from the equivalence relation defining $\mathbb{Q}$ that
$$\iota_{\mathbb{Q}} : \mathbb{Z} \longrightarrow \mathbb{Q}, \qquad a \longmapsto [a, 1_{\mathbb{Z}}]_{\mathbb{Q}},$$
is injective and preserves order.

Exercise 2.5. Use induction to prove
$$1 + 2 + \cdots + n = \frac{n(n + 1)}{2} \qquad (n \in \mathbb{N}).$$
State clearly the base case and the induction step.

Exercise 2.6. Give an explicit bijection between
$$\mathbb{N}$$
and the set of even positive integers.

Chapter 3

The Real Number System

3.1 Ordered Fields and the Problem to Be Solved

Chapter 2 constructs the arithmetic chain

$$\mathbb{N} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q}.$$

The first extension permits subtraction, and the second permits division by a nonzero element. The rational numbers nevertheless have gaps: as Chapter 1 observed, no rational number has square 2. This chapter completes the chain

$$\mathbb{N} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{R}$$

by constructing a system without those analytic and order-theoretic gaps. We begin by isolating the algebraic and order structure that the final system must possess.

Definition 3.1 (Field)**.** A **field** is a set F with elements $0, 1$ $(0 \neq 1)$ and operations $+$ and $\cdot$ such that, for all $a, b, c \in F$, addition and multiplication are associative and commutative, multiplication distributes over addition, $0 + a = a$, $1a = a$, every a has an additive inverse $-a$, and every nonzero a has a multiplicative inverse a^{-1}.

Definition 3.2 (Ordered field)**.** An **ordered field** is a field F equipped with a total order $\leq$, in the sense of Chapter 2, that is compatible with its field operations: for all $a, b, c \in F$,

$$a \leq b \quad \Longrightarrow \quad a + c \leq b + c,$$

and

$$0 \leq a \text{ and } 0 \leq b \quad \Longrightarrow \quad 0 \leq ab.$$

Proposition 3.3 (Positive-cone characterization)**.** Let F be a field. An ordered-field structure on F is equivalently specified by a subset

$$P = \{x \in F : x > 0\}$$

such that

$$P + P \subseteq P, \qquad PP \subseteq P,$$

where

$$P + P = \{p + q : p, q \in P\}, \qquad PP = \{pq : p, q \in P\},$$

and, for every $x \in F$, exactly one of

$$x = 0, \qquad x \in P, \qquad -x \in P$$

holds. Conversely, such a subset defines an ordered-field structure by

$$x \leq y \quad \Longleftrightarrow \quad x = y \text{ or } y - x \in P.$$

Proof. Suppose first that F is an ordered field. If $a, b > 0$, compatibility with addition gives $a + b \geq a > 0$; equality $a + b = a$ would force $b = 0$. Compatibility with multiplication gives $ab \geq 0$. If $ab = 0$, multiplication by a^{-1} gives $b = 0$, contradicting $b > 0$; hence $ab > 0$. Totality gives the displayed trichotomy.

Conversely, let P satisfy the stated conditions. The trichotomy for zero shows that $0 \notin P$. The displayed relation is reflexive. If both $x \leq y$ and $y \leq x$ hold and $x \neq y$, then both $y - x$ and $x - y$ belong to P, so closure under addition would put 0 in P, a contradiction; hence the relation is antisymmetric. Closure under addition also gives transitivity, and the trichotomy for $y - x$ gives totality. Its compatibility with addition is immediate from

$$(y + c) - (x + c) = y - x.$$

Finally, if $0 \leq a$ and $0 \leq b$, then either factor is zero or both belong to P, in which case $ab \in P$. Thus multiplication preserves nonnegativity, as required. □

Definition 3.4 (Absolute value)**.** In an ordered field, define $|x| = x$ if $x \geq 0$ and $|x| = -x$ if $x < 0$.

Proposition 3.5 (Basic order rules)**.** In an ordered field, $<$ is the strict order associated with a transitive total order. If $a < b$, then $a + c < b + c$ for every c; if $a < b$ and $0 < c$, then $ac < bc$. Also $a^2 \geq 0$ for every a, and

$$|a + b| \leq |a| + |b|, \qquad |ab| = |a||b|$$

for all a, b.

Proof. The total-order axioms give totality and transitivity. Addition compatibility gives $a + c \leq b + c$, and equality would imply $a = b$ after cancellation; hence $a + c < b + c$. If $a < b$ and $0 < c$, then

$$0 \leq (b - a)c.$$

If this product were zero, multiplication by $(b - a)^{-1}$ would give $c = 0$, a contradiction. Thus $bc - ac > 0$ and $ac < bc$. Totality gives either $a \geq 0$ or $-a \geq 0$, and multiplication compatibility then gives $a^2 \geq 0$. The four possible sign combinations for a and b prove the product formula for absolute values. Applying the order rules to

$$-|a| \leq a \leq |a|, \qquad -|b| \leq b \leq |b|$$

gives

$$-|a| - |b| \leq a + b \leq |a| + |b|,$$

which is the triangle inequality. □

Definition 3.6 (Upper bounds and completeness)**.** Let $S \subseteq F$. An element $u \in F$ is an **upper bound** of S if $s \leq u$ for every $s \in S$. A **least upper bound** of S is an upper bound u with $u \leq v$ for every upper bound v of S. An ordered field has the **least-upper-bound property** if every nonempty set having an upper bound has a least upper bound.

Definition 3.7 (Complete ordered field)**.** An ordered field having the least-upper-bound property is called a **complete ordered field**. This is a characterization, not a construction: it says what the real numbers are like, but does not yet exhibit them.

3.2 Rational Numbers, Completeness, and Uniqueness

The preceding axioms describe the target. Before constructing a model, we show that they already determine its arithmetic and order uniquely. The key is that every ordered field contains a forced copy of the rational numbers.

Definition 3.8 (Field homomorphisms and isomorphisms)**.** Let F and G be fields. A **field homomorphism** is a map

$$\phi : F \longrightarrow G$$

such that

$$\phi(1_F) = 1_G, \qquad \phi(x + y) = \phi(x) + \phi(y), \qquad \phi(xy) = \phi(x)\phi(y).$$

A bijective field homomorphism is a **field isomorphism**. Between ordered fields, a homomorphism is **order preserving** if

$$x < y \quad \Longrightarrow \quad \phi(x) < \phi(y).$$

An order-preserving field isomorphism is called an **ordered-field isomorphism**.

Theorem 3.9 (Canonical rational copy)**.** Every ordered field F has a unique order-preserving field homomorphism

$$\iota_F : \mathbb{Q} \longrightarrow F.$$

It is injective, and its image is an ordered subfield of F. Thus we identify $\mathbb{Q}$ with its image in F.

Proof. For $n \in \mathbb{N}$, let n_F be the sum of n copies of 1_F. For negative integers, use additive inverses to define m_F. Induction gives $n_F > 0$, hence $n_F \neq 0$. Define

$$\iota_F(m/n) = m_F(n_F)^{-1} \qquad (m \in \mathbb{Z},\ n \in \mathbb{N}).$$

To check well definedness, suppose

$$\frac{m}{n} = \frac{p}{q}.$$

Then $mq = np$, so the corresponding integer elements of F satisfy

$$m_F q_F = n_F p_F.$$

Multiplying by $(n_F)^{-1}(q_F)^{-1}$ gives

$$m_F(n_F)^{-1} = p_F(q_F)^{-1}.$$

Thus the definition does not depend on the chosen fraction. The field laws then directly show that ι_F preserves addition and multiplication.

If $u > 0$ in an ordered field, then $u^{-1} > 0$: it is nonzero, and if $u^{-1} < 0$, multiplication of this inequality by $u > 0$ would give

$$1 = u^{-1}u < 0,$$

contrary to $1 = 1^2 > 0$. A positive rational has a representation m/n with $m, n \in \mathbb{N}$, so $m_F > 0$ and $(n_F)^{-1} > 0$; hence its image is positive. Therefore ι_F is order preserving and thus injective. Any field homomorphism fixes 1, then every integer, and then every quotient of integers; uniqueness follows. □

Theorem 3.10 (Archimedean properties)**.** Let F be a complete ordered field. Then:

(a) for every $x \in F$, there is $N \in \mathbb{N}$ such that

$$x < N;$$

(b) for every $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that

$$\frac{1}{N} < \varepsilon.$$

Proof. Identify $\mathbb{Q}$ with its canonical copy. If $\mathbb{N}$ had an upper bound, let $s = \sup \mathbb{N}$. Since $s - 1$ is not an upper bound, some $n \in \mathbb{N}$ satisfies

$$s - 1 < n.$$

Then $s < n + 1$, contradicting that s is an upper bound. Thus $\mathbb{N}$ is unbounded above, which proves (a). Apply (a) to $1/\varepsilon$ to obtain $N > 1/\varepsilon$, proving (b). □

Theorem 3.11 (Rational density)**.** Let F be a complete ordered field and let $x < y$ in F. Then there is $q \in \mathbb{Q}$ such that

$$x < q < y.$$

Proof. Choose $N \in \mathbb{N}$ with

$$N(y - x) > 1.$$

Choose $M \in \mathbb{N}$ with $M > -Nx$. The set

$$E = \{n \in \mathbb{N} : n > Nx + M\}$$

is nonempty and has a least member k. Minimality gives

$$k - 1 \leq Nx + M < k,$$

and hence

$$Nx < k - M < Nx + 1 < Ny.$$

Divide by the positive integer N and take

$$q = \frac{k - M}{N}.$$

□

Lemma 3.12 (Rational cuts determine points)**.** If F is a complete ordered field and $x, y \in F$ satisfy

$$\{q \in \mathbb{Q} : q < x\} = \{q \in \mathbb{Q} : q < y\},$$

then $x = y$.

Proof. If $x < y$, rational density gives a rational strictly between them, which belongs to exactly one of the two sets. The case $y < x$ is symmetric. □

Theorem 3.13 (Uniqueness of complete ordered fields)**.** If F and G are complete ordered fields, there is a unique order-preserving field isomorphism

$$\Phi : F \longrightarrow G$$

which identifies their canonical rational copies.

Proof. Identify both rational copies with $\mathbb{Q}$. For $x \in F$, let

$$L_F(x) = \{q \in \mathbb{Q} : q < x\}.$$

The Archimedean property and rational density make this set nonempty and bounded above in G. Define

$$\Phi(x) = \sup_G L_F(x).$$

For every rational q,

$$q < \Phi(x) \iff q < x. \tag{*}$$

Indeed, if $q < x$, choose $r \in \mathbb{Q}$ with $q < r < x$, so $q < r \le \Phi(x)$. Conversely, if $x \le q$, then q is an upper bound of $L_F(x)$.

The equivalence (*) shows that Φ is strictly order preserving and fixes $\mathbb{Q}$. It also proves additivity: for a rational q,

$$q < x + y$$

holds exactly when there are rationals $p < x$ and $r < y$ with

$$q = p + r.$$

For the nontrivial implication, choose $p \in \mathbb{Q}$ between $q - y$ and x and put $r = q - p$. Apply (*) in both fields and then use the preceding lemma to conclude

$$\Phi(x + y) = \Phi(x) + \Phi(y).$$

For positive x, y, we next compare the lower cuts of the products. If

$$q < 0,$$

then automatically $q < xy$, and the same is true of

$$\Phi(x)\Phi(y).$$

Now let $q \in \mathbb{Q}$ satisfy

$$q \ge 0.$$

We claim that

$$q < xy$$

if and only if there are positive rationals p, r such that

$$0 < p < x, \qquad 0 < r < y, \qquad q < pr. \tag{**}$$

The reverse implication follows from the order rules. Conversely, if

$$q < xy,$$

then

$$\frac{q}{y} < x.$$

Rational density gives a positive rational p with

$$\frac{q}{y} < p < x.$$

Because $p > q/y$, one has $q/p < y$; rational density again gives a positive rational r with

$$\frac{q}{p} < r < y.$$

Thus $q < pr$, proving (**).

The equivalence (*) transfers the inequalities

$$p < x, \qquad r < y$$

exactly to

$$p < \Phi(x), \qquad r < \Phi(y).$$

Applying (**) in F and in G therefore shows that xy and $\Phi(x)\Phi(y)$ have the same rational lower cut. By Lemma 3.12,

$$\Phi(xy) = \Phi(x)\Phi(y)$$

when $x, y > 0$. Additivity gives

$$\Phi(-x) = -\Phi(x).$$

If one factor is zero, multiplicativity is immediate. If exactly one factor is negative, apply the positive case to its negative and use this identity; if both are negative, apply it twice. Hence multiplicativity holds for all

$$x, y \in F.$$

Thus Φ is an order-preserving field homomorphism.

Construct $\Psi : G \to F$ in the same way. By (*), the maps $\Psi\Phi$ and the identity of F have the same rational lower cuts, so they are equal by Lemma 3.12; similarly $\Phi\Psi$ is the identity of G. Hence Φ is an isomorphism. Finally, any order-preserving field isomorphism fixing $\mathbb{Q}$ has property (*), because it preserves rational lower cuts; therefore it equals Φ. □

Definition 3.14 (The real number system)**.** The **real number system**, denoted by $\mathbb{R}$, is a complete ordered field. The preceding theorem makes this definition unambiguous up to the unique order-preserving field isomorphism fixing $\mathbb{Q}$. The construction below proves that such a field exists.

3.3 Cauchy Sequences of Rational Numbers

The preceding discussion gives the intended abstract meaning of $\mathbb{R}$ and proves that such a field, if it exists, is unique. We now supply existence by constructing a concrete complete ordered field from rational Cauchy sequences. The uniqueness theorem will then identify this model with the real-number system just characterized.

We use the usual ordered-field properties of $\mathbb{Q}$, which Chapter 2 establishes from its construction as equivalence classes of integer pairs (p, q) with $q \neq 0$. No completeness property of $\mathbb{Q}$ will be used.

Definition 3.15 (Cauchy sequence in $\mathbb{Q}$)**.** A sequence $(q_n)_{n\in\mathbb{N}}$ in $\mathbb{Q}$ is **Cauchy** if, for every rational $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that $m, n \geq N$ implies $|q_m - q_n| < \varepsilon$.

Lemma 3.16 (Elementary Cauchy facts)**.** Let (q_n) and (r_n) be Cauchy sequences in $\mathbb{Q}$.

(a) Each sequence is bounded: there is $M \in \mathbb{Q}$ with $|q_n| \leq M$ for all n.

(b) The sequences $(q_n + r_n)$ and $(q_n r_n)$ are Cauchy.

(c) If $(q_n) \sim (q'_n)$ and $(r_n) \sim (r'_n)$, then

$$(q_n + r_n) \sim (q'_n + r'_n) \quad \text{and} \quad (q_n r_n) \sim (q'_n r'_n).$$

Proof. Choose N such that $m, n \geq N$ gives $|q_m - q_n| < 1$. Then $|q_n| \leq |q_N| + 1$ for $n \geq N$; enlarge this rational bound to cover the finitely many earlier terms. For sums, apply the triangle inequality. For products, take a common bound M for both sequences and use

$$|q_m r_m - q_n r_n| \leq |q_m||r_m - r_n| + |r_n||q_m - q_n| \leq M\big(|r_m - r_n| + |q_m - q_n|\big).$$

For the final assertion, the triangle inequality gives the sum statement. Choose a common rational bound M for the four sequences. The identity

$$q_n r_n - q'_n r'_n = q_n(r_n - r'_n) + r'_n(q_n - q'_n)$$

then gives

$$|q_n r_n - q'_n r'_n| \leq M|r_n - r'_n| + M|q_n - q'_n|,$$

which tends to 0. ☐

Definition 3.17 (Null equivalence)**.** For rational Cauchy sequences (q_n) and (r_n), write $(q_n) \sim (r_n)$ if, for every $\varepsilon \in \mathbb{Q}$ with $\varepsilon > 0$, there is N such that $n \geq N$ implies $|q_n - r_n| < \varepsilon$.

Lemma 3.18. The relation $\sim$ is an equivalence relation on the set of rational Cauchy sequences.

Proof. Reflexivity and symmetry are immediate. If $q \sim r$ and $r \sim s$, then for a given $\varepsilon > 0$, both $|q_n - r_n| < \varepsilon/2$ and $|r_n - s_n| < \varepsilon/2$ eventually hold. The triangle inequality gives $|q_n - s_n| < \varepsilon$, proving transitivity. ☐

Lemma 3.19 (Eventual separation)**.** Let (q_n) be a rational Cauchy sequence that is not null-equivalent to the zero sequence. Then there are $\delta > 0$ in $\mathbb{Q}$ and $N \in \mathbb{N}$ such that either $q_n \geq \delta$ for every $n \geq N$, or $q_n \leq -\delta$ for every $n \geq N$.

Proof. Since (q_n) is not null, there is $\eta > 0$ for which $|q_n| \geq \eta$ for infinitely many n. Choose N so that $|q_m - q_n| < \eta/2$ whenever $m, n \geq N$, and choose $k \geq N$ with $|q_k| \geq \eta$. If $q_k \geq \eta$, then $q_n > \eta/2$ for every $n \geq N$. If $q_k \leq -\eta$, then $q_n < -\eta/2$ for every $n \geq N$. Take $\delta = \eta/2$. ☐

Construction 3.20 (The set of real numbers)**.** Define $\mathbb{R}$ to be the set of equivalence classes $[(q_n)]$ under $\sim$. Define

$$[(q_n)] + [(r_n)] = [(q_n + r_n)], \qquad [(q_n)]\,[(r_n)] = [(q_n r_n)], \qquad -[(q_n)] = [(-q_n)].$$

For $a \in \mathbb{Q}$, define $\iota(a) = [(a, a, a, \ldots)]$.

Proposition 3.21 (Algebra of the construction)**.** The operations in construction 3.20 are well defined. With $0 = \iota(0)$ and $1 = \iota(1)$, they make $\mathbb{R}$ a field, and $\iota : \mathbb{Q} \to \mathbb{R}$ is an injective field homomorphism.

Proof. The last part of Lemma 3.16 proves well-definedness of addition and multiplication; it also proves that the displayed sequences are Cauchy. All associative, commutative, distributive, and identity laws hold term by term for rational sequences and therefore for their classes. Additive inverses are termwise. If $x = [(q_n)] \neq 0$, Lemma 3.19 gives $\delta > 0$ and N such that $|q_n| \geq \delta$ for every $n \geq N$. Define

$$p_n = \begin{cases} 1, & n < N, \\ q_n^{-1}, & n \geq N. \end{cases}$$

The estimate

$$|p_m - p_n| = \frac{|q_m - q_n|}{|q_m q_n|} \leq \frac{1}{\delta^2}|q_m - q_n|$$

for $m, n \geq N$ shows that (p_n) is Cauchy. Since $q_n p_n = 1$ for every $n \geq N$, its product class is 1; hence $[(p_n)]$ is the multiplicative inverse of x. Finally, constant-sequence operations show that ι is a homomorphism; if $\iota(a) = \iota(b)$, then the constant difference $a - b$ is smaller in absolute value than every positive rational, which is possible only when $a = b$. □

3.4 Order and Completeness

Definition 3.22 (Order on $\mathbb{R}$)**.** For $x = [(q_n)] \in \mathbb{R}$, write $x > 0$ if there are rational numbers $\delta > 0$ and N such that $q_n > \delta$ for every $n \geq N$. Define $x < y$ to mean $y - x > 0$.

Lemma 3.23. The preceding definition is independent of the chosen representative. It makes $\mathbb{R}$ an ordered field, and ι preserves order.

Proof. If $q_n > \delta$ eventually and $q \sim r$, then $|q_n - r_n| < \delta/2$ eventually, so $r_n > \delta/2$ eventually. Thus positivity is well defined. Positive eventual lower bounds are closed under addition and multiplication. For a nonzero class, Lemma 3.19 gives exactly one eventual sign. Therefore Proposition 3.3 makes the resulting relation a compatible total order, and Proposition 3.5 supplies its basic rules. Constant sequences visibly preserve rational order. □

Theorem 3.24 (Archimedean property)**.** For every $x \in \mathbb{R}$, there is $N \in \mathbb{N}$ with $x < N$. Consequently, for every $\varepsilon > 0$, there is $N \in \mathbb{N}$ with $1/N < \varepsilon$.

Proof. Let $x = [(q_n)]$. By Lemma 3.16, the rational sequence (q_n) has a rational upper bound M. Choose an integer $N > M$, using the Archimedean property of $\mathbb{Q}$. Then $q_n < N$ for all n, hence $x \leq N$; replacing N by $N + 1$ yields $x < N + 1$. For the consequence choose $N > 1/\varepsilon$. □

Lemma 3.25 (Integer-part lemma)**.** For every $x \in \mathbb{R}$, there is an integer $k \in \mathbb{Z}$ such that

$$k \leq x < k + 1.$$

Proof. By Theorem 3.24, choose $M \in \mathbb{N}$ with $M > -x$. The nonempty set

$$E = \{n \in \mathbb{N} : n > x + M\}$$

has a least element n by well ordering. Since $n - 1 \notin E$ when $n > 1$, one has $n - 1 \leq x + M < n$. The case $n = 1$ gives the same left inequality because $x + M > 0$. Thus $k = n - M - 1$ is an integer satisfying $k \leq x < k + 1$. □

Theorem 3.26 (Density of the rational numbers)**.** If $x, y \in \mathbb{R}$ and $x < y$, then there is $q \in \mathbb{Q}$ such that

$$x < \iota(q) < y.$$

Proof. By Theorem 3.24, choose $N \in \mathbb{N}$ with $N(y-x) > 1$. Apply Lemma 3.25 to Nx to obtain $k \in \mathbb{Z}$ with $k \leq Nx < k+1$. Then

$$Nx < k+1 < Nx+1 < Ny.$$

Dividing by the positive integer N gives the result for $q = (k+1)/N$. □

Definition 3.27 (Convergence of a real sequence)**.** Let (x_n) be a sequence in $\mathbb{R}$, and let $x \in \mathbb{R}$. We say that (x_n) **converges to** x, and write

$$\lim_{n\to\infty} x_n = x,$$

if for every $\varepsilon > 0$ there is $N \in \mathbb{N}$ such that

$$n \geq N \quad \Longrightarrow \quad |x_n - x| < \varepsilon.$$

Definition 3.28 (Cauchy sequence in $\mathbb{R}$)**.** A real sequence (x_n) is **Cauchy** if, for every $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that $m, n \geq N$ implies $|x_m - x_n| < \varepsilon$.

Lemma 3.29 (Tail estimates for classes)**.** Let (u_n) be a rational Cauchy sequence, let $u = [(u_n)]$, and let $e \in \mathbb{Q}$ satisfy $e > 0$.

(a) If $|u_n| < e$ for every sufficiently large n, then

$$|u| < \iota(e).$$

(b) If (u_n) is eventually bounded below by $a \in \mathbb{Q}$ and eventually bounded above by $b \in \mathbb{Q}$, then

$$\iota(a) \leq u \leq \iota(b).$$

Proof. For the first assertion, the inequalities $|u_n| < e$ imply

$$e - u_n > 0, \qquad e + u_n > 0$$

eventually. In fact both quantities are eventually greater than $e/2$. Thus $\iota(e) - u > 0$ and $\iota(e) + u > 0$, which says precisely that $|u| < \iota(e)$. For the second assertion, $u_n \geq a$ eventually implies that $\iota(a) - u$ cannot be positive: otherwise the definition of positivity would give $a - u_n > \delta > 0$ eventually, contradicting $u_n \geq a$. Hence $\iota(a) \leq u$. The upper estimate is analogous. □

Lemma 3.30 (Rational approximation from a representative)**.** Let $x = [(q_n)] \in \mathbb{R}$. Then

$$\lim_{n\to\infty} \iota(q_n) = x.$$

Consequently, for every $\varepsilon > 0$ there is a rational q such that

$$|\iota(q) - x| < \varepsilon.$$

Proof. Let $\varepsilon > 0$. By Theorem 3.26, choose $e \in \mathbb{Q}$ with

$$0 < \iota(e) < \varepsilon.$$

Cauchyness of (q_n) supplies N such that

$$m, n \geq N \quad \Longrightarrow \quad |q_m - q_n| < e.$$

For a fixed $n \geq N$, the real number $\iota(q_n) - x$ is represented by the sequence $(q_n - q_k)_{k\in\mathbb{N}}$, whose tail has absolute value less than e. By Lemma 3.29,

$$|\iota(q_n) - x| < \iota(e) < \varepsilon.$$

This proves the convergence. Taking one sufficiently large n proves the final assertion. □

Theorem 3.31 (Cauchy completeness of the construction)**.** Every Cauchy sequence in $\mathbb{R}$ converges to an element of $\mathbb{R}$.

Proof. Let (x_j) be Cauchy. By Lemma 3.30, choose $r_j \in \mathbb{Q}$ with

$$|\iota(r_j) - x_j| < \iota(2^{-j}).$$

We claim that (r_j) is a rational Cauchy sequence. Given $e \in \mathbb{Q}$ with $e > 0$, choose J with $2^{-J} < e/3$. Since (x_j) is Cauchy, there is K such that

$$j, k \geq K \quad \Longrightarrow \quad |x_j - x_k| < \iota(e/3).$$

Hence, for $j, k \geq \max\{J, K\}$,

$$|\iota(r_j) - \iota(r_k)| \leq |\iota(r_j) - x_j| + |x_j - x_k| + |x_k - \iota(r_k)| < \iota(e).$$

Because ι preserves order, this is exactly $|r_j - r_k| < e$.

Let $x = [(r_j)] \in \mathbb{R}$. Given $\varepsilon > 0$, choose a rational $e > 0$ with

$$\iota(3e) < \varepsilon.$$

The convergence in Lemma 3.30 supplies J such that

$$|\iota(r_J) - x| < \iota(e).$$

Increasing J if necessary, we may also arrange $2^{-J} < e$ and hence

$$|\iota(r_J) - x_J| < \iota(e).$$

Choose K so that

$$j \geq K \quad \Longrightarrow \quad |x_j - x_J| < \iota(e).$$

For $j \geq K$, the triangle inequality gives

$$|x_j - x| \leq |x_j - x_J| + |x_J - \iota(r_J)| + |\iota(r_J) - x| < \iota(3e) < \varepsilon.$$

Thus $x_j \to x$. □

Theorem 3.32 (Completeness of the constructed field)**.** The ordered field $\mathbb{R}$ of construction 3.20 has the least-upper-bound property.

Proof. Let $S \subseteq \mathbb{R}$ be nonempty and bounded above. Choose $s_0 \in S$ and an upper bound u_0. If s_0 is an upper bound, then it is a maximum of S and hence its least upper bound. Otherwise s_0 is not an upper bound. Since every element of S is at most u_0, one has $s_0 < u_0$. Bisect $[s_0, u_0]$: at stage n, let $m_n = (s_n + u_n)/2$; if m_n is an upper bound of S, put $(s_{n+1}, u_{n+1}) = (s_n, m_n)$, and otherwise put $(s_{n+1}, u_{n+1}) = (m_n, u_n)$. Then s_n is not an upper bound, u_n is an upper bound, and $u_n - s_n = 2^{-n}(u_0 - s_0)$.

The sequence (s_n) is Cauchy: for $m \geq n$, $0 \leq s_m - s_n \leq u_n - s_n = 2^{-n}(u_0 - s_0)$, and the right side tends to 0 by Theorem 3.24. By Theorem 3.31, let $s_n \to a$. The triangle inequality shows directly that $u_n \to a$, because $|u_n - a| \leq |u_n - s_n| + |s_n - a|$. Finally, for fixed n, the inequalities $s_n \leq s_m \leq u_m \leq u_n$ for $m \geq n$ pass to the limit: if, for example, $s_n > a$, then s_m cannot lie within $(s_n - a)/2$ of a for any $m \geq n$. Hence $s_n \leq a \leq u_n$.

If $t \in S$ and $t > a$, choose n with $u_n - s_n < t - a$; then $t > u_n$, contradicting that u_n is an upper bound. Thus a is an upper bound. If $v < a$, choose n with $u_n - s_n < a - v$; then $v < s_n$. Since s_n is not an upper bound, neither is v. Hence a is the least upper bound. □

The construction has now supplied the existence promised in Definition 3.14. Therefore the ordered field constructed in construction 3.20 is, up to the unique order-preserving field isomorphism fixing $\mathbb{Q}$, the real number system. The remaining results record useful equivalent forms and consequences of its completeness.

Other standard descriptions—Dedekind cuts, suitable decimal expansions, and nested-interval models—are not constructed in full here. The uniqueness theorem does not by itself prove that such descriptions are equivalent to this model: for each one, it remains necessary to prove the field, order, and completeness properties. Once that has been done, however, the uniqueness theorem supplies its canonical order-preserving field isomorphism with the Cauchy-sequence model and hence with $\mathbb{R}$.

Theorem 3.33 (Greatest-lower-bound property)**.** Every nonempty subset of $\mathbb{R}$ that is bounded below has a greatest lower bound.

Proof. Let $S \subseteq \mathbb{R}$ be nonempty and bounded below. Define

$$-S = \{-s : s \in S\}.$$

Then $-S$ is nonempty and bounded above. By Theorem 3.32, let $u = \sup(-S)$, and put $\ell = -u$. For every $s \in S$, one has $-s \leq u$, hence $\ell \leq s$; thus ℓ is a lower bound. If v is any lower bound of S, then $-s \leq -v$ for every $s \in S$, so $-v$ is an upper bound of $-S$. Therefore $u \leq -v$, or $v \leq \ell$. Hence ℓ is the greatest lower bound. □

Theorem 3.34 (Nested-interval property)**.** Let

$$[a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots$$

be nested closed intervals in $\mathbb{R}$, and suppose

$$\lim_{n\to\infty} (b_n - a_n) = 0.$$

Then there is exactly one real number belonging to every interval.

Proof. The sequence (a_n) is increasing and bounded above by b_1, so its range has a least upper bound c. For every fixed n, nesting gives $a_m \leq a_n \leq b_n$ when $m < n$, and $a_m \leq b_m \leq b_n$ when $m \geq n$. Thus b_n is an upper bound of the whole range, so $c \leq b_n$. Also $a_n \leq c$ by definition of upper bound. Hence $c \in [a_n, b_n]$ for every n.

If d belongs to every interval, then $|c-d| \leq b_n - a_n$ for every n. Taking the limit gives $|c-d| = 0$, so $c = d$. Thus the common point is unique. □

3.5 A Small Toolkit of Inequalities

Exact computation is often unnecessary in analysis. The decisive step is usually an estimate that replaces a complicated quantity by one already known to be small or bounded.

Proposition 3.35 (Basic absolute-value estimates)**.** For $x, y \in \mathbb{R}$,

$$|x + y| \leq |x| + |y|, \qquad ||x| - |y|| \leq |x - y|.$$

Consequently, $|x_1 + \cdots + x_n| \leq \sum_{j=1}^{n} |x_j|$.

Proof. The triangle inequality was established with the ordered-field absolute value. Applying it to $x = (x - y) + y$ gives $|x| - |y| \le |x - y|$; exchange x and y for the reverse bound. The finite form follows by induction. □

Proposition 3.36 (Quadratic and Bernoulli inequalities)**.** For $u, v \in \mathbb{R}$,

$$2uv \le u^2 + v^2.$$

If $t \ge -1$ and $n \in \mathbb{N}$, then

$$(1 + t)^n \ge 1 + nt.$$

Proof. The first inequality is $(u - v)^2 \ge 0$. For the second, induction begins at $n = 1$. If the assertion holds for n, multiplication by $1 + t \ge 0$ gives

$$(1 + t)^{n+1} \ge 1 + (n + 1)t + nt^2 \ge 1 + (n + 1)t.$$

□

Remark 3.37 (How estimates are used)**.** The triangle inequality splits an error into controllable pieces; the reverse triangle inequality keeps denominators away from zero; the quadratic estimate turns products into sums; and Bernoulli compares powers with linear growth. For nonnegative inputs, the quadratic estimate later yields the usual arithmetic–geometric mean inequality after roots are available. Chapter 10 adds Cauchy–Schwarz and the norm comparisons

$$|x_i| \le \|x\|_2 \le \sqrt{n}\, \|x\|_\infty$$

for multivariable estimates. Later examples will repeatedly seek a sufficiently strong bound rather than an exact formula.

Exercises

Exercise 3.1. Use the least-upper-bound property to prove the greatest-lower-bound property by applying it to

$$-S = \{-s : s \in S\}.$$

State precisely the hypotheses on S.

Exercise 3.2. Let

$$S = \left\{x \in \mathbb{R} : x^2 < 2,\ x \ge 0\right\}.$$

Use completeness and the algebraic order rules to prove that

$$\sup S$$

has square 2.

Exercise 3.3. Let F be a complete ordered field. Prove that every nonempty open interval in F contains a rational element.

Chapter 4

Sequences of Real Numbers

4.1 Convergence

In this chapter the real numbers are the ordered field constructed in Chapter 3. A sequence in $\mathbb{R}$ is a function $a : \mathbb{N} \to \mathbb{R}$; its value at n is written a_n.

Calculus repeatedly asks what happens after indefinitely many successive approximations. Convergence turns that informal question into a test with a clear responsibility: after a requested accuracy is named, all sufficiently late terms must meet it.

Definition 4.1 (Limit of a sequence)**.** Let (a_n) be a real sequence and let $L \in \mathbb{R}$. We say that (a_n) **converges to** L, and write

$$\lim_{n\to\infty} a_n = L,$$

or equivalently $a_n \to L$, if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that, for every $n \geq N$, one has $|a_n - L| < \varepsilon$. A sequence that converges to some real number is **convergent**; otherwise it is **divergent**.

The number N may depend on ε, but it must work for every $n \geq N$. This order of quantifiers is essential.

This definition also gives the eventual meaning of the decimal notation used informally in Chapter 2: once Chapter 3 has constructed $\mathbb{R}$, the notation

$$d = 0.d_1 d_2 \cdots$$

is interpreted through its finite truncations

$$s_n = 0.d_1 d_2 \cdots d_n.$$

It denotes the real number d when

$$\lim_{n\to\infty} s_n = d.$$

Lemma 4.2. If $x \in \mathbb{R}$ satisfies $|x| < \varepsilon$ for every $\varepsilon > 0$, then $x = 0$.

Proof. If $x \neq 0$, then $|x| > 0$. Taking $\varepsilon = |x|/2$ contradicts the hypothesis. $\square$

Theorem 4.3 (Uniqueness of limits)**.** A convergent real sequence has exactly one limit.

Proof. Suppose $a_n \to L$ and $a_n \to M$. Given $\varepsilon > 0$, choose N_1, N_2 such that $|a_n - L| < \varepsilon/2$ for $n \geq N_1$ and $|a_n - M| < \varepsilon/2$ for $n \geq N_2$. For $n \geq \max\{N_1, N_2\}$, the triangle inequality gives

$$|L - M| \leq |L - a_n| + |a_n - M| < \varepsilon.$$

By Lemma 4.2, $L - M = 0$. □

Example 4.4. The sequence $a_n = 1/n$ converges to 0. For a given $\varepsilon > 0$, Theorem 3.24 provides $N > 1/\varepsilon$, and then $n \geq N$ implies

$$0 < \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

Definition 4.5 (Divergence to infinity)**.** We write

$$a_n \longrightarrow +\infty$$

if, for every $M \in \mathbb{R}$, there is $N \in \mathbb{N}$ such that

$$n \geq N \quad \Longrightarrow \quad a_n > M.$$

We write $a_n \to -\infty$ if, for every $M \in \mathbb{R}$, there is N such that

$$n \geq N \quad \Longrightarrow \quad a_n < M.$$

These statements describe two particular forms of divergence. Neither $+\infty$ nor $-\infty$ is a real number, and neither statement says that (a_n) converges in $\mathbb{R}$.

Example 4.6 (Examples and nonexamples)**.** The sequence n^2 tends to $+\infty$: given $M \in \mathbb{R}$, choose $N > \sqrt{\max\{M, 0\}}$. The sequence $-n$ tends to $-\infty$. By contrast, $(-1)^n n$ is unbounded in both directions but tends to neither infinity; for every proposed stage, later terms occur with each sign. Thus "unbounded" is much weaker than "eventually larger than every real bound."

Remark 4.7 (A useful proof pattern)**.** To prove $a_n \to +\infty$, start with an arbitrary target height M and solve the inequality $a_n > M$ for a sufficient condition on n. This is the infinite-limit analogue of solving $|a_n - L| < \varepsilon$ in an ordinary ε–N proof.

4.2 Algebra and Order of Limits

Once limits have been defined, the next question is whether familiar algebra survives passage to the limit. The following rules justify the calculations that will later occur inside functions, series, derivatives, and integrals.

Theorem 4.8 (Algebra of limits)**.** Let (a_n) and (b_n) be real sequences with $a_n \to a$ and $b_n \to b$, and let $c \in \mathbb{R}$. Then

$$\lim_{n\to\infty}(a_n + b_n) = a + b, \qquad \lim_{n\to\infty} ca_n = ca, \qquad \lim_{n\to\infty} a_n b_n = ab.$$

If $b \neq 0$, then

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{a}{b}$$

whenever $b_n \neq 0$ for all sufficiently large n.

Proof. For the sum, choose stages at which $|a_n - a| < \varepsilon/2$ and $|b_n - b| < \varepsilon/2$, then apply the triangle inequality. Scalar multiplication follows from $|ca_n - ca| = |c||a_n - a|$; for $c = 0$ it is immediate, and otherwise use tolerance $\varepsilon/|c|$.

Convergent sequences are eventually bounded: choosing N with $|a_n - a| < 1$ gives $|a_n| \le |a| + 1$ for $n \ge N$, and the finitely many earlier terms can be included in a larger bound. Let M bound both $|a_n|$ and $|b|$. Then

$$|a_n b_n - ab| \le |a_n||b_n - b| + |b||a_n - a| \le M\big(|b_n - b| + |a_n - a|\big),$$

which proves the product assertion.

For division, choose N_0 so that $|b_n - b| < |b|/2$ for $n \ge N_0$. Then $|b_n| \ge |b|/2$, so such b_n are nonzero. Thus

$$|\frac{a_n}{b_n} - \frac{a}{b}| \le \frac{2}{|b|}|a_n - a| + \frac{2|a|}{|b|^2}|b_n - b|.$$

Choose stages making the two terms on the right sum to less than ε. $\square$

Theorem 4.9 (Order preservation)**.** Let $a_n \to a$ and $b_n \to b$. If $a_n \le b_n$ for every sufficiently large n, then $a \le b$.

Proof. If $a > b$, let $\varepsilon = (a - b)/3 > 0$. For all sufficiently large n, $|a_n - a| < \varepsilon$ and $|b_n - b| < \varepsilon$. Hence $a_n > a - \varepsilon = b + 2\varepsilon > b_n$, a contradiction. $\square$

Corollary 4.10 (Squeeze theorem)**.** If $a_n \le b_n \le c_n$ eventually and $a_n \to L$, $c_n \to L$, then $b_n \to L$.

Example 4.11 (Algebraic simplification and rationalization)**.** The candidate limit in

$$a_n = \frac{3n^2 - n}{2n^2 + 5}$$

becomes visible after division by n^2; the limit laws then give $a_n \to 3/2$. For

$$b_n = \sqrt{n^2 + n} - n,$$

direct substitution hides the answer. Multiplying by the conjugate gives

$$b_n = \frac{n}{\sqrt{n^2 + n} + n} = \frac{1}{\sqrt{1 + 1/n} + 1} \longrightarrow \frac{1}{2}.$$

The algebra does not itself prove a limit; it rewrites the expression into terms to which already proved limit laws apply.

Remark 4.12 (Strategy for sequence limits)**.** Separate discovery from proof. First simplify, compute a few terms, or compare dominant terms to guess a candidate. Then prove it using the definition, the limit laws, a squeeze estimate, or monotonicity and boundedness. To prove that no finite limit exists, look for incompatible subsequences, failure of the Cauchy condition, or divergence to $\pm\infty$. A failed attempt to find a limit is not a proof of nonexistence.

Proof. For large n, $0 \le b_n - a_n \le c_n - a_n$. The limit algebra gives $c_n - a_n \to 0$. Given $\varepsilon > 0$, eventually $c_n - a_n < \varepsilon$, and then $|b_n - L| \le |b_n - a_n| + |a_n - L| < \varepsilon/2 + \varepsilon/2$ after increasing the stage if necessary. $\square$

4.3 Subsequences and Elementary Divergence

A subsequence samples an infinite process along a thinner infinite set of stages. It is especially useful for detecting divergence: incompatible long-term behaviors cannot belong to one convergent sequence.

Definition 4.13 (Subsequence)**.** Let (a_n) be a sequence. A **subsequence** is a sequence $(a_{n_k})_{k\in\mathbb{N}}$, where $n_1 < n_2 < \cdots$ are natural numbers.

Proposition 4.14 (Subsequences preserve limits)**.** If $a_n \to L$, then every subsequence of (a_n) converges to L.

Proof. For $n_1 < n_2 < \cdots$, induction gives $n_k \geq k$. Given $\varepsilon > 0$, choose N such that $n \geq N$ implies $|a_n - L| < \varepsilon$. Then $k \geq N$ implies $n_k \geq N$, hence $|a_{n_k} - L| < \varepsilon$. ☐

Corollary 4.15. If a sequence has two subsequences converging to distinct limits, then it is divergent.

Proof. Otherwise Proposition 4.14 would make both subsequences converge to the sequence's limit, contradicting Theorem 4.3. ☐

Example 4.16. The sequence $(-1)^n$ diverges. Its even-indexed subsequence is constantly 1, while its odd-indexed subsequence is constantly -1.

4.4 Completeness Consequences

The order completeness of the real numbers now becomes a practical tool. It converts bounded monotone approximation into a limit, prevents Cauchy approximations from leaving a gap, and forces bounded sequences to have convergent subsequences.

Definition 4.17 (Monotone sequence)**.** A sequence (a_n) is **increasing** if $a_n \leq a_{n+1}$ for all n, **decreasing** if $a_{n+1} \leq a_n$ for all n, and **monotone** if it is increasing or decreasing.

Theorem 4.18 (Monotone convergence)**.** An increasing sequence converges if and only if it is bounded above; in that event, its limit is $\sup\{a_n : n \in \mathbb{N}\}$. A decreasing sequence converges if and only if it is bounded below, and its limit is the infimum of its range.

Proof. If $a_n \to L$, then $a_n \leq L + 1$ eventually, and the finitely many earlier terms have a common upper bound. Conversely, let $s = \sup\{a_n : n \in \mathbb{N}\}$. Given $\varepsilon > 0$, $s - \varepsilon$ is not an upper bound, so $a_N > s - \varepsilon$ for some N. For $n \geq N$, $s - \varepsilon < a_N \leq a_n \leq s$, and hence $|a_n - s| < \varepsilon$. The decreasing statement follows on applying the increasing statement to $(-a_n)$. ☐

Theorem 4.19 (Cauchy criterion)**.** A real sequence converges if and only if it is Cauchy.

Proof. If $a_n \to L$, choose N with $|a_n - L| < \varepsilon/2$ for $n \geq N$ and use the triangle inequality. Conversely, the rational-Cauchy construction of $\mathbb{R}$ proves exactly this assertion in Theorem 3.31. ☐

Theorem 4.20 (Bolzano–Weierstrass)**.** Every bounded real sequence has a convergent subsequence.

Proof. Let all terms lie in $[a, b]$. Bisect this interval and retain a closed half containing infinitely many terms; repeat indefinitely, obtaining nested closed intervals $I_n = [a_n, b_n]$ of length $(b-a)2^{-n}$, each containing infinitely many sequence terms. Choose indices $k_1 < k_2 < \cdots$ with $a_{k_n} \in I_n$; this is possible because each I_n contains infinitely many terms. The left endpoints are increasing and bounded above, so by Theorem 4.18 they converge to some x. Since $0 \le b_n - a_n \to 0$, also $b_n \to x$. For $m \ge n$, $a_{k_m} \in I_n$, so $|a_{k_m} - x| \le b_n - a_n$. This tends to zero, proving $a_{k_m} \to x$. ☐

Definition 4.21 (Limsup and liminf)**.** Let (a_n) be bounded. For $n \in \mathbb{N}$, put

$$s_n = \sup\{a_k : k \ge n\}, \qquad i_n = \inf\{a_k : k \ge n\}.$$

Define

$$\lim_{n\to\infty} \sup a_n = \lim_{n\to\infty} s_n, \qquad \lim_{n\to\infty} \inf a_n = \lim_{n\to\infty} i_n.$$

Theorem 4.22 (Basic limsup and liminf facts)**.** For every bounded sequence, limsup and liminf exist and

$$\lim_{n\to\infty} \inf a_n \le \lim_{n\to\infty} \sup a_n.$$

Moreover, $a_n \to L$ if and only if the two displayed limits both equal L.

Proof. The sequence (s_n) is decreasing and bounded below, while (i_n) is increasing and bounded above. Their limits therefore exist by Theorem 4.18; since $i_n \le s_n$, their limits have the stated order by Theorem 4.9. If $a_n \to L$, then for every $\varepsilon > 0$, all sufficiently late terms lie in

$$\left(L - \frac{\varepsilon}{2}, L + \frac{\varepsilon}{2}\right).$$

For every tail beginning after that stage, its infimum is at least $L - \varepsilon/2$ and its supremum is at most $L + \varepsilon/2$. Their limits therefore equal L. Conversely, if both limits equal L, then for a given $\varepsilon > 0$, eventually $i_n > L - \varepsilon$ and $s_n < L + \varepsilon$. Since $i_n \le a_n \le s_n$, the squeeze theorem gives $a_n \to L$. ☐

Why boundedness remains in this definition. One can develop conventions for limsup and liminf of unbounded sequences, but they are unnecessary here. We retain real-valued suprema and infima and use Definition 4.5 when a sequence eventually escapes every real bound.

4.5 Equivalent Completeness Principles

The least-upper-bound property is not the only possible formulation of completeness. The equivalences below explain why several apparently different principles may all serve as the decisive feature distinguishing the real numbers from the rationals.

Theorem 4.23 (Equivalence of standard completeness principles)**.** Let F be an Archimedean ordered field. The following assertions are equivalent.

(a) Every nonempty subset of F having an upper bound has a least upper bound.

(b) Every Cauchy sequence in F converges.

(c) Every bounded sequence in F has a convergent subsequence.

(d) Every increasing sequence in F that is bounded above converges.

(e) Whenever
$$[a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots$$
and
$$\lim_{n\to\infty} (b_n - a_n) = 0,$$
the intervals have a common point.

Consequently, each assertion may be used as a completeness axiom for an Archimedean ordered field.

Proof. All definitions of convergence, Cauchyness, boundedness, and intervals, as well as the elementary order-limit lemmas used below, are obtained verbatim by replacing $\mathbb{R}$ with F; their proofs use only the ordered-field laws.

We prove a cycle of implications. Suppose first that (1) holds. The proof of Theorem 4.18 uses only (1), so (4) follows. A Cauchy sequence is bounded: choose N so that
$$m, n \geq N \quad \Longrightarrow \quad |x_m - x_n| < 1,$$
and bound the tail by $|x_N| + 1$ and the finitely many earlier terms separately. For the tail sets
$$T_n = \{x_k : k \geq n\},$$
put
$$s_n = \sup T_n, \qquad i_n = \inf T_n.$$
The sequences (s_n) and (i_n) are respectively decreasing and increasing, so (4) gives limits s and i. Given $\varepsilon > 0$, Cauchyness gives N such that every two terms of T_N differ in absolute value by less than ε. Hence
$$s_N - i_N \leq 2\varepsilon.$$
Passing to the limit in this inequality gives
$$0 \leq s - i \leq 2\varepsilon.$$
Since this holds for every $\varepsilon > 0$, the small-positive-number argument shows $s = i$. Since $i_n \leq x_n \leq s_n$, the squeeze theorem gives $x_n \to s$. Thus (1) implies (2).

Assume (2), and let S be nonempty and bounded above. Choose $s_0 \in S$ and an upper bound u_0. If s_0 is already an upper bound, it is the maximum and hence the least upper bound. Otherwise, bisect the interval $[s_0, u_0]$ repeatedly: retain its lower half when its midpoint is an upper bound of S, and retain its upper half otherwise. This produces numbers s_n, u_n such that
$$s_n \text{ is not an upper bound}, \qquad u_n \text{ is an upper bound},$$
and
$$u_n - s_n = 2^{-n}(u_0 - s_0).$$
The Archimedean property implies that $2^{-n} \to 0$: choose an integer larger than any prescribed positive reciprocal, and use $2^n \geq n + 1$. Thus (s_n) is Cauchy, since
$$0 \leq s_m - s_n \leq u_n - s_n \qquad (m \geq n).$$
Let $s_n \to c$ by (2). The same estimate gives $u_n \to c$. If $t \in S$ were greater than c, then for sufficiently large n one would have
$$u_n - s_n < t - c,$$

and the order-limit argument gives

$$s_n \le c \le u_n.$$

These inequalities force $t > u_n$, a contradiction. Hence c is an upper bound. If $v < c$, then the same inequality with $c - v$ in place of $t - c$, together with $s_n \le c \le u_n$, gives $v < s_n$ for a sufficiently large n; since s_n is not an upper bound, neither is v. Therefore $c = \sup S$, proving (1).

Still under (1), the nested-interval argument in Theorem 3.34 proves (5): the supremum of the left endpoints belongs to every interval. The bisection argument used for Theorem 4.20, together with this nested-interval conclusion, proves (3). More explicitly, repeatedly retain a half of a bounded interval containing infinitely many terms, choose a strictly increasing sequence of indices from the retained intervals, and use the common point of the nested intervals as the limit of that subsequence.

Assume now (3), and let (a_n) be increasing and bounded above. A subsequence (a_{n_k}) converges, say to L. For a fixed n, all sufficiently late members of this subsequence are at least a_n; Theorem 4.9 gives $a_n \le L$. Given $\varepsilon > 0$, choose k with

$$|a_{n_k} - L| < \varepsilon.$$

Then, for every $n \ge n_k$,

$$L - \varepsilon < a_{n_k} \le a_n \le L,$$

so $a_n \to L$. This proves (4).

Assume (4). The bisection construction in the second paragraph gives an increasing, bounded-above sequence (s_n), hence a limit c. The same shrinking-width argument proves that c is the least upper bound of S. Thus (4) implies (1).

Finally, assume (5). Apply the same bisection construction to a nonempty bounded-above set S. Its intervals are nested and their lengths tend to zero, so they have a common point c. The two final paragraphs of the bisection argument show that c is the least upper bound of S. Thus (5) implies (1), completing the proof. □

Exercises

Exercise 4.1. Let (a_n) be increasing and bounded above, and let s be the supremum of its range. Adapt the proof of Theorem 4.18 to show directly that

$$\lim_{n\to\infty} a_n = s.$$

Identify the exact use of the least-upper-bound property.

Exercise 4.2. Give an example of an unbounded sequence for which

$$\lim_{n\to\infty} \sup a_n$$

cannot be interpreted as a real number. Explain the purpose of the postponed extended convention.

Exercise 4.3. Assuming the least-upper-bound property, prove that every decreasing sequence that is bounded below converges, without appealing to the increasing case as a black box.

Chapter 5

Limits and Continuity of Functions

5.1 Limits at a Point

Sequence limits describe processes indexed by natural numbers. A function limit asks the same question when the input can approach a point through all nearby domain values, so the value assigned at the target itself must be kept logically separate.

Definition 5.1 (Limit point)**.** Let $D \subseteq \mathbb{R}$ and $a \in \mathbb{R}$. The point a is a **limit point** of D if, for every $\delta > 0$, there is an $x \in D$ such that $0 < |x - a| < \delta$.

The condition excludes $x = a$: a function's value at a should not affect its limit at a.

Definition 5.2 (Function limit)**.** Let $f : D \to \mathbb{R}$, let a be a limit point of D, and let $L \in \mathbb{R}$. We write

$$\lim_{x \to a} f(x) = L$$

if, for every $\varepsilon > 0$, there is $\delta > 0$ such that every $x \in D$ satisfying $0 < |x - a| < \delta$ also satisfies $|f(x) - L| < \varepsilon$.

Theorem 5.3 (Uniqueness of function limits)**.** Under the hypotheses of Definition 5.2, there is at most one value L for which the displayed limit holds.

Proof. Suppose L and M both satisfy the definition. Given $\varepsilon > 0$, obtain positive numbers δ_L, δ_M for tolerance $\varepsilon/2$. Since a is a limit point, choose $x \in D$ with

$$0 < |x - a| < \min\{\delta_L, \delta_M\}.$$

Then

$$|L - M| \leq |L - f(x)| + |f(x) - M| < \varepsilon.$$

By Lemma 4.2, $L = M$. □

Theorem 5.4 (Sequential criterion)**.** Let $f : D \to \mathbb{R}$ and let a be a limit point of D. Then

$$\lim_{x \to a} f(x) = L$$

if and only if, for every sequence (x_n) in $D \setminus \{a\}$ with $x_n \to a$, one has $f(x_n) \to L$.

Proof. Suppose first that the function limit exists. Given $\varepsilon > 0$, take δ from Definition 5.2. Since $x_n \to a$, eventually $|x_n - a| < \delta$, so eventually $|f(x_n) - L| < \varepsilon$.

Conversely, suppose the sequential statement holds but the function limit fails. Then there is $\varepsilon_0 > 0$ such that, for every $n \in \mathbb{N}$, one can choose $x_n \in D$ with

$$0 < |x_n - a| < 1/n \qquad \text{and} \qquad |f(x_n) - L| \geq \varepsilon_0.$$

The first inequality and the Archimedean property imply $x_n \to a$, while the second prevents $f(x_n)$ from converging to L, a contradiction. □

Definition 5.5 (One-sided and infinite limits)**.** Let $f : D \to \mathbb{R}$. The assertion

$$\lim_{x \to a^+} f(x) = L$$

means that for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$x \in D, \quad 0 < x - a < \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

The left-hand limit is defined with $0 < a - x < \delta$. We write $f(x) \to +\infty$ as $x \to a^+$ if for every $M \in \mathbb{R}$ there is $\delta > 0$ such that the same hypotheses imply $f(x) > M$; replace the last inequality by $f(x) < M$ for a limit of $-\infty$. The analogous two-sided definitions use $0 < |x - a| < \delta$.

At an unbounded end of the domain,

$$\lim_{x \to +\infty} f(x) = L$$

means that for every $\varepsilon > 0$ there is $A \in \mathbb{R}$ such that $x \in D$ and $x > A$ imply $|f(x) - L| < \varepsilon$. Replacing this conclusion by $f(x) > M$ defines $f(x) \to +\infty$ as $x \to +\infty$; changing the direction of the input or output inequalities gives the other cases.

Remark 5.6 (Infinity is behavior, not a value)**.** The notation $f(x) \to +\infty$ does not assert convergence to a member of $\mathbb{R}$. It abbreviates an eventual-inequality statement. For example, $1/x \to +\infty$ as $x \to 0^+$, but the two-sided limit at 0 does not tend to either sign of infinity because $1/x \to -\infty$ from the left.

Example 5.7 (Direct proof and rationalization)**.** To prove

$$\lim_{x \to 2} (3x - 1) = 5,$$

observe that $|(3x - 1) - 5| = 3|x - 2|$ and choose $\delta = \varepsilon/3$. For a less direct example,

$$\lim_{x \to 0} \frac{\sqrt{1 + x} - 1}{x} = \lim_{x \to 0} \frac{1}{\sqrt{1 + x} + 1} = \frac{1}{2},$$

where the equality for $x \neq 0$ follows by rationalization and the last step follows from continuity and the quotient law.

Example 5.8 (One-sided limits detect nonexistence)**.** For $f(x) = x/|x|$ on $\mathbb{R} \setminus \{0\}$,

$$\lim_{x \to 0^-} f(x) = -1, \qquad \lim_{x \to 0^+} f(x) = 1.$$

Hence the two-sided limit does not exist. Indeed, the sequences $x_n = -1/n$ and $y_n = 1/n$ approach the same point while their images have different limits.

Remark 5.9 (A working strategy for function limits)**.** First discover a plausible answer by substitution, simplification, a graph, or comparison. Next prove it: use limit laws only after their hypotheses are met; cancel only on the punctured neighborhood; rationalize radicals; or bound the error by a simpler expression and squeeze. For nonexistence, compare one-sided limits or produce two approaching sequences with different image limits. The exact value $f(a)$ is irrelevant to $\lim_{x \to a} f(x)$.

5.2 Asymptotic Notation

Limit arguments often need only a comparison of sizes rather than an exact formula. The following compact notation records two common kinds of comparison without replacing the underlying estimates.

Definition 5.10 (Big-O and little-o)**.** Let f, g be real-valued functions defined sufficiently near a. We write

$$f(x) = O(g(x)) \qquad (x \to a)$$

if there are $C > 0$ and a punctured neighborhood of a on which

$$|f(x)| \leq C|g(x)|.$$

We write

$$f(x) = o(g(x)) \qquad (x \to a)$$

if $g(x) \neq 0$ sufficiently near a and

$$\frac{f(x)}{g(x)} \longrightarrow 0.$$

For sequences, $a_n = O(b_n)$ as $n \to \infty$ and $a_n = o(b_n)$ as $n \to \infty$ have the identical eventual-bound and quotient-limit meanings.

Thus $O(g)$ means "eventually controlled by a constant multiple of g," whereas $o(g)$ means "negligible compared with g." For example,

$$x^2 = o(x), \qquad x = O(x), \qquad x \neq o(x) \qquad (x \to 0),$$

and

$$\frac{1}{n^2} = o\left(\frac{1}{n}\right) \qquad (n \to \infty).$$

The notation always abbreviates a concrete bound or limit; its usefulness is that it keeps the main comparison visible in a longer calculation.

5.3 Continuity

Continuity is the condition that removes that separation: the local limiting value agrees with the value of the function. Open and closed sets provide a second language for this stability and prepare for compactness.

Definition 5.11 (Neighborhoods, open sets, and closed sets)**.** For $a \in \mathbb{R}$ and $r > 0$, the open interval

$$B(a, r) = \{x \in \mathbb{R} : |x - a| < r\}$$

is the **open ball** of radius r about a. A set $U \subseteq \mathbb{R}$ is **open** if every $a \in U$ has some $r > 0$ for which

$$B(a, r) \subseteq U.$$

A set $C \subseteq \mathbb{R}$ is **closed** if its complement is open.

Theorem 5.12 (Sequential characterization of closed sets)**.** A set $C \subseteq \mathbb{R}$ is closed if and only if every convergent sequence in C has its limit in C.

Proof. Suppose first that C is closed and that $x_n \in C$ converges to x. If $x \notin C$, then the open set C^c contains a ball

$$B(x, r).$$

Eventually $x_n \in B(x, r)$, contradicting $x_n \in C$. Conversely, suppose the sequential condition holds and let $x \in C^c$. If no ball about x were contained in C^c, then for every n one could choose

$$x_n \in C, \qquad |x_n - x| < 1/n.$$

Then $x_n \to x$, contradicting the sequential condition. Hence C^c is open. □

Definition 5.13 (Continuity)**.** Let $f : D \to \mathbb{R}$ and let $a \in D$. The function f is **continuous at** a if, for every $\varepsilon > 0$, there is $\delta > 0$ such that every $x \in D$ satisfying $|x - a| < \delta$ satisfies $|f(x) - f(a)| < \varepsilon$. It is **continuous on** D if it is continuous at every point of D.

If a is a limit point of D, continuity at a is equivalent to the statement that the limit of $f(x)$ as x tends to a equals $f(a)$. The definition also handles isolated points: every function is continuous at an isolated point of its domain.

Theorem 5.14 (Sequential criterion for continuity)**.** Let $f : D \to \mathbb{R}$ and $a \in D$. Then f is continuous at a if and only if every sequence (x_n) in D satisfying $x_n \to a$ also satisfies $f(x_n) \to f(a)$.

Proof. The proof is the same as Theorem 5.4, except that terms equal to a require no exclusion: for such a term, $|f(x_n) - f(a)| = 0$. Conversely, failure of continuity gives, for each n, an $x_n \in D$ with $|x_n - a| < 1/n$ and $|f(x_n) - f(a)| \geq \varepsilon_0$. □

Theorem 5.15 (Algebra of continuous functions)**.** Let $f, g : D \to \mathbb{R}$ be continuous at $a \in D$, and let $c \in \mathbb{R}$. Then $f + g$, cf, and fg are continuous at a. If $g(a) \neq 0$, then f/g is continuous at a on the set where it is defined.

Proof. For every sequence $x_n \to a$ in D, continuity gives $f(x_n) \to f(a)$ and $g(x_n) \to g(a)$. Apply the sequence limit laws of Theorem 4.8 and then Theorem 5.14. □

Example 5.16. Every polynomial is continuous on $\mathbb{R}$: constant and identity functions are continuous directly from Definition 5.13, and repeated application of Theorem 5.15 proves the claim. Every rational function is continuous wherever its denominator is nonzero.

5.4 Uniform Continuity

Ordinary continuity permits the required input tolerance to vary from point to point. Uniform continuity asks for one tolerance that works everywhere on the domain, precisely the stronger control needed for approximation and integration.

Definition 5.17 (Uniform continuity)**.** Let $f : D \to \mathbb{R}$. The function f is **uniformly continuous on** D if, for every $\varepsilon > 0$, there is $\delta > 0$ such that, for all $x, y \in D$,

$$|x - y| < \delta \Longrightarrow |f(x) - f(y)| < \varepsilon.$$

The difference from ordinary continuity is that δ may depend on ε, but not on the point of D. Schematically, the quantifiers are

$$\text{continuity:} \quad \forall a \in D\, \forall \varepsilon > 0\, \exists \delta = \delta(a, \varepsilon) > 0,$$

whereas

$$\text{uniform continuity:}\quad \forall\varepsilon > 0\ \exists\delta = \delta(\varepsilon) > 0\ \forall x, y \in D.$$

Moving the point quantifiers past $\exists\delta$ is the whole strengthening.

Proposition 5.18 (Lipschitz functions)**.** Let $f : D \to \mathbb{R}$. If there is $K \geq 0$ such that

$$|f(x) - f(y)| \leq K|x - y| \qquad \text{for all } x, y \in D,$$

then f is uniformly continuous on D.

Proof. If $K = 0$, the function is constant. If $K > 0$, choose $\delta = \varepsilon/K$. Then the displayed inequality gives the conclusion of Definition 5.17. □

The displayed hypothesis defines a **Lipschitz function**, and K is a Lipschitz constant. Thus

$$\text{Lipschitz} \Longrightarrow \text{uniformly continuous} \Longrightarrow \text{continuous.}$$

Neither converse holds in general.

Example 5.19 (The three levels of control)**.** The function $x \mapsto x^2$ is not uniformly continuous on $\mathbb{R}$: with $x_n = n$ and $y_n = n + 1/n$, the inputs approach one another while

$$y_n^2 - x_n^2 = 2 + 1/n^2.$$

On $[-A, A]$, however,

$$|x^2 - y^2| \leq 2A|x - y|,$$

so it is Lipschitz there. The function $\sqrt{x}$ on $[0, 1]$ is uniformly continuous because

$$|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|},$$

but it is not Lipschitz: if a constant K worked, then $\sqrt{x} \leq Kx$ for every $x > 0$, equivalently $1/\sqrt{x} \leq K$, which fails near 0.

Example 5.20 (Continuity need not be uniform)**.** The function $f : (0, 1) \to \mathbb{R}$ given by $f(x) = 1/x$ is continuous by Theorem 5.15, but not uniformly continuous. Indeed, put $x_n = 1/n$ and $y_n = 1/(n + 1)$. Then $|x_n - y_n| \to 0$, whereas $|f(x_n) - f(y_n)| = 1$ for every n.

On $[a, \infty)$ with $a > 0$, the same function $1/x$ is Lipschitz, since

$$|\frac{1}{x} - \frac{1}{y}| = \frac{|x - y|}{xy} \leq \frac{1}{a^2}|x - y|.$$

This contrast shows that uniform continuity depends on the domain as well as on the formula.

5.5 Intermediate Values

Continuity has a qualitative consequence beyond limiting estimates: it prevents a function from jumping over a value. Completeness supplies the point at which this intuitive crossing must occur.

Theorem 5.21 (Intermediate value theorem)**.** Let $f : [a, b] \to \mathbb{R}$ be continuous, with $a < b$. If u lies between $f(a)$ and $f(b)$, then there is $c \in [a, b]$ such that $f(c) = u$.

Proof. It is enough to treat $f(a) \le u \le f(b)$. Let

$$S = \{x \in [a,b] : f(x) \le u\}.$$

The set is nonempty and bounded above, so let $c = \sup S$. If $f(c) < u$, then $c \ne b$ because

$$f(b) \ge u.$$

Continuity gives a positive δ such that $f(x) < u$ whenever

$$|x - c| < \delta.$$

Choose

$$0 < r < \min(\delta, b - c).$$

Then $c + r \in S$, contradicting that c is an upper bound of S. If $f(c) > u$, then $c \ne a$ because

$$f(a) \le u.$$

Continuity gives a positive δ such that $f(x) > u$ whenever

$$|x - c| < \delta.$$

By the defining property of the supremum, there is

$$x \in S$$

with

$$c - \delta < x \le c,$$

which contradicts $f(x) \le u$. Therefore $f(c) = u$. □

Theorem 5.22 (Existence and uniqueness of roots)**.** For every $c \ge 0$ and $n \in \mathbb{N}$, there is a unique $r \ge 0$ such that $r^n = c$.

Proof. The polynomial $p(t) = t^n - c$ is continuous. On $[0, \max\{1, c\}]$ its endpoint values have opposite weak signs, so the Intermediate Value Theorem gives a zero. If $0 \le r < s$, then

$$s^n - r^n = (s - r)(s^{n-1} + s^{n-2}r + \cdots + r^{n-1}) > 0,$$

so $t \mapsto t^n$ is strictly increasing on $[0, \infty)$ and the root is unique. □

Remark 5.23 (Looking back at familiar radicals)**.** The notation $\sqrt[n]{c}$ now names the unique root supplied by the theorem; it was not justified merely by writing the symbol. For even n, a real nth root requires $c \ge 0$. For odd n, a negative input has the unique negative root $-\sqrt[n]{-c}$. Chapter 7 uses these roots to define rational powers consistently, and Chapter 8 later defines arbitrary real powers of positive bases through exp and log.

5.6 Compact Intervals

Closed bounded intervals are the domains on which local information can be made global. Compactness will turn subsequential control into maxima, minima, and uniform continuity.

Definition 5.24 (Sequential compactness)**.** A set $K \subseteq \mathbb{R}$ is **sequentially compact** if every sequence in K has a subsequence converging to an element of K.

Theorem 5.25 (Compactness of closed bounded intervals)**.** If $a \leq b$, then $[a, b]$ is sequentially compact.

Proof. Let (x_n) be a sequence in $[a, b]$. It is bounded, so Theorem 4.20 gives a subsequence converging to some $x \in \mathbb{R}$. Since $a \leq x_{n_k} \leq b$ for every k, Theorem 4.9 gives $a \leq x \leq b$. Thus $x \in [a, b]$. □

Theorem 5.26 (Extreme value theorem)**.** Let $f : [a, b] \to \mathbb{R}$ be continuous, where $a \leq b$. Then there are points $x_{\min}, x_{\max} \in [a, b]$ such that

$$f(x_{\min}) \leq f(x) \leq f(x_{\max}) \qquad \text{for every } x \in [a, b].$$

Proof. Let $s = \sup f([a, b])$. To justify that this supremum exists, suppose first that $f([a, b])$ were unbounded above. For each n, choose $x_n \in [a, b]$ with $f(x_n) > n$. By Theorem 5.25, a subsequence x_{n_k} converges to $x \in [a, b]$. Continuity gives $f(x_{n_k}) \to f(x)$, but $f(x_{n_k}) \geq n_k \to \infty$, an impossibility. Hence the image is bounded above, and it is nonempty. For each n, choose $x_n \in [a, b]$ with $f(x_n) > s - 1/n$. A convergent subsequence has limit $x_{\max} \in [a, b]$. Continuity and the squeeze theorem give $f(x_{\max}) = s$. Applying the same argument to $-f$ gives $x_{\min}$. □

Theorem 5.27 (Heine–Cantor theorem)**.** Every continuous function $f : [a, b] \to \mathbb{R}$, where $a \leq b$, is uniformly continuous.

Proof. Suppose not. Then some $\varepsilon_0 > 0$ has the property that, for every n, there are $x_n, y_n \in [a, b]$ such that

$$|x_n - y_n| < 1/n \qquad \text{and} \qquad |f(x_n) - f(y_n)| \geq \varepsilon_0.$$

By Theorem 5.25, a subsequence x_{n_k} converges to $x \in [a, b]$. The first inequality gives $y_{n_k} \to x$ as well. Continuity gives both $f(x_{n_k}) \to f(x)$ and $f(y_{n_k}) \to f(x)$, so their difference tends to 0, contradicting the second inequality. □

Exercises

Exercise 5.1. Use the sequential criterion to prove that

$$f(x) = x^2$$

is continuous at every real number.

Exercise 5.2. Let

$$f(x) = \begin{cases} 0, & x \leq 0, \\ 1, & x > 0. \end{cases}$$

Determine all points at which f is continuous, and explain your answer using sequences.

Exercise 5.3. Let $f : [a, b] \to \mathbb{R}$ be continuous and suppose

$$f(a)f(b) < 0.$$

Use the Intermediate Value Theorem to prove that f has a zero in (a, b).

Chapter 6

Sequences and Families of Functions

6.1 Pointwise and Uniform Convergence

For functions, convergence has two distinct quantifier patterns. Pointwise convergence lets the stage of approximation depend on the input; uniform convergence does not. This distinction explains exactly when a limit may inherit analytic properties from its approximants.

Definition 6.1 (Pointwise convergence)**.** Let $D \subseteq \mathbb{R}$, and let $f_n : D \to \mathbb{R}$ and $f : D \to \mathbb{R}$. We say that (f_n) **converges pointwise** to f on D if, for every $x \in D$,

$$\lim_{n\to\infty} f_n(x) = f(x).$$

Definition 6.2 (Uniform convergence)**.** Under the same hypotheses, (f_n) **converges uniformly** to f on D if, for every $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that, for every $n \geq N$ and every $x \in D$,

$$|f_n(x) - f(x)| < \varepsilon.$$

Definition 6.3 (Sup norm)**.** If $h : D \to \mathbb{R}$ is bounded, define

$$\|h\|_\infty = \sup_{x\in D} |h(x)|.$$

Whenever $f_n - f$ is bounded, uniform convergence is equivalently

$$\|f_n - f\|_\infty \longrightarrow 0.$$

Indeed, the inequality $\|f_n - f\|_\infty < \varepsilon$ gives the same error bound at every x, while a uniform error bound of $\varepsilon/2$ forces the supremum to be at most $\varepsilon/2 < \varepsilon$.

The dependence of the index is worth displaying:

$$\begin{aligned} &\text{pointwise:} && \forall x \in D\ \forall \varepsilon > 0\ \exists N = N(x, \varepsilon), \\ &\text{uniform:} && \forall \varepsilon > 0\ \exists N = N(\varepsilon)\ \forall x \in D. \end{aligned}$$

Uniform convergence implies pointwise convergence, because one may fix x after choosing the common stage N. The converse fails: $f_n(x) = x^n$ on $[0, 1]$ converges pointwise to the discontinuous function described in Chapter 1, and hence cannot converge uniformly by Theorem 6.5 below.

More directly, the pointwise limit of x^n on $[0,1]$ equals 0 on $[0,1)$ and 1 at 1. For every n, choosing $x_n = 2^{-1/n}$ gives $x_n < 1$ and hence $f(x_n) = 0$, but

$$|f_n(x_n) - f(x_n)| = \frac{1}{2}.$$

Thus the sup-norm error never tends to zero. In contrast, $g_n(x) = x/n$ on $[0,1]$ converges uniformly to zero because

$$\|g_n\|_\infty = \frac{1}{n}.$$

Proposition 6.4 (Uniform Cauchy criterion)**.** Let $f_n : D \to \mathbb{R}$. There is a function $f : D \to \mathbb{R}$ to which f_n converges uniformly if and only if, for every $\varepsilon > 0$, there is N such that, for all $m, n \geq N$ and all $x \in D$,

$$|f_n(x) - f_m(x)| < \varepsilon.$$

Proof. If $f_n \to f$ uniformly, apply the triangle inequality with tolerance $\varepsilon/2$. Conversely, the displayed condition makes $(f_n(x))$ a Cauchy real sequence for each fixed x. By Theorem 4.19, define $f(x)$ to be its limit. Fix $\varepsilon > 0$ and choose N from the displayed condition. For $n \geq N$, take the limit as $m \to \infty$ in the inequality with fixed n, x; the order preservation of limits gives $|f_n(x) - f(x)| \leq \varepsilon$. Repeating with $\varepsilon/2$ proves the strict inequality required by Definition 6.2. □

6.2 Continuous Limits

The principal benefit of uniform convergence is that one approximation index works simultaneously near every point. The three-term estimate below makes this control visible and explains why pointwise convergence is insufficient.

Theorem 6.5 (Uniform limit theorem)**.** Let $D \subseteq \mathbb{R}$. If every $f_n : D \to \mathbb{R}$ is continuous at $a \in D$ and $f_n \to f$ uniformly on D, then f is continuous at a.

Proof. Given $\varepsilon > 0$, choose N such that $|f_N(x) - f(x)| < \varepsilon/3$ for every $x \in D$. Continuity of f_N at a supplies $\delta > 0$ such that $|x - a| < \delta$ implies $|f_N(x) - f_N(a)| < \varepsilon/3$. For such x,

$$|f(x) - f(a)| \leq |f(x) - f_N(x)| + |f_N(x) - f_N(a)| + |f_N(a) - f(a)| < \varepsilon.$$

□

Remark 6.6 (What uniform convergence permits)**.** Uniform limits of continuous functions are continuous by the theorem above. Chapter 9 proves that a uniform limit may also be passed through a Riemann integral. Differentiation is more delicate: uniform convergence of f_n alone does not control the difference quotients, so Chapter 8 imposes uniform convergence on derivatives and fixes one anchor value. The general lesson is to identify the estimate needed by the operation before interchanging two limiting processes.

Proposition 6.7 (Uniform convergence respects algebra)**.** Let $f_n \to f$ and $g_n \to g$ uniformly on D. Then $f_n + g_n \to f + g$ uniformly. If both (f_n) and (g_n) are uniformly bounded, then

$$f_n g_n \to fg$$

uniformly on D.

Proof. The sum follows from the triangle inequality. Let

$$|f_n(x)| \le M, \qquad |g_n(x)| \le K \qquad (n \in \mathbb{N},\ x \in D).$$

Taking pointwise limits gives

$$|g(x)| \le K \qquad (x \in D).$$

Then

$$|f_n(x)g_n(x) - f(x)g(x)| \le M|g_n(x) - g(x)| + K|f_n(x) - f(x)|.$$

Choose a common stage making both terms uniformly small. □

Proposition 6.8 (Uniform limits preserve boundedness)**.** Let $f_n \to f$ uniformly on D. If one function f_N is bounded on D, then f is bounded on D.

Proof. Choose N so that

$$|f(x) - f_N(x)| < 1 \qquad (x \in D).$$

If $|f_N(x)| \le M$ throughout D, then

$$|f(x)| \le M + 1 \qquad (x \in D).$$

□

Example 6.9 (Pointwise limits need not preserve boundedness)**.** On

$$D = (0, 1],$$

define

$$f_n(x) = \begin{cases} n, & 0 < x \le 1/n, \\ 1/x, & 1/n < x \le 1. \end{cases}$$

Each f_n is bounded by n, while for every $x \in D$,

$$\lim_{n\to\infty} f_n(x) = \frac{1}{x}.$$

The pointwise limit is unbounded. Thus the uniform hypothesis in Proposition 6.8 cannot be weakened to pointwise convergence.

Example 6.10 (Two limits need not be interchanged)**.** For

$$f_n(x) = x^n \qquad (0 \le x < 1),$$

one has

$$\lim_{n\to\infty} \left(\lim_{x\to 1^-} f_n(x) \right) = 1,$$

whereas

$$\lim_{x\to 1^-} \left(\lim_{n\to\infty} f_n(x) \right) = 0.$$

The convergence in n is pointwise but not uniform near $x = 1$. This is the basic obstruction behind unjustified interchanges of limits.

6.3 Polynomial Approximation

Polynomials are among the simplest continuous functions: they can be added, multiplied, and differentiated exactly. A natural question is how much of $C([a,b])$ they occupy. The answer is strikingly strong: every continuous function can be approximated everywhere on a compact interval by one polynomial with one common error bound.

Theorem 6.11 (Weierstrass approximation theorem)**.** If $f \in C([a,b])$, then for every $\varepsilon > 0$ there is a polynomial p such that

$$\|f - p\|_\infty < \varepsilon.$$

Equivalently, polynomials are uniformly dense in $C([a,b])$.

Proof. We first work on $[0,1]$. For $f \in C([0,1])$, define its nth Bernstein polynomial by

$$B_n(f)(x) = \sum_{k=0}^{n} f(k/n) \binom{n}{k} x^k (1-x)^{n-k}.$$

It is a polynomial. Its nonnegative weights have sum 1 by the binomial theorem, so $B_n(f)(x)$ is a weighted average of sampled values $f(k/n)$. The weights concentrate near $k/n = x$: direct differentiation of the finite binomial sum, or the usual binomial identities, gives

$$\sum_{k=0}^{n} \left(\frac{k}{n} - x\right)^2 \binom{n}{k} x^k (1-x)^{n-k} = \frac{x(1-x)}{n} \le \frac{1}{4n}.$$

Let $M = \|f\|_\infty$. Given $\varepsilon > 0$, uniform continuity of f supplies $\delta > 0$ such that $|s-t| < \delta$ implies $|f(s) - f(t)| < \varepsilon/2$. Split the defining weighted average for $B_n(f)(x) - f(x)$ according as $|k/n - x| < \delta$ or not. The first part has absolute value at most $\varepsilon/2$. On the second part, the error is at most $2M$, while

$$\sum_{|k/n-x|\ge\delta} \binom{n}{k} x^k (1-x)^{n-k} \le \frac{1}{\delta^2} \sum_{k=0}^{n} \left(\frac{k}{n} - x\right)^2 \binom{n}{k} x^k (1-x)^{n-k} \le \frac{1}{4n\delta^2}.$$

For sufficiently large n, this second contribution is below $\varepsilon/2$, uniformly in x. Hence $B_n(f) \to f$ uniformly on $[0,1]$.

For a general interval $[a,b]$, apply this result to $F(t) = f(a + (b-a)t)$ and substitute $t = (x-a)/(b-a)$. The resulting expression is a polynomial in x and has the required uniform error. □

Example 6.12 (Approximating a corner)**.** The function $f(x) = |x|$ on $[-1,1]$ is not a polynomial and is not differentiable at 0. Nevertheless, the theorem provides polynomials p_n with $\|p_n - |x|\|_\infty \to 0$. This does not contradict the fact that uniform limits of differentiable functions need not be differentiable: uniform convergence preserves continuity, but derivatives require the stronger hypotheses discussed in Chapter 8.

Remark 6.13 (Optional perspective: Stone–Weierstrass)**.** For a compact Hausdorff space X, the real Stone–Weierstrass theorem says that a subalgebra of $C(X,\mathbb{R})$ which contains the constants and separates points is dense in the sup norm. On $[a,b]$, polynomials have these properties, so the classical theorem is its first example. The general result belongs naturally to functional analysis and is not used here.

6.4 Equicontinuity

Uniform convergence controls variation in the sequence index. Equicontinuity instead controls variation in the input uniformly across a family; together with compactness it is a central compactness principle for function spaces.

Definition 6.14 (Equicontinuity)**.** Let $\mathcal{F}$ be a family of functions $D \to \mathbb{R}$. It is **equicontinuous at** $a \in D$ if, for every $\varepsilon > 0$, there is $\delta > 0$ such that every $f \in \mathcal{F}$ satisfies

$$|x - a| < \delta \Longrightarrow |f(x) - f(a)| < \varepsilon \qquad (x \in D).$$

It is equicontinuous on D if it is equicontinuous at every point of D.

Proposition 6.15. If $f_n \to f$ uniformly on D and the family $\{f_n : n \in \mathbb{N}\}$ is equicontinuous at a, then f is continuous at a.

Proof. Given $\varepsilon > 0$, choose N such that $|f_N - f| < \varepsilon/3$ throughout D. Equicontinuity supplies δ for f_N and tolerance $\varepsilon/3$. The three-term estimate in Theorem 6.5 proves continuity of f. ☐

Lemma 6.16 (Uniform equicontinuity on a compact interval)**.** Let $\mathcal{F}$ be an equicontinuous family of functions

$$[a, b] \longrightarrow \mathbb{R}.$$

Then, for every $\varepsilon > 0$, there is $\delta > 0$ such that

$$|x - y| < \delta \quad \Longrightarrow \quad |f(x) - f(y)| < \varepsilon$$

for every $f \in \mathcal{F}$ and every $x, y \in [a, b]$.

Proof. If not, there would be $\varepsilon > 0$, functions $h_n \in \mathcal{F}$, and points $x_n, y_n \in [a, b]$ such that

$$|x_n - y_n| < \frac{1}{n}, \qquad |h_n(x_n) - h_n(y_n)| \geq \varepsilon.$$

By compactness of $[a, b]$, pass to a subsequence with $x_n \to x$. Then also $y_n \to x$. Relabel this subsequence as (x_n, y_n, h_n). Equicontinuity at x, with tolerance $\varepsilon/3$, makes both $|h_n(x_n) - h_n(x)|$ and $|h_n(y_n) - h_n(x)|$ less than $\varepsilon/3$ for all sufficiently large n. The triangle inequality gives a contradiction. ☐

Theorem 6.17 (Arzelà–Ascoli on an interval)**.** Let (f_n) be an equicontinuous sequence in $C([a, b])$ for which there is $M > 0$ satisfying

$$|f_n(x)| \leq M \qquad (n \in \mathbb{N},\ x \in [a, b]).$$

Then it has a subsequence that converges uniformly on $[a, b]$ to a continuous function.

Proof. The idea is to first obtain convergence on a countable dense set and then use equicontinuity to make that convergence uniform. Enumerate a countable dense subset of $[a, b]$ as (q_j). Uniform boundedness and Bolzano–Weierstrass allow us to choose successively a subsequence converging at q_1, then a subsequence of it converging at q_2, and so on. The diagonal subsequence (g_k) converges at every q_j.

Given $\varepsilon > 0$, Lemma 6.16 supplies $\delta > 0$ such that

$$|x - y| < \delta \quad \Longrightarrow \quad |g_k(x) - g_k(y)| < \varepsilon/3 \qquad (k \in \mathbb{N}).$$

Using density and a finite partition of $[a, b]$, choose finitely many of the q_j so that every $x \in [a, b]$ lies within δ of one of them. Since (g_k) is Cauchy at each of these finitely many points, there is N such that for $r, s \geq N$ its values there differ by less than $\varepsilon/3$. The triangle inequality then gives

$$|g_r(x) - g_s(x)| < \varepsilon \qquad (x \in [a, b]).$$

Thus (g_k) is uniformly Cauchy, so Proposition 6.4 gives a uniform limit. It is continuous by Theorem 6.5. □

Exercises

Exercise 6.1. On

$$[0, 1],$$

let

$$f_n(x) = x^n.$$

Determine its pointwise limit and prove directly from Definition 6.2 that the convergence is not uniform.

Exercise 6.2. Let

$$f_n(x) = \frac{x}{n} \qquad (x \in \mathbb{R}).$$

Prove that f_n converges pointwise to zero but not uniformly on $\mathbb{R}$.

Exercise 6.3. Suppose that

$$f_n \to f$$

uniformly on D and that every f_n is bounded by the same number M. Prove that

$$|f(x)| \leq M \qquad (x \in D).$$

Chapter 7

Infinite Series

7.1 Series, Partial Sums, and the Cauchy Criterion

An infinite sum is never formed by adding infinitely many terms at once. Its mathematical meaning is the limit of finite partial sums, so all questions about series begin with the sequence theory already developed.

Definition 7.1 (Series)**.** Let (a_n) be a real sequence. Its Nth **partial sum** is

$$s_N = \sum_{n=1}^{N} a_n.$$

The **series** with terms (a_n) is denoted by

$$\sum_{n=1}^{\infty} a_n.$$

It **converges** when (s_N) converges; its sum is then

$$\sum_{n=1}^{\infty} a_n = \lim_{N\to\infty} s_N.$$

Theorem 7.2 (Cauchy criterion for series)**.** The series

$$\sum_{n=1}^{\infty} a_n$$

converges if and only if, for every $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that

$$m > n \geq N \quad \Longrightarrow \quad |\sum_{k=n+1}^{m} a_k| < \varepsilon.$$

Proof. For $m > n$,

$$s_m - s_n = \sum_{k=n+1}^{m} a_k.$$

Thus the stated condition says exactly that the partial sums are Cauchy. Apply Theorem 4.19. □

Theorem 7.3 (Weierstrass M-test)**.** Let $f_n : D \to \mathbb{R}$ be functions, and suppose that there are numbers $M_n \geq 0$ such that

$$|f_n(x)| \leq M_n \qquad (x \in D, n \in \mathbb{N}).$$

If

$$\sum_{n=1}^{\infty} M_n$$

converges, then

$$\sum_{n=1}^{\infty} f_n$$

converges uniformly and absolutely on D.

Proof. For $q \geq p$,

$$|\sum_{n=p}^{q} f_n(x)| \leq \sum_{n=p}^{q} |f_n(x)| \leq \sum_{n=p}^{q} M_n.$$

The final bound is independent of x and tends uniformly to zero as $p, q \to \infty$ by the numerical Cauchy criterion. The uniform Cauchy criterion of Chapter 6 proves uniform convergence. The same estimate applied to $|f_n|$ proves absolute convergence at every point. □

Example 7.4 (A uniform majorant)**.** On $[-1, 1]$,

$$|\frac{x^n}{n^2}| \leq \frac{1}{n^2}.$$

Since $\sum_{n=1}^{\infty} 1/n^2$ converges, the M-test shows that

$$\sum_{n=1}^{\infty} \frac{x^n}{n^2}$$

converges uniformly and absolutely on $[-1, 1]$.

Proposition 7.5 (Finite changes and telescoping)**.** Changing finitely many terms of a series does not affect convergence. If

$$a_n = b_n - b_{n+1},$$

then

$$\sum_{n=1}^{N} a_n = b_1 - b_{N+1}.$$

Consequently, this telescoping series converges if and only if (b_n) converges, and its sum is

$$b_1 - \lim_{n\to\infty} b_n.$$

Proof. Partial sums after a finite change differ by a constant from some point onward. The displayed finite identity follows by cancellation; its limit proves the last assertion. □

Proposition 7.6 (Necessary condition)**.** If a series converges, then its terms tend to zero.

Proof. If $s_N \to s$, then

$$a_N = s_N - s_{N-1} \qquad (N \geq 2).$$

The algebra of limits gives $a_N \to 0$. □

Remark 7.7 (Preliminary strategy for choosing a test)**.** Before trying a sophisticated theorem, check the necessary condition $a_n \to 0$; failure settles divergence, although success proves nothing by itself. Next look for exact geometric or telescoping structure. For nonnegative terms, compare with an already understood benchmark. Factorials and exponentials often favor the Ratio Test, while expressions raised to the nth power often favor the Root Test. Alternating signs suggest checking absolute convergence first and then the Alternating-Series Test. An inconclusive test is information about the test, not a verdict about the series.

7.2 Nonnegative Series and Comparison

When terms are nonnegative, partial sums can only increase. This lets order and completeness convert estimates by simpler benchmark series into decisive convergence tests.

Theorem 7.8 (Comparison test)**.** Let

$$0 \le a_n \le b_n \qquad (n \in \mathbb{N}).$$

If

$$\sum_{n=1}^{\infty} b_n$$

converges, then

$$\sum_{n=1}^{\infty} a_n$$

converges. If the latter series diverges, then the former diverges.

Proof. The partial sums of (a_n) are increasing and bounded above by the sum of the series with terms b_n. Apply Theorem 4.18. The second statement is the contrapositive. □

Theorem 7.9 (Limit comparison test)**.** Let $a_n, b_n > 0$ eventually, and suppose

$$\lim_{n\to\infty} \frac{a_n}{b_n} = L$$

with

$$0 < L < \infty.$$

Then the series with terms a_n converges if and only if the series with terms b_n converges.

Proof. Eventually,

$$\frac{L}{2} b_n < a_n < \frac{3L}{2} b_n.$$

The conclusion follows from two applications of Theorem 7.8 and Proposition 7.5. □

Theorem 7.10 (Geometric series)**.** For $r \in \mathbb{R}$, the series with terms r^n, beginning at $n = 0$, converges if and only if

$$|r| < 1.$$

In that case,

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}.$$

Proof. For $r \neq 1$,

$$\sum_{n=0}^{N} r^n = \frac{1 - r^{N+1}}{1 - r}.$$

If $r = 0$, the conclusion is immediate. If $0 < r < 1$, write $r = 1/(1 + h)$ with $h > 0$. Bernoulli's inequality gives

$$(1 + h)^N \geq 1 + Nh,$$

so

$$0 \leq r^N \leq \frac{1}{1 + Nh} \longrightarrow 0.$$

The same conclusion holds whenever $|r| < 1$, because

$$|r^N| = |r|^N.$$

Taking limits proves the formula. If $|r| \geq 1$, the terms do not tend to zero, so Proposition 7.6 proves divergence. □

Theorem 7.11 (Cauchy condensation test)**.** Let (b_n) be a nonincreasing sequence of nonnegative real numbers. Then

$$\sum_{n=1}^{\infty} b_n$$

converges if and only if

$$\sum_{k=0}^{\infty} 2^k b_{2^k}$$

converges.

Proof. For every $k \geq 0$,

$$2^k b_{2^{k+1}} \leq \sum_{n=2^k}^{2^{k+1}-1} b_n \leq 2^k b_{2^k}.$$

Summing the right inequalities proves that convergence of the condensed series implies convergence of the original series. Summing the left inequalities gives

$$\frac{1}{2} \sum_{k=1}^{m+1} 2^k b_{2^k} \leq \sum_{n=1}^{2^{m+1}-1} b_n.$$

Hence boundedness of the original partial sums implies boundedness of the condensed partial sums. Both series have nonnegative terms, so monotone convergence completes the proof. □

7.3 Absolute Convergence and Products

Signs permit cancellation, which is useful but dangerous: a series can look small only because large positive and negative tails offset each other. Absolute convergence rules out that instability and will justify changing orders and multiplying series.

Definition 7.12 (Absolute and conditional convergence)**.** A series with real terms a_n **converges absolutely** if

$$\sum_{n=1}^{\infty} |a_n|$$

converges. It **converges conditionally** if it converges but does not converge absolutely.

Absolute convergence controls the total size of the terms; this is why it will justify rearrangement. Conditional convergence is weaker, so changing the order of its terms requires separate care.

Theorem 7.13 (Absolute convergence test)**.** Absolute convergence implies convergence.

Proof. For $m > n$,

$$\left|\sum_{k=n+1}^{m} a_k\right| \le \sum_{k=n+1}^{m} |a_k|.$$

The Cauchy criterion for the absolute-value series and Theorem 7.2 show that the original partial sums are Cauchy. □

Theorem 7.14 (Rearrangement of an absolutely convergent series)**.** Let

$$\sum_{n=1}^{\infty} a_n$$

converge absolutely, and let $(a_{\sigma(n)})$ be a rearrangement, where $\sigma : \mathbb{N} \to \mathbb{N}$ is a bijection. Then the rearranged series converges absolutely and has the same sum.

Proof. Let

$$A = \sum_{n=1}^{\infty} |a_n|.$$

Every finite rearranged absolute partial sum is at most A, so the rearranged absolute-value series converges. Given $\varepsilon > 0$, choose N such that

$$\sum_{n=N+1}^{\infty} |a_n| < \frac{\varepsilon}{2}.$$

Choose M so that

$$\{1, \ldots, N\} \subseteq \{\sigma(1), \ldots, \sigma(M)\}.$$

For $m \ge M$, cancellation of the common first N terms shows that the difference between the mth rearranged partial sum and the original sum has absolute value less than ε. Thus the sums agree. □

Theorem 7.15 (Cauchy product)**.** Suppose that

$$\sum_{n=0}^{\infty} a_n \quad \text{and} \quad \sum_{n=0}^{\infty} b_n$$

converge absolutely. Define

$$c_n = \sum_{k=0}^{n} a_k b_{n-k}.$$

Then

$$\sum_{n=0}^{\infty} c_n$$

converges absolutely and

$$\left(\sum_{n=0}^{\infty} a_n\right)\left(\sum_{n=0}^{\infty} b_n\right) = \sum_{n=0}^{\infty} c_n.$$

Proof. The partial sums of

$$\sum_{n=0}^{\infty}|c_n|$$

are bounded by the product of the two absolute sums, because

$$\sum_{n=0}^{N}|c_n| \leq \sum_{\substack{j,k\geq 0\\ j+k\leq N}} |a_j||b_k|.$$

Thus the Cauchy product converges absolutely. Let $\varepsilon > 0$. Choose K so large that

$$\left(\sum_{j=K+1}^{\infty}|a_j|\right)\left(\sum_{k=0}^{\infty}|b_k|\right) + \left(\sum_{j=0}^{\infty}|a_j|\right)\left(\sum_{k=K+1}^{\infty}|b_k|\right) < \varepsilon.$$

For $N \geq 2K$, every pair (j,k) with

$$0 \leq j,k \leq N, \qquad j+k > N$$

has either $j > K$ or $k > K$. Thus the difference between the finite product and the Nth Cauchy-product partial sum is less than ε in absolute value. Passing to limits proves the formula. □

7.4 Ratio, Root, and Alternating Tests

Many series are best compared not term by term with a fixed model, but by examining how consecutive terms or their roots behave. The following tests make that asymptotic comparison precise.

Definition 7.16 (Positive roots)**.** For $x \geq 0$ and $n \in \mathbb{N}$, $n \geq 1$, write

$$x^{1/n}$$

for the unique nonnegative number whose nth power is x. Existence follows from the Intermediate Value Theorem applied to

$$t \longmapsto t^n$$

on a sufficiently large compact interval; uniqueness follows from strict increase on $[0,\infty)$.

Definition 7.17 (Rational powers)**.** Let

$$x > 0, \qquad p \in \mathbb{Q}.$$

Choose integers a and b with

$$b > 0, \qquad p = \frac{a}{b},$$

and define

$$x^p = (x^a)^{1/b}.$$

The next lemma proves that this does not depend on the chosen representation of p.

Lemma 7.18 (Elementary laws of rational powers)**.** For $x > 0$ and $p, q \in \mathbb{Q}$, rational powers are well defined and satisfy

$$x^{p+q} = x^p x^q, \qquad (x^m)^p = x^{mp} \qquad (m \in \mathbb{Z}).$$

If

$$0 < x < y \qquad \text{and} \qquad 0 < p,$$

then

$$x^p < y^p.$$

Proof. Suppose

$$\frac{a}{b} = \frac{c}{d} \qquad (b, d > 0).$$

If

$$u = (x^a)^{1/b}, \qquad v = (x^c)^{1/d},$$

then $u, v > 0$ and

$$u^{bd} = x^{ad} = x^{bc} = v^{bd}.$$

Uniqueness of positive roots gives $u = v$, so the definition is independent of the representation. Writing rational exponents over a common positive denominator and raising both sides to that denominator proves the two displayed laws, again by uniqueness of positive roots.

For the order statement, put

$$z = \frac{y}{x} > 1.$$

The power laws give

$$\frac{y^p}{x^p} = z^p.$$

Writing

$$p = \frac{a}{b} \qquad (a, b \in \mathbb{N}),$$

one has

$$(z^p)^b = z^a > 1.$$

Thus $z^p > 1$, for otherwise its positive bth power would be at most 1. Hence $y^p > x^p$. $\square$

Theorem 7.19 (Rational p-series test)**.** Let

$$p \in \mathbb{Q}.$$

Then

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges if and only if

$$p > 1.$$

Proof. Suppose first that

$$p > 0.$$

By Lemma 7.18, the sequence

$$b_n = n^{-p}$$

is nonincreasing. The Cauchy Condensation Test gives convergence precisely when

$$\sum_{k=0}^{\infty} 2^k (2^k)^{-p} = \sum_{k=0}^{\infty} 2^{k(1-p)}$$

converges. This is a geometric series with positive ratio

$$2^{1-p};$$

by the order statement in Lemma 7.18, that ratio is less than 1 exactly when

$$p > 1.$$

If

$$p \le 0,$$

then

$$n^{-p} \ge 1 \qquad (n \in \mathbb{N}),$$

so the terms do not tend to zero. The necessary condition in Proposition 7.6 proves divergence. □

Remark 7.20 (From calculus to analysis: an area preview)**.** The familiar calculus picture compares a positive decreasing sequence $a_n = f(n)$ with rectangles under and over the graph of f. It suggests that

$$\sum_{n=1}^{\infty} f(n) \quad \text{and} \quad \int_1^{\infty} f(x)\,\mathrm{d}x$$

should have the same convergence behavior. This is only a preview here: improper integrals have not yet been defined, so the picture is not used as a theorem. Chapter 9 defines the integral as a limit of proper integrals, proves the Integral Test, and recovers the full real p-series rule

$$\sum \frac{1}{n^p} \text{ converges exactly when } p > 1.$$

The preceding theorem has already established the rational-exponent case without integration.

Theorem 7.21 (Ratio test)**.** Let $a_n \neq 0$ eventually. If there are $r < 1$ and N such that

$$|a_{n+1}/a_n| \le r \qquad (n \ge N),$$

then the series with terms a_n converges absolutely. If there are $r > 1$ and N such that

$$|a_{n+1}/a_n| \ge r \qquad (n \ge N),$$

then the series diverges.

Proof. The first inequality gives

$$|a_{N+k}| \le |a_N| r^k,$$

so geometric comparison applies. The second makes the absolute values eventually increasing and nonzero, so the terms cannot tend to zero. □

Theorem 7.22 (Root test)**.** Let (a_n) be a real sequence. If there are r and N such that

$$0 \le r < 1$$

and

$$|a_n|^{1/n} \le r \qquad (n \ge N),$$

then the series with terms a_n converges absolutely. If there are $r > 1$ and N such that

$$|a_n|^{1/n} \ge r \qquad (n \ge N),$$

then the series diverges.

Proof. The first inequality gives

$$|a_n| \le r^n \qquad (n \ge N),$$

so geometric comparison applies. The second gives

$$|a_n| \ge r^n \ge 1$$

eventually, so the terms do not tend to zero. □

Example 7.23 (When a test says nothing)**.** For $a_n = 1/n^2$,

$$\frac{a_{n+1}}{a_n} = \left(\frac{n}{n+1}\right)^2 \longrightarrow 1,$$

so the common limit form of the Ratio Test is inconclusive. The series nevertheless converges by the rational p-series theorem. The same ratio limit is 1 for the divergent harmonic series. A value of 1 therefore cannot distinguish the two cases; comparison or the p-series theorem is the appropriate tool.

Theorem 7.24 (Alternating-series test)**.** Let $b_n \ge 0$ decrease to 0. Then

$$\sum_{n=1}^{\infty} (-1)^{n+1} b_n$$

converges.

Proof. The even partial sums are increasing and the odd partial sums are decreasing. Each even sum is at most each later odd sum, so both are bounded. Their limits differ by b_{2n+1}, which tends to zero. Hence all partial sums have one common limit. □

Forward reference: the Integral Test and real exponents. The Integral Test compares a decreasing sequence of positive terms with an improper integral, so it belongs in Chapter 9. The rational p-series criterion above uses only Chapter 7 material. Arbitrary real powers have not yet been defined; after Chapter 8 constructs them from exp and log, Chapter 9 gives an analytic rederivation of the p-series criterion for every real exponent.

Remark 7.25 (Expanded test-selection guide)**.** Once the full toolkit is available, rational functions of n usually invite comparison with the power determined by their leading degrees. A positive decreasing term given by a manageable function invites the Integral Test. Factorials and exponentials invite ratios; nth powers invite roots; signs invite absolute convergence and then alternation. Limit comparison is often cleaner than forcing a termwise inequality. Always record the hypotheses (positivity, monotonicity, or eventual nonvanishing) before computing, and switch tests when the conclusion is inconclusive.

7.5 Power Series

Power series are infinite polynomials. Their special feature is a sharp radius inside which convergence is uniform on smaller intervals, allowing algebra and later differentiation to be recovered from finite polynomials.

Definition 7.26 (Power series)**.** For real coefficients (c_n) and center $a \in \mathbb{R}$, a **power series** is

$$\sum_{n=0}^{\infty} c_n (x-a)^n.$$

Theorem 7.27 (Radius of convergence)**.** Every power series has a number

$$R \in [0, \infty]$$

such that it converges absolutely for

$$|x-a| < R$$

and diverges for

$$|x-a| > R.$$

Proof. Let E be the set of $t \geq 0$ for which

$$\sum_{n=0}^{\infty} |c_n| t^n$$

converges, with $0^0 = 1$. Then $0 \in E$. If $t \in E$ and $0 \leq u < t$, comparison shows that $u \in E$. Let

$$R = \sup E,$$

with $R = \infty$ if E is unbounded. If $|x-a| < R$, choose $t \in E$ with

$$|x-a| < t;$$

comparison gives absolute convergence.

Conversely, suppose the series converges at a point with

$$z = |x-a| > 0.$$

Its terms tend to zero and are eventually bounded by some M. Thus, for $0 \leq t < z$ and all sufficiently large n,

$$|c_n| t^n \leq M \left(\frac{t}{z}\right)^n.$$

Geometric comparison gives $t \in E$. If $z > R$, choosing t strictly between R and z contradicts the definition of R. Hence the series diverges outside its radius. $\square$

The theorem intentionally makes no assertion at points satisfying

$$|x-a| = R.$$

Endpoint behavior depends on the particular coefficients and must be examined separately.

Theorem 7.28 (Uniform convergence inside the radius)**.** Let a power series have radius R, and let

$$0 \le \rho < R.$$

Then it converges uniformly on

$$[a-\rho, a+\rho].$$

Proof. Choose t with

$$\rho < t < R.$$

The series

$$\sum_{n=0}^{\infty} |c_n| t^n$$

converges. For $|x-a| \le \rho$,

$$|c_n(x-a)^n| \le |c_n| t^n.$$

The Weierstrass M-test, Theorem 7.3, proves the assertion. □

Power-series operations deferred. On every closed interval strictly inside its radius, a power series is uniformly convergent. Termwise differentiation and integration require additional theorems and are proved in Chapters 8 and 9.

Exercises

Exercise 7.1. Use Theorem 7.19 to prove that

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

converges and that the harmonic series diverges. Identify the dyadic blocks responsible for the latter conclusion.

Exercise 7.2. Use Theorem 7.24 and Theorem 7.19 to show that

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$$

converges conditionally.

Exercise 7.3. Find the radius of convergence of

$$\sum_{n=0}^{\infty} 3^n (x-2)^n.$$

What does Theorem 7.27 leave undecided at the two endpoints?

Exercise 7.4. Let

$$\sum_{n=0}^{\infty} c_n (x-a)^n$$

have radius of convergence $R > 0$. Prove that it converges absolutely at every point satisfying

$$|x-a| < R.$$

Which step in the proof of Theorem 7.27 gives this conclusion?

Chapter 8

Differentiation

8.1 Derivative and Differentiability

The derivative formalizes instantaneous rate of change and the slope of the best local linear approximation. Its definition is a function limit, so the earlier theory of limits and continuity supplies its logical foundation.

Definition 8.1 (Derivative)**.** Let $f : D \to \mathbb{R}$ and let a be a limit point of $D \cap \{x : x \neq a\}$. The function is **differentiable at** a if the following limit exists:

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}.$$

The number $f'(a)$ is the **derivative** of f at a.

Theorem 8.2 (Differentiability implies continuity)**.** If f is differentiable at a, then it is continuous at a.

Proof. For $x \neq a$,

$$f(x) - f(a) = (x - a)\frac{f(x) - f(a)}{x - a}.$$

The first factor tends to zero and the second tends to $f'(a)$, so the product tends to zero by the algebra of limits. Thus $f(x) \to f(a)$. $\square$

Theorem 8.3 (Derivative rules)**.** Let $f, g : D \to \mathbb{R}$ be differentiable at a, and let $c \in \mathbb{R}$. Then $f + g$, cf, and fg are differentiable at a, with

$$(f + g)'(a) = f'(a) + g'(a), \qquad (cf)'(a) = cf'(a),$$
$$(fg)'(a) = f'(a)g(a) + f(a)g'(a).$$

If $g(a) \neq 0$, then f/g is differentiable at a and

$$\left(\frac{f}{g}\right)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{g(a)^2}.$$

Proof. Expand each difference quotient. For products, use

$$f(x)g(x) - f(a)g(a) = f(x)\big(g(x) - g(a)\big) + g(a)\big(f(x) - f(a)\big).$$

Divide by $x - a$, then use continuity of differentiable functions and the algebra of limits. The quotient formula follows by applying the product formula to $f \cdot (1/g)$ and differentiating the identity $g \cdot (1/g) = 1$. $\square$

Example 8.4 (Exceptional points require the definition)**.** Away from 0, the absolute-value function is given by a polynomial and has derivative 1 or -1. At 0, however,

$$\frac{|h| - |0|}{h} = \begin{cases} 1, & h > 0, \\ -1, & h < 0, \end{cases}$$

so the one-sided derivatives disagree and $|x|$ is not differentiable there. It is nevertheless continuous. This shows both that continuity is necessary but insufficient and that a formula valid on separate pieces does not settle a junction point.

Example 8.5 (A piecewise differentiability check)**.** Let

$$f(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Temporarily using the familiar prior-calculus bound $|\sin t| \leq 1$ (proved later from the power-series construction),

$$\left| \frac{f(h) - f(0)}{h} \right| = |h \sin(1/h)| \leq |h| \longrightarrow 0.$$

Thus $f'(0) = 0$. The correct workflow is to use differentiation rules where their hypotheses apply and return to the limit definition at exceptional points.

8.2 Chain Rule and Mean Value Theorems

Derivative rules explain how local linear approximations combine. The mean value theorems add a global principle: a net change across an interval is recorded by a derivative at some interior point.

Theorem 8.6 (Chain rule)**.** Let $f : D \to \mathbb{R}$ be differentiable at a, let $g : E \to \mathbb{R}$ be differentiable at $f(a)$, and suppose $f(x) \in E$ for all $x \in D$ sufficiently near a. Then $g \circ f$ is differentiable at a, with

$$(g \circ f)'(a) = g'(f(a))f'(a).$$

Proof. For $f(x) \neq f(a)$, factor the difference quotient as

$$\frac{g(f(x)) - g(f(a))}{f(x) - f(a)} \, \frac{f(x) - f(a)}{x - a}.$$

The first factor tends to $g'(f(a))$ along points where it is defined, and the second tends to $f'(a)$. If $f(x) = f(a)$, define the first factor to be $g'(f(a))$; the product identity still holds because the second factor is zero. The limit algebra now gives the formula. □

Theorem 8.7 (Rolle's theorem)**.** Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b), where $a < b$. If $f(a) = f(b)$, then some $c \in (a, b)$ satisfies $f'(c) = 0$.

Proof. By Theorem 5.26, f attains a maximum and a minimum. If both values equal $f(a)$, then f is constant and every interior point works. Otherwise an extremum occurs at an interior point c. At an interior maximum, the difference quotient is nonnegative for $x < c$ and nonpositive for $x > c$; since its two-sided limit exists, it is zero. The minimum case is identical with signs reversed. □

Theorem 8.8 (Mean value theorem)**.** Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b), where $a < b$. Then some $c \in (a, b)$ satisfies

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. Define

$$h(x) = f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a).$$

Then $h(a) = h(b) = 0$. Apply Theorem 8.7 to h and compute $h'(c) = 0$. □

Definition 8.9 (Monotonicity)**.** Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$. The function f is **increasing** (or **nondecreasing**) if

$$x < y \quad \Longrightarrow \quad f(x) \leq f(y),$$

and **strictly increasing** if the final inequality is strict. It is **decreasing** (or **nonincreasing**) if

$$x < y \quad \Longrightarrow \quad f(x) \geq f(y),$$

and **strictly decreasing** if the final inequality is strict. A function is **monotone** if it is increasing or decreasing, and **strictly monotone** if it is strictly increasing or strictly decreasing. Throughout these notes, "increasing" and "decreasing" are used in the non-strict sense; strict inequalities are indicated by "strictly increasing" and "strictly decreasing."

Corollary 8.10 (Bounded derivative criterion for Lipschitz continuity)**.** If f is continuous on an interval I, differentiable in its interior, and $|f'| \leq M$ there, then

$$|f(y) - f(x)| \leq M|y - x| \qquad (x, y \in I).$$

Hence f is Lipschitz, and therefore uniformly continuous, on I.

Proof. Apply the Mean Value Theorem on the closed interval with endpoints x and y; endpoint cases follow by continuity. □

The ordinary Mean Value Theorem compares the net change of one function with one derivative. Cauchy's version compares the changes of two functions at once, which is exactly the structure needed later when L'Hôpital's Rule relates a quotient of function increments to a quotient of derivatives.

Theorem 8.11 (Cauchy's mean value theorem)**.** Let $f, g : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b), where $a < b$. Then some $c \in (a, b)$ satisfies

$$\big(f(b) - f(a)\big)g'(c) = \big(g(b) - g(a)\big)f'(c).$$

Proof. Apply Theorem 8.7 to

$$h(x) = \big(f(b) - f(a)\big)g(x) - \big(g(b) - g(a)\big)f(x).$$

The endpoint values of h agree, and $h'(c) = 0$ is exactly the displayed identity. □

Infinite limits. We use the eventual-inequality definitions from Chapter 5. In particular, expressions such as ∞/∞ name an **indeterminate limiting form**; they are not arithmetic quotients of numbers called infinity. The absolute-divergence formulation permits either sign for each function.

Theorem 8.12 (L'Hôpital's rule: the 0/0 and ∞/∞ forms)**.** Let $f, g : (a, b) \to \mathbb{R}$ be differentiable, where $a < b$. Assume that

$$g'(x) \neq 0 \qquad \text{for every } x \in (a, b),$$

and that one of the following alternatives holds:

(a) f and g extend continuously to a with $f(a) = g(a) = 0$;

(b)

$$\lim_{x\to a^+} |f(x)| = \infty \qquad \text{and} \qquad \lim_{x\to a^+} |g(x)| = \infty.$$

If $L \in \mathbb{R}$ and

$$\lim_{x\to a^+} \frac{f'(x)}{g'(x)} = L.$$

then the quotient is defined sufficiently close to a and

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = L.$$

Proof. First suppose that $f(a) = g(a) = 0$. If $g(x) = 0$ for some $x > a$, Rolle's theorem applied to g on $[a, x]$ would give a point where g' vanishes. Thus the quotient is defined. Fix $x \in (a, b)$. Cauchy's theorem on $[a, x]$ gives $c_x \in (a, x)$ such that

$$\frac{f(x)}{g(x)} = \frac{f'(c_x)}{g'(c_x)}.$$

As $x \to a^+$, one has $a < c_x < x$, hence $c_x \to a^+$. The assumed derivative-quotient limit therefore gives the conclusion.

Now suppose that both absolute values diverge. Let $\varepsilon > 0$. Choose $\delta > 0$ so that

$$|\frac{f'(t)}{g'(t)} - L| < \frac{\varepsilon}{4} \qquad \text{whenever} \qquad a < t < a + \delta,$$

and choose $c \in (a, a + \delta)$. For $x \in (a, c)$, Rolle's theorem shows that $g(x) \neq g(c)$, and Cauchy's Mean Value Theorem on $[x, c]$ gives a point $\xi_x \in (x, c)$ for which

$$\frac{f(x) - f(c)}{g(x) - g(c)} = \frac{f'(\xi_x)}{g'(\xi_x)}.$$

Thus this quotient differs from L by less than $\varepsilon/4$. Since $|g(x)| \to \infty$, for all x sufficiently close to a we also have

$$|\frac{g(x) - g(c)}{g(x)}| \leq \frac{3}{2} \qquad \text{and} \qquad |\frac{f(c) - Lg(c)}{g(x)}| < \frac{\varepsilon}{2}.$$

Writing the desired difference as

$$\frac{f(x)}{g(x)} - L = \frac{g(x) - g(c)}{g(x)} \left(\frac{f(x) - f(c)}{g(x) - g(c)} - L \right) + \frac{f(c) - Lg(c)}{g(x)},$$

the triangle inequality proves that its absolute value is less than ε. $\square$

Infinite derivative-quotient variants. Under the same hypotheses, if the derivative quotient tends to $+\infty$ (respectively, $-\infty$), then so does $f(x)/g(x)$. For the $0/0$ form this follows from the displayed Cauchy-Mean-Value identity in the proof. For the ∞/∞ form, use the second displayed decomposition there: the first factor tends to 1, while the second term tends to 0. Taking the derivative quotient arbitrarily large positive (respectively, negative) therefore gives the stated conclusion.

Other domains and forms. The same proof works for limits as $x \to a^-$ and as $x \to \pm\infty$, after the evident changes of domain. For example,

$$\lim_{x\to\infty} \frac{3x^2 + x}{2x^2 - 1} = \lim_{x\to\infty} \frac{6x+1}{4x} = \frac{3}{2}.$$

Here both the numerator and denominator have absolute value tending to infinity; their signs can affect the resulting quotient, but they require no separate theorem. L'Hôpital's Rule applies directly only to quotient forms of type $0/0$ and ∞/∞. Forms such as $0 \cdot \infty$, $\infty - \infty$, 1^∞, 0^0, and ∞^0 must first be transformed into a suitable quotient or logarithmic form.

Example 8.13 (Two legitimate applications)**.** Once the elementary-function derivatives below have been established, L'Hôpital's Rule gives

$$\lim_{x\to 0} \frac{e^x - 1}{x} = \lim_{x\to 0} \frac{e^x}{1} = 1$$

from the $0/0$ form, and

$$\lim_{x\to+\infty} \frac{\log x}{x} = \lim_{x\to+\infty} \frac{1/x}{1} = 0$$

from the ∞/∞ form. In each calculation the equal sign relates two limits under the theorem's hypotheses; it is not the pointwise identity $f/g = f'/g'$.

Remark 8.14 (When not to apply the rule)**.** The quotient $(x^2+1)/x$ tends to $+\infty$ as $x \to 0^+$, but it is not a $0/0$ or ∞/∞ form, so differentiating numerator and denominator would incorrectly suggest the limit 0. For $x^2 \sin(1/x)/x$ at 0, the derivative quotient is more complicated than the original expression, while the estimate $|x \sin(1/x)| \leq |x|$ settles the limit immediately. Finally, if f'/g' has no limit, the theorem gives no conclusion; the original quotient may still have a limit by another method. Always identify the form and verify the hypotheses before differentiating.

8.3 Definitions of Elementary Functions

Remark 8.15 (From familiar formulas to rigorous objects)**.** Polynomials

$$a_0 + a_1 x + \cdots + a_n x^n$$

are available as soon as the field operations on $\mathbb{R}$ are available, and rational functions are defined wherever their denominators are nonzero. Roots require the existence and uniqueness theorem proved from continuity and completeness in Chapter 5; rational powers then use those roots. The expressions a^x, $\log x$, $\sin x$, and $\cos x$ for arbitrary real x require more analysis. Earlier calculus treated them as familiar computational objects; here we now give definitions and prove the identities and derivative formulas previously used informally.

Power series provide one coherent construction that does not appeal to geometry, areas, or already-known calculus formulas. Their continuity, identities, periodicity, and derivatives will be theorems rather than parts of the definitions.

Proposition 8.16 (Convergence of the elementary power series)**.** For every $x \in \mathbb{R}$, the three series

$$\sum_{n=0}^{\infty} \frac{x^n}{n!}, \qquad \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \qquad \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

converge absolutely.

Proof. For the exponential series, the ratio of consecutive absolute terms is

$$\frac{|x|}{n+1} \longrightarrow 0.$$

For the sine and cosine series, the corresponding ratios are

$$\frac{|x|^2}{(2n+2)(2n+3)} \longrightarrow 0, \qquad \frac{|x|^2}{(2n+1)(2n+2)} \longrightarrow 0.$$

The Ratio Test proves the claim in all three cases. □

Definition 8.17 (Exponential and trigonometric functions)**.** For $x \in \mathbb{R}$, define

$$\exp(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \qquad \sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!},$$

$$\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}.$$

The base-e exponential function is defined by $e^x = \exp(x)$.

8.4 Differentiating Limits and Power Series

Differentiating a limit term by term is not automatic. Uniform control of the derivative series, together with one anchored value, supplies exactly the hypothesis that prevents differentiation from being an unjustified interchange of limits.

Theorem 8.18 (Termwise differentiation)**.** Let

$$I = [\alpha, \beta]$$

be a compact interval, let (f_n) be differentiable functions on I, and let $x_0 \in I$. Suppose that

$$\sum_{n=0}^{\infty} f_n(x_0)$$

converges and that

$$\sum_{n=0}^{\infty} f_n'$$

converges uniformly on I to a function g. Then

$$f(x) = \sum_{n=0}^{\infty} f_n(x)$$

converges uniformly on I, is differentiable on the interior of I, and satisfies

$$f'(x) = g(x).$$

Proof. Let

$$F_N = \sum_{n=0}^{N} f_n.$$

Uniform Cauchyness of (F_N') and the Mean Value Theorem imply that, for m, n sufficiently large,

$$|(F_m(x) - F_n(x)) - (F_m(x_0) - F_n(x_0))| \leq \frac{\varepsilon}{1+\beta-\alpha}|x - x_0| < \varepsilon$$

throughout I. Since $(F_N(x_0))$ converges, this proves that (F_N) is uniformly Cauchy, hence uniformly convergent, to f.

Fix an interior point x. Given $\varepsilon > 0$, choose N such that

$$|F_m'(u) - F_n'(u)| < \varepsilon$$

for all $m, n \geq N$ and all $u \in I$. The Mean Value Theorem, followed by $m \to \infty$, gives

$$\left| \frac{f(x+h) - f(x)}{h} - \frac{F_N(x+h) - F_N(x)}{h} \right| \leq \varepsilon$$

whenever $x + h \in I$ and $h \neq 0$. Letting $h \to 0$ and then using $F_N'(x) \to g(x)$ proves

$$f'(x) = g(x).$$

□

Theorem 8.19 (Power-series differentiation)**.** Let

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n$$

have radius of convergence R. Then, for

$$|x - a| < R,$$

the function f is differentiable and

$$f'(x) = \sum_{n=1}^{\infty} n c_n (x-a)^{n-1}.$$

The derivative series has the same radius of convergence. Repeated termwise differentiation is valid at every point strictly inside the radius.

Proof. Fix numbers ρ, s, t with

$$0 < \rho < s < t < R.$$

The numerical series

$$\sum_{n=0}^{\infty} |c_n| t^n$$

converges. Since the sequence

$$n\left(\frac{\rho}{t}\right)^{n-1}$$

is bounded, comparison gives uniform convergence of the derivative series on

$$[a-\rho, a+\rho].$$

The original series converges at a, so Theorem 8.18 gives the formula on that interval. Since every point with distance less than R from a lies in such an interval, the formula holds throughout the open convergence interval. Applying the same argument to the differentiated coefficients proves the final statement and the equality of radii. □

This theorem is the analytic justification for formal power-series differentiation: the formal derivative represents the derivative of the summed function only after uniform convergence of the derivative series has been verified on smaller closed intervals.

Definition 8.20 (Higher derivatives)**.** If f' is differentiable, write

$$f'' = (f')'.$$

Inductively, $f^{(n)}$ denotes the nth derivative when it exists, with

$$f^{(0)} = f.$$

Successive derivatives provide increasingly accurate local polynomial models:

$$f(x) \approx f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots .$$

Taylor's theorem makes this familiar calculus approximation rigorous. Its new content is the remainder term, which quantifies the error and foreshadows the local linearization estimates used in multivariable analysis.

Theorem 8.21 (Taylor's theorem with Lagrange remainder)**.** Let $f : [a,b] \to \mathbb{R}$ have $n+1$ continuous derivatives on $[a,b]$, where $n \geq 0$. For every $x \in [a,b]$ there is a number ξ between a and x such that

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1}.$$

Proof. The case $x = a$ is immediate. For $x \neq a$, subtract the Taylor polynomial of degree n from f and choose a constant C so that

$$H(t) = f(t) - \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(t-a)^k - C(t-a)^{n+1}$$

satisfies

$$H(a) = H(x) = 0.$$

By the definition of the Taylor polynomial,

$$H(a) = H'(a) = \cdots = H^{(n)}(a) = 0.$$

Rolle's theorem first gives a zero of H' between a and x. Together with $H'(a) = 0$, that gives two zeros of H'; applying Rolle again gives a zero of H''. Continuing in this way is legitimate because each derivative through order n is continuous on the closed interval between a and x and differentiable in its interior. After $n+1$ applications, there is a point ξ between a and x for which

$$H^{(n+1)}(\xi) = 0.$$

Solving this identity for C gives

$$C = \frac{f^{(n+1)}(\xi)}{(n+1)!}.$$

The equation $H(x) = 0$ is the required formula. □

8.5 The Standard Elementary Functions

The preceding machinery now turns the power-series definitions into the familiar exponential, logarithmic, trigonometric, and inverse trigonometric functions. Their standard properties are consequences, not assumptions.

Theorem 8.22 (Derivatives of the power-series definitions)**.** The functions exp, sin, and cos defined above satisfy
$$\exp'(x) = \exp(x), \qquad \sin'(x) = \cos(x), \qquad \cos'(x) = -\sin(x).$$

Proof. The preceding proposition shows that all three series have infinite radius of convergence. Theorem 8.19 now gives the three derivative formulas. □

Theorem 8.23 (Addition formulas and fundamental identities)**.** For all $x, y \in \mathbb{R}$,

$$\exp(x + y) = \exp(x)\exp(y),$$

$$\sin(x + y) = \sin x \cos y + \cos x \sin y,$$
$$\cos(x + y) = \cos x \cos y - \sin x \sin y,$$

and
$$\sin^2 x + \cos^2 x = 1.$$

Proof. The Cauchy Product Theorem and the binomial theorem give the exponential identity. Fix y and let the difference between the two sides of the proposed sine addition formula be $h(x)$. By Theorem 8.22,
$$h'(x) = \cos(x + y) + \sin x \sin y - \cos x \cos y$$
is the difference between the two sides of the proposed cosine addition formula. Differentiating the latter difference gives the negative of h. Thus the sum of the squares of these two differences is constant. At $x = 0$, both differences vanish, because

$$\sin 0 = 0, \qquad \cos 0 = 1,$$

as follows from their series. Hence both differences vanish identically. Finally, the derivative of

$$\sin^2 x + \cos^2 x$$

is zero, and its value at 0 is 1; the Mean Value Theorem proves the last identity. □

Theorem 8.24 (Exponential bijection)**.** The function exp is a strictly increasing bijection from $\mathbb{R}$ onto
$$(0, \infty).$$

Proof. The exponential identity gives

$$\exp(x/2)\exp(-x/2) = \exp(0) = 1.$$

Thus $\exp(x/2) \neq 0$, and consequently

$$\exp(x) = \exp(x/2)^2 > 0.$$

Its derivative is positive, so the Mean Value Theorem makes it strictly increasing. Its series gives

$$\exp(n) \geq 1 + n \qquad (n \in \mathbb{N}),$$

so monotonicity implies that $\exp(x)$ tends to infinity as $x \to \infty$. For $t > 0$, the addition law gives

$$\exp(-t) = \frac{1}{\exp(t)},$$

and hence $\exp(x)$ tends to 0 as $x \to -\infty$. If $y > 0$, choose real numbers $A < B$ such that

$$\exp(A) < y < \exp(B).$$

Continuity and the Intermediate Value Theorem then give an $x \in (A, B)$ with $\exp(x) = y$. Positivity shows that no other values occur, proving the asserted range. $\square$

Definition 8.25 (Natural logarithm)**.** Define

$$\log : (0, \infty) \to \mathbb{R}, \qquad \log = (\exp)^{-1}.$$

Theorem 8.26 (Inverse derivative theorem)**.** Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \to \mathbb{R}$ be continuously differentiable and strictly monotone. If

$$f'(x) \neq 0 \qquad (x \in I),$$

then its inverse on $f(I)$ is differentiable, with

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}.$$

In particular,

$$\log'(x) = \frac{1}{x} \qquad (x > 0).$$

Proof. Continuity and strict monotonicity make f^{-1} continuous on its image. For $y = f(x)$ and $z = f(u)$, the Mean Value Theorem gives a point c between x and u such that

$$\frac{f^{-1}(z) - f^{-1}(y)}{z - y} = \frac{u - x}{f(u) - f(x)} = \frac{1}{f'(c)}.$$

As $z \to y$, continuity of the inverse gives $u \to x$, hence $c \to x$; continuity of f' proves the formula. Apply it to $f = \exp$ to obtain

$$\log'(x) = \frac{1}{\exp(\log x)} = \frac{1}{x}.$$

$\square$

Definition 8.27 (Real powers of positive numbers)**.** For $x > 0$ and $p \in \mathbb{R}$, define

$$x^p = \exp(p \log x).$$

Theorem 8.28 (Derivative of a real power)**.** For every $p \in \mathbb{R}$, the function

$$x \longmapsto x^p$$

is differentiable on $(0, \infty)$ and

$$\frac{\mathrm{d}}{\mathrm{d}x} x^p = p x^{p-1}.$$

Proof. The chain rule and Theorem 8.26 give

$$\frac{\mathrm{d}}{\mathrm{d}x}\exp(p\log x) = \exp(p\log x)\frac{p}{x} = px^{p-1}.$$

□

Theorem 8.29 (Construction of π and the trigonometric ranges)**.** There is a least positive zero c of cos. Put

$$\pi = 2c.$$

Then

$$\cos c = 0, \quad \sin c = 1, \quad \cos\pi = -1, \quad \sin\pi = 0,$$

and

$$\cos(x+2\pi) = \cos x, \qquad \sin(x+2\pi) = \sin x.$$

Moreover, $\cos x > 0$ for $-c < x < c$, and sin maps $[-c, c]$ continuously and strictly increasingly onto $[-1, 1]$.

Proof. Taylor's theorem applied to cos at 0 with degree 3 gives, for some ξ between 0 and 3,

$$\cos 3 = 1 - \frac{3^2}{2} + \frac{\cos\xi}{4!}3^4.$$

Since the identity in Theorem 8.23 gives $|\cos\xi| \le 1$, the right side is at most

$$1 - \frac{9}{2} + \frac{81}{24} < 0.$$

Thus cos has a zero in $(0, 3)$ by the Intermediate Value Theorem. Since $\cos 0 = 1$, continuity gives a number

$$\delta > 0$$

such that

$$\cos x > 0 \qquad (0 \le x \le \delta).$$

Consequently the set

$$Z = \{x \in [\delta, 3] : \cos x = 0\}$$

is nonempty: the zero just obtained cannot lie in $[0, \delta]$. Let

$$c = \inf Z.$$

For every $n \in \mathbb{N}$, the defining property of the infimum gives a point

$$z_n \in Z$$

with

$$c \le z_n < c + \frac{1}{n}.$$

Thus $z_n \to c$, and continuity gives

$$\cos c = \lim_{n\to\infty} \cos z_n = 0.$$

Also $c \ge \delta > 0$. Every positive zero of cosine is at least c: a zero larger than 3 is certainly so, and a zero in $(0, 3]$ belongs to Z. Thus c is the least positive zero of cosine.

If $0 \le x < c$ and $\cos x \le 0$, the Intermediate Value Theorem applied on $[0, x]$ would give a zero smaller than c. Thus $\cos x > 0$ on $[0, c)$. The power-series definitions give that cosine is even and sine is odd. Hence cosine is positive on $(-c, c)$. By the Mean Value Theorem, sine is strictly increasing on $[-c, c]$. The identity

$$\sin^2 c + \cos^2 c = 1$$

and monotonicity give $\sin c = 1$. The addition formulas now give

$$\sin(2c) = 0, \qquad \cos(2c) = -1,$$

and, after one further doubling,

$$\sin(4c) = 0, \qquad \cos(4c) = 1.$$

The addition formulas therefore give the stated $2\pi = 4c$ periodicity. Finally, oddness and the endpoint values show that the continuous strictly increasing function sine maps $[-c, c]$ onto $[-1, 1]$. □

Definition 8.30 (Tangent)**.** On

$$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right),$$

define

$$\tan x = \frac{\sin x}{\cos x}.$$

Theorem 8.31 (Trigonometric restrictions)**.** The function

$$\sin : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \to [-1, 1]$$

is continuous, strictly increasing, and onto. The function

$$\cos : [0, \pi] \to [-1, 1]$$

is continuous, strictly decreasing, and onto. Tangent is continuous, strictly increasing, and maps

$$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

onto $\mathbb{R}$.

Proof. The sine assertion is part of Theorem 8.29. For cosine on $(0, \pi)$, sine is positive: on

$$\left(0, \frac{\pi}{2}\right)$$

this follows from strict increase, and on

$$\left(\frac{\pi}{2}, \pi\right)$$

it follows from

$$\sin x = \sin(\pi - x).$$

Hence

$$\cos' = -\sin < 0,$$

so the endpoint values and the Intermediate Value Theorem give the cosine assertion.

The preceding theorem gives positivity of cosine on the tangent domain, and hence the quotient rule gives

$$\tan'(x) = \frac{1}{\cos^2 x} > 0.$$

Thus tangent is strictly increasing. As $x \to (\pi/2)^-$, continuity gives $\sin x \to 1$ and $\cos x \to 0^+$. Given $M > 0$, choose x sufficiently close to $\pi/2$ that

$$\sin x > \frac{1}{2}, \qquad 0 < \cos x < \frac{1}{2M};$$

then $\tan x > M$. Oddness gives the corresponding limit $-\infty$ at $-\pi/2$. The Intermediate Value Theorem proves that tangent is onto $\mathbb{R}$. □

Definition 8.32 (Inverse trigonometric functions)**.** Define

$$\arcsin : [-1,1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \qquad \arcsin = (\sin|_{[-\pi/2,\pi/2]})^{-1},$$

$$\arccos : [-1,1] \to [0,\pi], \qquad \arccos = (\cos|_{[0,\pi]})^{-1},$$

and

$$\arctan : \mathbb{R} \to \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \qquad \arctan = (\tan|_{(-\pi/2,\pi/2)})^{-1}.$$

Theorem 8.33 (Derivatives of the inverse trigonometric functions)**.** The derivative formulas are

$$(\arcsin)'(x) = \frac{1}{\sqrt{1-x^2}}, \qquad (\arccos)'(x) = -\frac{1}{\sqrt{1-x^2}} \qquad (-1 < x < 1),$$

and

$$(\arctan)'(x) = \frac{1}{1+x^2} \qquad (x \in \mathbb{R}).$$

Proof. The inverse-derivative theorem and

$$\cos(\arcsin x) > 0, \qquad \sin(\arccos x) > 0$$

give the first two formulas using $\sin^2 + \cos^2 = 1$. The tangent formula follows from the inverse-derivative theorem and

$$\tan'(x) = 1 + \tan^2 x.$$

□

8.6 Convexity and a Continuous Nondifferentiable Example

Definition 8.34 (Convex function)**.** Let I be an interval. A function $f : I \to \mathbb{R}$ is **convex** if

$$f(tx + (1-t)y) \le tf(x) + (1-t)f(y)$$

for all $x, y \in I$ and $0 \le t \le 1$. Geometrically, the graph lies below each of its chords.

Theorem 8.35 (Derivative criteria for convexity)**.** Let f be differentiable on an open interval I. Then f is convex if and only if f' is nondecreasing. In that case every tangent line supports the graph:

$$f(y) \geq f(x) + f'(x)(y - x) \qquad (x, y \in I).$$

If f'' exists throughout I and $f'' \geq 0$, then f is convex.

Proof. For $x < z < y$, convexity rearranges to

$$\frac{f(z) - f(x)}{z - x} \leq \frac{f(y) - f(x)}{y - x} \leq \frac{f(y) - f(z)}{y - z}.$$

Letting endpoints approach z shows that f' is nondecreasing. Conversely, the Mean Value Theorem applied on the two sides of $z = tx + (1 - t)y$ compares the corresponding secant slopes and yields the chord inequality. Letting $z \to x$ in the first secant comparison gives the tangent-line inequality. Finally, $f'' \geq 0$ makes f' nondecreasing by the Mean Value Theorem. □

Thus x^2, e^x, and $-\log x$ are convex on their natural domains, by their nonnegative second derivatives. The function $-x^2$ is not convex; at the midpoint of -1 and 1 its value 0 lies above the chord value -1. Applying the tangent inequality to e^x at 0 gives the useful estimate

$$e^y \geq 1 + y.$$

Example 8.36 (Weierstrass's warning)**.** For suitable constants $0 < a < 1$ and odd integer b (for example under the classical sufficient condition $ab > 1 + 3\pi/2$), the series

$$W(x) = \sum_{n=0}^{\infty} a^n \cos(\pi b^n x)$$

converges uniformly and hence defines a continuous function on $\mathbb{R}$, yet W is differentiable nowhere. Uniform convergence follows immediately from $|a^n \cos(\pi b^n x)| \leq a^n$ and the geometric series. The nowhere-differentiability proof requires a delicate oscillation estimate and is not needed later, so it is omitted. The example decisively separates continuity from differentiability: continuity alone supplies no hidden smoothness.

Exercises

Exercise 8.1. Use the Mean Value Theorem to prove that if

$$|f'(x)| \leq M \qquad (x \in (a, b)),$$

where f is continuous on $[a, b]$ and differentiable on (a, b), then

$$|f(y) - f(x)| \leq M|y - x| \qquad (x, y \in [a, b]).$$

Exercise 8.2. Use Theorem 8.19 to differentiate

$$\sum_{n=0}^{\infty} (x - 1)^n$$

on its interval of convergence. Explain why this is an analytic theorem rather than merely a formal manipulation of symbols.

Exercise 8.3. Prove from the definitions that

$$\arcsin(-x) = -\arcsin(x) \qquad (-1 \leq x \leq 1).$$

Chapter 9

Riemann Integration and the Fundamental Theorem of Calculus

9.1 Partitions and Riemann Sums

Integration formalizes approximation of accumulated quantities by finite sums. Darboux upper and lower sums avoid choosing sample points: they record the best guaranteed under- and over-approximations on every subinterval.

Definition 9.1 (Partition)**.** Let $a < b$. A **partition** of $[a, b]$ is a finite ordered set

$$P = \{x_0, x_1, \dots, x_n\}$$

with $a = x_0 < x_1 < \cdots < x_n = b$. Its norm is

$$\|P\| = \max_{1 \le i \le n} (x_i - x_{i-1}).$$

Definition 9.2 (Upper and lower sums)**.** Let $f : [a, b] \to \mathbb{R}$ be bounded and let P be a partition. Put

$$M_i = \sup \{f(x) : x \in [x_{i-1}, x_i]\}, \qquad m_i = \inf \{f(x) : x \in [x_{i-1}, x_i]\}.$$

The **upper sum** and **lower sum** are

$$U(f, P) = \sum_{i=1}^{n} M_i(x_i - x_{i-1}), \qquad L(f, P) = \sum_{i=1}^{n} m_i(x_i - x_{i-1}).$$

Definition 9.3 (Riemann integrability)**.** Let $f : [a, b] \to \mathbb{R}$ be bounded. It is **Riemann integrable** if

$$\inf_P U(f, P) = \sup_P L(f, P),$$

where the infimum and supremum range over all partitions of $[a, b]$. Their common value is the **Riemann integral** of f and is denoted by

$$\int_a^b f(x)\, \mathrm{d}x.$$

Proposition 9.4 (Darboux criterion)**.** A bounded function $f : [a, b] \to \mathbb{R}$ is Riemann integrable if and only if, for every $\varepsilon > 0$, there is a partition P with

$$U(f, P) - L(f, P) < \varepsilon.$$

Proof. Every lower sum is at most every upper sum: refine both partitions by taking their union, then compare each subinterval. Thus the lower supremum is at most the upper infimum. If they are equal to I, choose partitions P, Q with $U(f,P) < I + \varepsilon/2$ and $L(f,Q) > I - \varepsilon/2$. Their common refinement R satisfies $U(f,R) \le U(f,P)$ and $L(f,R) \ge L(f,Q)$, giving the claim. Conversely, the displayed condition forces the gap between the lower supremum and upper infimum to be at most every positive ε, hence zero by Lemma 4.2. □

9.2 Continuous Functions Are Integrable

Continuity prevents large oscillation on sufficiently short intervals. Uniform continuity on a compact interval makes this local control uniform, which is exactly what the Darboux criterion requires.

Theorem 9.5 (Integrability of continuous functions)**.** Every continuous function $f : [a,b] \to \mathbb{R}$ is Riemann integrable.

Proof. By Theorem 5.27, f is uniformly continuous. Given $\varepsilon > 0$, choose $\delta > 0$ such that

$$|x - y| < \delta \Longrightarrow |f(x) - f(y)| < \frac{\varepsilon}{b-a}.$$

Choose a partition with norm less than δ. On every subinterval, $M_i - m_i \le \varepsilon/(b-a)$; hence

$$U(f,P) - L(f,P) = \sum_{i=1}^{n} (M_i - m_i)(x_i - x_{i-1}) \le \varepsilon.$$

The Darboux criterion proves integrability. □

Theorem 9.6 (Basic properties of the integral)**.** If $f, g : [a,b] \to \mathbb{R}$ are Riemann integrable and $c \in \mathbb{R}$, then $f + g$ and cf are Riemann integrable, and

$$\int_a^b (f+g)\ \mathrm{d}x = \int_a^b f\ \mathrm{d}x + \int_a^b g\ \mathrm{d}x, \qquad \int_a^b cf\ \mathrm{d}x = c\int_a^b f\ \mathrm{d}x.$$

Moreover, $f \le g$ implies that their integrals have the same order.

Proof. Fix $\varepsilon > 0$. By the Darboux criterion, choose partitions P_f and P_g such that

$$U(f,P_f) - L(f,P_f) < \frac{\varepsilon}{2}, \qquad U(g,P_g) - L(g,P_g) < \frac{\varepsilon}{2}.$$

Let P be their common refinement. Refinement can only decrease upper sums and increase lower sums. On each interval of P,

$$\sup(f+g) - \inf(f+g) \le \big(\sup f - \inf f\big) + \big(\sup g - \inf g\big).$$

Equivalently,

$$U(f+g,P) \le U(f,P) + U(g,P), \qquad L(f+g,P) \ge L(f,P) + L(g,P)$$

and therefore

$$U(f+g,P) - L(f+g,P) < \varepsilon.$$

Thus $f + g$ is integrable. Write

$$I_f = \int_a^b f\ \mathrm{d}x, \qquad I_g = \int_a^b g\ \mathrm{d}x.$$

The same inequalities, together with

$$L(f,P) > I_f - \frac{\varepsilon}{2}, \qquad L(g,P) > I_g - \frac{\varepsilon}{2},$$

give

$$\int_a^b (f+g)\ \mathrm{d}x > I_f + I_g - \varepsilon.$$

Similarly, using the upper sums gives

$$\int_a^b (f+g)\ \mathrm{d}x < I_f + I_g + \varepsilon.$$

Since this holds for every $\varepsilon > 0$, Lemma 4.2 gives the asserted equality.

For a scalar $c > 0$, the identities

$$U(cf,P) = cU(f,P), \qquad L(cf,P) = cL(f,P)$$

prove both integrability and the scalar formula by the Darboux criterion. If $c < 0$, then

$$U(cf,P) = cL(f,P), \qquad L(cf,P) = cU(f,P),$$

and the same conclusion follows; the case $c = 0$ is immediate. Finally, if $f \le g$, then

$$U(f,P) \le U(g,P)$$

for every partition P. Taking infima and using integrability gives

$$\int_a^b f\ \mathrm{d}x \le \int_a^b g\ \mathrm{d}x.$$

□

Theorem 9.7 (Additivity and absolute-value estimates)**.** Let $a < c < b$.

(a) If $f : [a,b] \to \mathbb{R}$ is Riemann integrable, then its restrictions to $[a,c]$ and $[c,b]$ are Riemann integrable and
$$\int_a^b f(x)\ \mathrm{d}x = \int_a^c f(x)\ \mathrm{d}x + \int_c^b f(x)\ \mathrm{d}x.$$

(b) If $f : [a,b] \to \mathbb{R}$ is Riemann integrable, then $|f|$ is Riemann integrable and
$$\left|\int_a^b f(x)\ \mathrm{d}x\right| \le \int_a^b |f(x)|\ \mathrm{d}x.$$

(c) If f is Riemann integrable and
$$|f(x)| \le M \qquad (x \in [a,b]),$$
then
$$\left|\int_a^b f(x)\ \mathrm{d}x\right| \le M(b-a).$$

Proof. For (1), insert c into a partition. Upper and lower sums then split exactly into the corresponding sums on the two subintervals. Refining partitions and applying Proposition 9.4 proves integrability of the restrictions and the displayed equality.

For (2), the inequality

$$\big||u| - |v|\big| \le |u - v|$$

shows on every partition interval that the oscillation of $|f|$ is at most the oscillation of f. The Darboux criterion proves integrability of $|f|$. Since

$$-|f| \le f \le |f|,$$

Theorem 9.6 gives the stated inequality. Part (3) follows from (2) and

$$0 \le |f| \le M.$$

□

Theorem 9.8 (Integrability of monotone functions)**.** Every monotone function $f : [a, b] \to \mathbb{R}$ is Riemann integrable.

Proof. It is enough to consider an increasing function; the decreasing case follows by applying the result to $-f$. For a partition P, monotonicity gives

$$M_i = f(x_i), \qquad m_i = f(x_{i-1}).$$

Thus

$$U(f, P) - L(f, P) = \sum_{i=1}^{n} (f(x_i) - f(x_{i-1}))(x_i - x_{i-1}) \le \|P\|(f(b) - f(a)).$$

Choose a partition with norm less than

$$\frac{\varepsilon}{f(b) - f(a)}$$

when $f(b) > f(a)$; if $f(b) = f(a)$, the function is constant and the upper and lower sums are equal. The Darboux criterion proves integrability. □

9.3 Uniform Limits and Integration

An integral is itself a limit, so a second limiting process must be handled with care. In the Riemann theory, uniform convergence is the clean condition that permits a limit to pass through an integral: one error bound works on the whole interval and hence controls the accumulated error.

Theorem 9.9 (Uniform interchange of limits and Riemann integrals)**.** Let

$$f_n : [a, b] \longrightarrow \mathbb{R}$$

be Riemann integrable for every n, and suppose that f_n converges uniformly on $[a, b]$ to a function f. Then f is Riemann integrable and

$$\lim_{n\to\infty} \int_a^b f_n(x)\ \mathrm{d}x = \int_a^b f(x)\ \mathrm{d}x.$$

In fact,

$$\lim_{n\to\infty} \int_a^b |f_n(x) - f(x)|\ \mathrm{d}x = 0.$$

Proof. Fix $\varepsilon > 0$. Choose n such that

$$\sup_{x\in[a,b]} |f_n(x) - f(x)| < \frac{\varepsilon}{4(b-a)}.$$

Choose a partition P for which the Darboux gap of f_n is less than $\varepsilon/2$. On each subinterval of P, the oscillation of f is at most the oscillation of f_n plus twice the displayed uniform error. Consequently,

$$U(f,P) - L(f,P) < \varepsilon,$$

and f is integrable by Proposition 9.4.

The basic integral estimates give

$$\left| \int_a^b f_n \, \mathrm{d}x - \int_a^b f \, \mathrm{d}x \right| \le \int_a^b |f_n - f| \, \mathrm{d}x \le (b-a) \sup_{x\in[a,b]} |f_n(x) - f(x)|.$$

The right-hand side tends to zero. The same final estimate proves the last assertion. □

Corollary 9.10 (Uniform convergence under an integrable domination)**.** Under the hypotheses of Theorem 9.9, suppose in addition that there is a nonnegative Riemann-integrable function

$$g : [a,b] \longrightarrow \mathbb{R}$$

such that

$$|f_n(x)| \le g(x) \qquad (x \in [a,b],\ n \in \mathbb{N}).$$

Then f is Riemann integrable,

$$|f(x)| \le g(x) \qquad (x \in [a,b]),$$

and the conclusions of Theorem 9.9 hold.

Proof. This is an immediate application of Theorem 9.9. The domination is often useful for obtaining bounds in applications, but uniform convergence, rather than domination, is what justifies the interchange in this Riemann statement. It is included here primarily to foreshadow the stronger Lebesgue Dominated Convergence Theorem. For each fixed x, taking limits in $|f_n(x)| \le g(x)$ gives the displayed bound for f. □

Remark 9.11 (The stronger Lebesgue theorem)**.** The Lebesgue dominated convergence theorem is substantially stronger. In its usual form, pointwise almost-everywhere convergence of measurable functions, together with domination by one integrable function, implies convergence of the integrals; uniform convergence is not required. This result depends on measure and Lebesgue integration and is therefore not proved or used in these notes.

Pointwise convergence is generally too weak to preserve continuity or to justify passing a limit through an integral. Dini's theorem is a striking special case in which compactness, continuity, and monotonicity cooperate to upgrade pointwise convergence to uniform convergence.

Theorem 9.12 (Dini's theorem)**.** Let

$$f_n : [a,b] \longrightarrow \mathbb{R}$$

be continuous functions such that

$$f_n(x) \le f_{n+1}(x) \qquad (x \in [a,b],\ n \in \mathbb{N}),$$

and suppose that $f_n(x)$ converges pointwise to a continuous function f on $[a, b]$. Then f_n converges uniformly to f, and hence

$$\lim_{n\to\infty} \int_a^b f_n(x)\ \mathrm{d}x = \int_a^b f(x)\ \mathrm{d}x.$$

The same conclusion holds for a pointwise decreasing sequence.

Proof. Consider the increasing case and put

$$h_n = f - f_n.$$

Each h_n is continuous, nonnegative, and the sequence (h_n) is pointwise decreasing to zero. If convergence were not uniform, some $\varepsilon > 0$ and points

$$x_n \in [a, b]$$

would satisfy

$$h_n(x_n) \geq \varepsilon \qquad (n \in \mathbb{N}).$$

By Theorem 5.25, a subsequence x_{n_j} converges to some $x \in [a, b]$. For every fixed m, eventually $n_j \geq m$, and monotonicity gives

$$h_m(x_{n_j}) \geq h_{n_j}(x_{n_j}) \geq \varepsilon.$$

Continuity of h_m implies $h_m(x) \geq \varepsilon$. This holds for every m, contradicting $h_m(x) \to 0$. Thus convergence is uniform, and Theorem 9.9 proves the integral formula. Apply the increasing case to $-f_n$ for a decreasing sequence. □

Remark 9.13 (Dini versus Lebesgue monotone convergence)**.** Dini's theorem uses a compact interval, continuity of every f_n and of the limit, and monotone pointwise convergence to produce **uniform** convergence. The preceding integral formula follows only after applying Theorem 9.9. By contrast, the Lebesgue Monotone Convergence Theorem concerns nonnegative measurable functions and gives convergence of integrals without continuity or uniform convergence. It is a measure-theoretic result and is not used in the Riemann development.

9.4 The Fundamental Theorem of Calculus

Differentiation measures local change, while integration accumulates local change. The Fundamental Theorem identifies these two operations under the continuity hypotheses that make both constructions stable.

Theorem 9.14 (Fundamental theorem of calculus, first part)**.** Let $f : [a, b] \to \mathbb{R}$ be continuous and define

$$F(x) = \int_a^x f(t)\ \mathrm{d}t.$$

Then F is differentiable on (a, b) and $F'(x) = f(x)$.

Proof. For $h > 0$ with $x + h \in [a, b]$, integral additivity gives

$$\frac{F(x+h) - F(x)}{h} - f(x) = \frac{1}{h} \int_x^{x+h} \big(f(t) - f(x)\big)\ \mathrm{d}t.$$

For $h < 0$, additivity instead gives

$$\frac{F(x+h) - F(x)}{h} - f(x) = \frac{1}{-h} \int_{x+h}^{x} \big(f(t) - f(x)\big)\ \mathrm{d}t.$$

In either case, the absolute value is bounded by the maximum of

$$|f(t) - f(x)|$$

over the interval with endpoints x and $x + h$, which tends to zero by continuity at x. Thus the difference quotient tends to $f(x)$. $\square$

Theorem 9.15 (Fundamental theorem of calculus, second part)**.** Let $F : [a, b] \to \mathbb{R}$ be differentiable on (a, b) and continuous on $[a, b]$. Suppose that its derivative extends to a continuous function $f : [a, b] \to \mathbb{R}$, so that

$$F'(x) = f(x) \qquad \text{for every } x \in (a, b).$$

Then

$$\int_a^b f(x)\ \mathrm{d}x = F(b) - F(a).$$

Proof. Define

$$G(x) = \int_a^x f(t)\ \mathrm{d}t.$$

By Theorem 9.14, $G'(x) = f(x)$ for $x \in (a, b)$. Therefore the function $H = F - G$ is continuous on $[a, b]$, differentiable on (a, b), and satisfies $H'(x) = 0$ throughout (a, b). For $x = a$, the identity

$$H(x) = H(a)$$

is immediate. If $x \in (a, b]$, the Mean Value Theorem applied on $[a, x]$ gives a point $c \in (a, x)$ for which

$$H(x) - H(a) = H'(c)(x - a) = 0.$$

Thus H is constant. Since $G(a) = 0$, evaluating at b gives

$$F(b) - F(a) = G(b) = \int_a^b f(x)\ \mathrm{d}x.$$

$\square$

Definition 9.16 (Oriented interval integrals)**.** If f is Riemann integrable on the closed interval with endpoints r and s, retain the preceding Riemann integral when

$$r \leq s,$$

and, when

$$s < r,$$

define

$$\int_r^s f(x)\ \mathrm{d}x = -\int_s^r f(x)\ \mathrm{d}x.$$

In particular,

$$\int_r^r f(x)\ \mathrm{d}x = 0.$$

Thus reversing endpoints reverses the orientation of an integral.

Theorem 9.17 (Substitution)**.** Let

$$\phi : [\alpha, \beta] \longrightarrow \mathbb{R}$$

be of class C^1, and let f be continuous on an open interval containing

$$\phi([\alpha, \beta]).$$

Then

$$\int_\alpha^\beta f(\phi(t))\phi'(t)\ \mathrm{d}t = \int_{\phi(\alpha)}^{\phi(\beta)} f(u)\ \mathrm{d}u.$$

Proof. Choose c in the open interval on which f is defined and set

$$F(x) = \int_c^x f(u)\ \mathrm{d}u,$$

using Definition 9.16. The first part of the Fundamental Theorem of Calculus, applied locally on closed subintervals, gives

$$F'(x) = f(x).$$

The chain rule therefore gives

$$(F \circ \phi)'(t) = f(\phi(t))\phi'(t).$$

Applying Theorem 9.15 to $F \circ \phi$ yields

$$\int_\alpha^\beta f(\phi(t))\phi'(t)\ \mathrm{d}t = F(\phi(\beta)) - F(\phi(\alpha)) = \int_{\phi(\alpha)}^{\phi(\beta)} f(u)\ \mathrm{d}u.$$

□

Remark 9.18 (Substitution, Jacobians, and pullbacks)**.** In one variable, substitution is a short consequence of the chain rule and the Fundamental Theorem of Calculus; no injectivity of ϕ is required. In several variables, the analogous transformation factor is

$$\phi'(t) \quad \longrightarrow \quad \det D\Phi(x),$$

and its rigorous Riemann/Jordan treatment is the genuinely nontrivial change-of-variables theorem of Chapter 12. Chapter 13 then interprets both formulas as pullback: one-variable substitution leads to multivariable change of variables and then to the pullback of differential forms.

Theorem 9.19 (Integration by parts)**.** Let

$$u, v : [a, b] \longrightarrow \mathbb{R}$$

be of class C^1. Then

$$\int_a^b u(x)v'(x)\ \mathrm{d}x = [u(x)v(x)]_{x=a}^{x=b} - \int_a^b u'(x)v(x)\ \mathrm{d}x.$$

Proof. The product rule gives

$$(uv)' = u'v + uv'.$$

Integrating and applying Theorem 9.15 to uv gives

$$u(b)v(b) - u(a)v(a) = \int_a^b u'(x)v(x)\ \mathrm{d}x + \int_a^b u(x)v'(x)\ \mathrm{d}x.$$

Rearrangement proves the formula. The mnemonic notation

$$\int u\ \mathrm{d}v = uv - \int v\ \mathrm{d}u$$

records this identity but is not its formal statement. □

Example 9.20 (A short integration-by-parts computation)**.** Taking

$$u(x) = x, \qquad v(x) = \exp x$$

on $[0, 1]$ gives

$$\int_0^1 x \exp x \, \mathrm{d}x = \left[x \exp x\right]_{x=0}^{x=1} - \int_0^1 \exp x \, \mathrm{d}x = 1.$$

Theorem 9.21 (Mean value theorem for integrals)**.** Let $f : [a, b] \to \mathbb{R}$ be continuous, where $a < b$. There is a point

$$c \in [a, b]$$

such that

$$\int_a^b f(x) \, \mathrm{d}x = f(c)(b - a).$$

Proof. By the Extreme Value Theorem, f has a minimum m and maximum M. Order preservation of the integral gives

$$m(b - a) \leq \int_a^b f(x) \, \mathrm{d}x \leq M(b - a).$$

Dividing by $b - a > 0$ gives a number between m and M. The Intermediate Value Theorem supplies $c \in [a, b]$ at which f takes that value. □

9.5 Improper Integrals

Finite-interval Riemann integration does not directly cover infinite domains or unbounded behavior. Improper integrals retain the finite theory and add an explicit limiting process whose convergence must be established.

Definition 9.22 (Improper Riemann integral)**.** Let $f : [a, \infty) \to \mathbb{R}$ be Riemann integrable on every finite interval $[a, b]$. The **improper integral** of f converges if

$$\int_a^\infty f(x) \, \mathrm{d}x = \lim_{b \to \infty} \int_a^b f(x) \, \mathrm{d}x$$

exists as a finite real number. Define an improper integral at a finite endpoint analogously by a one-sided limit.

Thus an improper integral is not a new kind of finite-interval integral: it is a limit of ordinary Riemann integrals whose existence must be proved.

At a singular left endpoint, the definition is

$$\int_a^b f(x) \, \mathrm{d}x = \lim_{c \to a^+} \int_c^b f(x) \, \mathrm{d}x,$$

when this limit exists finitely; a right-endpoint singularity is analogous. An interior singularity at c requires two separately convergent integrals,

$$\int_a^c f + \int_c^b f,$$

and cancellation between divergent sides is not allowed. Likewise,

$$\int_{-\infty}^\infty f = \int_{-\infty}^c f + \int_c^\infty f$$

means that both one-sided improper integrals converge, independently of the chosen finite splitting point c. Infinity is never an endpoint in $\mathbb{R}$.

Example 9.23 (p-integrals at infinity and at zero)**.** For real p, direct evaluation of the truncated proper integrals gives

$$\int_1^\infty x^{-p}\ \mathrm{d}x \text{ converges exactly when } p > 1,$$

whereas

$$\int_0^1 x^{-p}\ \mathrm{d}x \text{ converges exactly when } p < 1.$$

In particular,

$$\int_0^1 \frac{1}{\sqrt{x}}\ \mathrm{d}x = \lim_{c\to 0^+} 2(1-\sqrt{c}) = 2,$$

while $\int_0^1 x^{-1}\ \mathrm{d}x$ and $\int_1^\infty x^{-1}\ \mathrm{d}x$ diverge. The opposite thresholds are a useful reminder to identify which endpoint causes the improper behavior.

Definition 9.24 (Absolute convergence of an improper integral)**.** An improper integral of f is **absolutely convergent** if the corresponding improper integral of $|f|$ converges. Absolute convergence implies convergence, because the Cauchy estimate for any truncated tail is bounded by the corresponding tail integral of $|f|$.

Theorem 9.25 (Comparison for improper integrals)**.** Let $0 \le f \le g$ on $[a,\infty)$, with both functions Riemann integrable on finite intervals. If the improper integral of g converges, then that of f converges.

Proof. The truncated integrals of f are increasing and bounded above by those of g. Monotone convergence supplies their finite limit. □

Corollary 9.26 (Limit comparison for improper integrals)**.** Suppose $f, g > 0$ eventually on $[a,\infty)$ and

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = L, \qquad 0 < L < \infty.$$

Then their improper integrals have the same convergence behavior.

Proof. Eventually $(L/2)g \le f \le (3L/2)g$. Apply comparison in both directions; changing a finite initial interval does not affect convergence. □

Theorem 9.27 (Integral test)**.** Let $f : [1,\infty) \to \mathbb{R}$ be continuous, nonnegative, and decreasing, and put $a_n = f(n)$. Then

$$\sum_{n=1}^\infty a_n$$

converges if and only if the improper integral

$$\int_1^\infty f(x)\ \mathrm{d}x$$

converges. More precisely, for every $N \in \mathbb{N}$,

$$\int_1^{N+1} f(x)\ \mathrm{d}x \le \sum_{n=1}^N f(n) \le f(1) + \int_1^N f(x)\ \mathrm{d}x.$$

Proof. For $x \in [n, n+1]$, monotonicity gives

$$f(n+1) \le f(x) \le f(n).$$

Integrating these inequalities over the unit intervals and summing gives the displayed bounds. The partial sums and truncated integrals are increasing, so each is bounded above if and only if the other is bounded above. Monotone convergence then proves the equivalence. □

Theorem 9.28 (p-series test: analytic rederivation)**.** For $p \in \mathbb{R}$, the series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges if and only if $p > 1$.

Proof. For rational p, this conclusion was already proved without integration in Theorem 7.19. We now give an analytic proof that extends the criterion to every real exponent, using the real powers constructed in Chapter 8.

For $p > 0$, the function

$$f(x) = x^{-p}$$

is continuous, nonnegative, and decreasing on $[1, \infty)$ by Theorem 8.28. If $p \neq 1$, that theorem gives

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{x^{1-p}}{1-p}\right) = x^{-p};$$

if $p = 1$, Theorem 8.26 gives

$$\frac{\mathrm{d}}{\mathrm{d}x} \log x = \frac{1}{x}.$$

Thus Theorem 9.15 gives

$$\int_1^b x^{-p}\,\mathrm{d}x = \begin{cases} \dfrac{b^{1-p}-1}{1-p}, & p \neq 1, \\ \log b, & p = 1. \end{cases}$$

Thus the improper integral converges exactly when $p > 1$, and Theorem 9.27 gives the conclusion. If $p \le 0$, the terms do not tend to zero, so the series diverges by Proposition 7.6. □

9.6 The Lebesgue Criterion for Riemann Integrability

Continuity is sufficient for Riemann integrability, but it is not necessary. The natural question is how large a discontinuity set Riemann integration can tolerate. Darboux sums measure oscillation on intervals; the criterion below answers the question by asking whether discontinuities fit inside interval covers of arbitrarily small total length. This will make the Dirichlet–Thomae comparison decisive: density alone does not decide integrability. We use only countable interval covers, not a formal theory of Lebesgue measure.

Definition 9.29 (Oscillation and Riemann-negligible sets)**.** Let $f : [a, b] \to \mathbb{R}$ be bounded. For each $x \in [a, b]$, define

$$\omega_f(x) = \inf_{\delta>0} \sup\{|f(y) - f(z)| : y, z \in [a, b],\ |y - x| < \delta,\ |z - x| < \delta\}.$$

A set $E \subseteq [a, b]$ is **Riemann-negligible** if, for every

$$\eta > 0,$$

there are open intervals $I_1, I_2, \ldots$ such that

$$E \subseteq \bigcup_{j=1}^{\infty} I_j \qquad \text{and} \qquad \sum_{j=1}^{\infty} \text{length}(I_j) < \eta.$$

Finite covers are allowed as a special case.

Remark 9.30 (Terminology)**.** The temporary term **Riemann-negligible** refers only to the preceding covering property. In measure theory, these are precisely the sets of Lebesgue measure zero. No measure is being introduced or used here; the interval-covering property is all that is needed below.

Proposition 9.31 (Oscillation and continuity)**.** Let $f : [a, b] \to \mathbb{R}$ be bounded. Then

$$\omega_f(x) = 0$$

if and only if f is continuous at x.

Proof. If f is continuous at x, then, for a given $\varepsilon > 0$, all values in a sufficiently small relative neighborhood of x lie within $\varepsilon/2$ of $f(x)$. Their pairwise differences are therefore less than ε, so the defining infimum is zero. Conversely, if $\omega_f(x) = 0$, then the defining infimum supplies a $\delta > 0$ for which the pairwise oscillation in the relative δ-neighborhood of x is less than ε. Taking one of the two points to be x gives

$$|f(y) - f(x)| < \varepsilon$$

whenever $|y - x| < \delta$. □

Proposition 9.32 (Closed oscillation sets)**.** Let $f : [a, b] \to \mathbb{R}$ be bounded and let $\varepsilon > 0$. Then

$$D_\varepsilon = \{x \in [a, b] : \omega_f(x) \geq \varepsilon\}$$

is closed in $[a, b]$.

Proof. If $\omega_f(x) < \varepsilon$, then some neighborhood of x has pairwise oscillation less than ε. If this neighborhood has radius δ, then the relative $\delta/2$-neighborhood of every sufficiently nearby point is contained in it. Thus

$$\omega_f(y) < \varepsilon$$

for every y sufficiently near x. Hence the complement of D_ε is open. □

Proposition 9.33 (Countable sets are Riemann-negligible)**.** Every countable subset of

$$[a, b]$$

is Riemann-negligible.

Proof. Let

$$E \subseteq \{x_1, x_2, \ldots\},$$

where repetitions are allowed if E is finite. Given $\eta > 0$, cover x_n by an open interval of length

$$\frac{\eta}{2^{n+1}}.$$

These intervals cover E, and their total length is less than η. □

Lemma 9.34 (Countable unions of negligible sets)**.** If

$$E_1, E_2, \ldots \subseteq [a, b]$$

are Riemann-negligible, then

$$\bigcup_{m=1}^{\infty} E_m$$

is Riemann-negligible.

Proof. Given $\eta > 0$, cover E_m by open intervals having total length less than

$$\frac{\eta}{2^{m+1}}.$$

The combined countable family covers the displayed union and has total length less than η. □

Lemma 9.35 (Lebesgue number for a closed interval)**.** Let $\mathcal{U}$ be an open cover of $[a, b]$. There is a number $\lambda > 0$ such that every interval contained in $[a, b]$ and having length less than λ is contained in a member of $\mathcal{U}$.

Proof. Suppose no such λ exists. For every $n \in \mathbb{N}$, choose an interval

$$J_n \subseteq [a, b]$$

of length less than $1/n$ that is contained in no member of $\mathcal{U}$, and choose $x_n \in J_n$. By Theorem 4.20, some subsequence of (x_n) converges to a point $x \in [a, b]$. Choose $U \in \mathcal{U}$ containing x. Since U is open, there is $r_0 > 0$ such that

$$(x - r_0, x + r_0) \cap [a, b] \subseteq U.$$

For sufficiently large terms of the convergent subsequence, both the distance from x_n to x and the length of J_n are less than $r_0/2$. Then

$$J_n \subseteq U,$$

a contradiction. □

Corollary 9.36 (Finite subcovers of closed subsets)**.** Let $K \subseteq [a, b]$ be closed in $[a, b]$. Every cover of K by open intervals has a finite subcover.

Proof. Let $\mathcal{V}$ be an open cover of K. For every point of

$$[a, b] \setminus K,$$

there is an open interval in $\mathbb{R}$ whose intersection with K is empty and which contains that point. Let W be the union of all open intervals with this property. Then

$$\mathcal{V} \cup \{W\}$$

is an open cover of $[a, b]$. By Lemma 9.35, a finite partition of $[a, b]$ having sufficiently small mesh has each of its intervals contained in one member of this cover. For every partition interval meeting K, that member cannot be W, so it belongs to $\mathcal{V}$. The finitely many members thereby selected cover K. □

Theorem 9.37 (Lebesgue criterion for Riemann integrability)**.** Let $f : [a, b] \to \mathbb{R}$ be bounded. The following are equivalent.

(a) f is Riemann integrable.

(b) For every
$$\varepsilon > 0,$$
the set
$$D_\varepsilon = \{x \in [a,b] : \omega_f(x) \geq \varepsilon\}$$
is Riemann-negligible.

(c) The discontinuity set of f is Riemann-negligible.

Remark 9.38 (Terminology)**.** This result is commonly called **Lebesgue's criterion for Riemann integrability**. Some sources use a name such as the **Riemann–Lebesgue criterion**; it must not be confused with the Riemann–Lebesgue lemma from Fourier analysis.

Proof. Assume (1), and let $\varepsilon, \eta > 0$. Choose a partition P satisfying
$$U(f,P) - L(f,P) < \frac{\varepsilon\eta}{4}.$$
Except for the finitely many partition points, D_ε lies in partition intervals having oscillation at least ε: otherwise a small neighborhood of such a point contained in its partition interval would give oscillation less than ε. Since
$$U(f,P) - L(f,P)$$
is the sum of each partition interval's oscillation times its length, the total length of the indicated intervals is less than $\eta/4$. Enlarge the finitely many closed partition intervals in question to open intervals whose total length is less than $\eta/2$. Cover the finitely many partition points by open intervals having total length less than $\eta/2$. These two finite families form an open cover of D_ε of total length less than η. Thus (2) holds.

Conversely, let
$$|f(x)| \leq M \qquad (x \in [a,b]),$$
where $M \geq 1$, and fix $\varepsilon > 0$. Put
$$\delta = \frac{\varepsilon}{2(b-a+1)}.$$
By (2), cover D_δ by open intervals whose total length is less than
$$\frac{\varepsilon}{8M+4}.$$
By Proposition 9.32 and Corollary 9.36, finitely many of these intervals still cover D_δ. Let O be their union. For every
$$x \in [a,b] \setminus O,$$
one has $\omega_f(x) < \delta$, so there is an open interval V_x containing x on which the oscillation of f is less than δ. The selected intervals together with all the intervals V_x form an open cover of $[a,b]$. Apply Lemma 9.35, and choose a partition of mesh less than the resulting number λ.

Classify a partition interval as bad if it is contained in one of the selected intervals covering D_δ; classify all other partition intervals as good. A bad interval has oscillation at most $2M$, and the sum of the lengths of all bad intervals is at most the sum of the lengths of the selected intervals. Hence the bad intervals contribute less than
$$2M\frac{\varepsilon}{8M+4} < \frac{\varepsilon}{2}$$

to the Darboux gap. Every good interval lies in one of the intervals V_x and therefore has oscillation less than δ. Thus the good intervals contribute less than

$$\delta(b-a) < \frac{\varepsilon}{2}.$$

The Darboux criterion proves (1).

By Proposition 9.31, the discontinuity set is

$$\bigcup_{m=1}^{\infty} D_{1/m}.$$

By Lemma 9.34, (2) implies (3). Conversely,

$$D_\varepsilon$$

is contained in the discontinuity set, so (3) implies (2). □

Example 9.39 (The Dirichlet function)**.** For

$$\mathbf{1}_{\mathbb{Q}} : [0,1] \to \mathbb{R},$$

every neighborhood contains rational and irrational points. Hence

$$\omega_{\mathbf{1}_{\mathbb{Q}}}(x) = 1 \qquad (x \in [0,1]),$$

so

$$D_{1/2} = [0,1]$$

is not Riemann-negligible.

Example 9.40 (Thomae's function)**.** For $x \in [0,1]$, define

$$T(x) = \begin{cases} 1/q, & x = p/q \text{ is rational and written in lowest terms,} \\ 0, & x \text{ is irrational.} \end{cases}$$

This is commonly called **Thomae's function**, or the **popcorn function**; some texts call it Riemann's function, a name best avoided because it is ambiguous in analysis. The function T is continuous at every irrational point and discontinuous at every rational point. Consequently,

$$\mathrm{Disc}(T) = \mathbb{Q} \cap [0,1],$$

T is Riemann integrable, and

$$\int_0^1 T(x)\ \mathrm{d}x = 0.$$

Proof. Let x be irrational, and fix $\varepsilon > 0$. There are only finitely many rational numbers in $[0,1]$ that, in lowest terms, have denominator

$$q \le \frac{1}{\varepsilon};$$

indeed, there are only finitely many possible numerators for each such denominator. Let F_ε be this finite set. Since

$$x \notin F_\varepsilon,$$

there is a neighborhood of x avoiding F_ε. At every point y in this neighborhood, either y is irrational and $T(y) = 0$, or its reduced denominator satisfies

$$q > \frac{1}{\varepsilon},$$

so

$$T(y) = \frac{1}{q} < \varepsilon.$$

Since $T(x) = 0$, this proves that T is continuous at x.

Because $(0, 1)$ is uncountable by Theorem 2.26, whereas $\mathbb{Q}$ is countable, choose an irrational number

$$\alpha \in (0, 1).$$

If

$$x = p/q$$

is rational in lowest terms and $x < 1$, choose $N \in \mathbb{N}$ such that

$$\frac{\alpha}{N} < 1 - x,$$

and set

$$y_n = x + \frac{\alpha}{N + n}.$$

Then

$$y_n \in [0, 1] \setminus \mathbb{Q} \qquad \text{and} \qquad \lim_{n\to\infty} y_n = x.$$

If $x = 1$, the sequence

$$y_n = 1 - \frac{\alpha}{n + 1}$$

has the same two properties. In either formula, rationality of y_n would force α to be rational, so the asserted irrationality is justified. Thus, in every case, rational x is approached by irrational points y_n with

$$\lim_{n\to\infty} T(y_n) = 0 \neq \frac{1}{q} = T(x).$$

Thus T is discontinuous at every rational point, which proves the displayed description of its discontinuity set.

The set

$$\mathbb{Q} \cap [0, 1]$$

is countable and therefore Riemann-negligible by Proposition 9.33. The criterion proves that T is integrable. Finally, every nondegenerate subinterval of $[0, 1]$ contains an irrational point: it is uncountable by an affine rescaling of Theorem 2.26, while its rational points form a countable set. The infimum of T on it is therefore zero. Every lower Darboux sum is zero; since T is integrable, its integral is zero. □

Remark 9.41 (The Dirichlet–Thomae contrast)**.** Both functions are defined using the dense subset $\mathbb{Q}$ of $\mathbb{R}$, but density alone does not decide Riemann integrability:

$$\begin{aligned} &\text{Dirichlet:} \qquad \operatorname{Disc}(\mathbf{1}_{\mathbb{Q}}) = [0, 1] \quad \Longrightarrow \quad \text{not Riemann integrable,} \\ &\text{Thomae:} \qquad \operatorname{Disc}(T) = \mathbb{Q} \cap [0, 1] \quad \Longrightarrow \quad \text{Riemann integrable.} \end{aligned}$$

The criterion measures the size of the discontinuity set in the covering sense, not whether that set is dense.

9.7 Limits of the Riemann Theory

Riemann integration is deliberately tied to partitions of intervals and to the size of a function's oscillation on each subinterval. This makes it excellent for continuous functions and for much of elementary calculus, but it also has sharp limitations. The contrast between Example 9.39 and Example 9.40 identifies the next obstacle precisely: dense sets can still be negligible, so a more systematic notion of the size of a subset of the real line is beginning to be required.

The Dirichlet function is not pathological from the viewpoint of total amount: the rational numbers are sparse, yet the Riemann criterion cannot ignore them because they occur in every interval. More generally, Riemann integration is awkward for functions with discontinuities distributed throughout an interval, for limits of functions, and for integrals on complicated sets. Improper integrals also force one to treat unbounded behavior and infinite domains by separate limiting conventions.

Remark 9.42 (Riemann–Stieltjes integration)**.** Another classical extension replaces the ordinary Riemann sums

$$\sum_i f(t_i)(x_i - x_{i-1})$$

by sums of the form

$$\sum_i f(t_i)\big(\alpha(x_i) - \alpha(x_{i-1})\big),$$

whose limiting value, when it exists, is denoted by

$$\int_a^b f \, \mathrm{d}\alpha.$$

The ordinary Riemann integral is recovered when

$$\alpha(x) = x.$$

This Riemann–Stieltjes integral records accumulation relative to changes in another function: increasing or step-function integrators can encode weighted or discrete contributions, and the theory leads naturally to functions of bounded variation. Under suitable regularity hypotheses, it has the later integration-by-parts identity

$$\int_a^b f \, \mathrm{d}\alpha + \int_a^b \alpha \, \mathrm{d}f = f(b)\alpha(b) - f(a)\alpha(a),$$

which generalizes Theorem 9.19. No Riemann–Stieltjes theory is developed here. The measure-theoretic viewpoint later subsumes many such integrals by associating suitable measures to increasing functions; it gives the broader framework previewed again in Section 13.7.

Measure theory introduces a systematic notion of size for sets that is fine enough to distinguish a sparse set from an interval. The Lebesgue integral then organizes summation by the values of a function and by the sizes of their level sets, rather than solely by partitions of the domain. This gives stronger convergence theorems and integrates many functions beyond the Riemann class. We will not define measures or the Lebesgue integral in these notes; the point here is motivational: the Lebesgue criterion already shows that Riemann integrability is controlled by the size of the discontinuity set, while the limitations above indicate why a more flexible theory is valuable.

Exercises

Exercise 9.1. Use Theorem 9.37 to prove that a bounded function with finitely many discontinuities is Riemann integrable.

Exercise 9.2. Let $f : [a, b] \to \mathbb{R}$ be monotone. Show that, for each

$$\varepsilon > 0,$$

the set

$$D_\varepsilon$$

is finite.

Exercise 9.3. Prove directly from the Darboux criterion that a function having only finitely many discontinuities on

$$[a, b]$$

and otherwise constant on finitely many intervals is Riemann integrable.

Exercise 9.4. Determine whether

$$\int_1^\infty \frac{1}{x^2}\ \mathrm{d}x$$

and

$$\int_1^\infty \frac{1}{x}\ \mathrm{d}x$$

converge. Use Theorem 9.15, then compare the conclusion with the two corresponding series.

Exercise 9.5. Use Theorem 9.17 to evaluate

$$\int_0^1 2t \exp(t^2)\ \mathrm{d}t.$$

State the substitution and identify the endpoint values.

Exercise 9.6. Use Theorem 9.19 to evaluate

$$\int_0^1 x \sin x\ \mathrm{d}x.$$

Exercise 9.7. Explain why the function

$$\mathbf{1}_{\mathbb{Q}}$$

on

$$[0, 1]$$

has the same upper and lower sums for no partition. Which density facts from earlier chapters are being used?

Part II

Multivariable Calculus and Analysis

Chapter 10

Necessary Topology and Linear Algebra

10.1 Metric Spaces and Topological Spaces

Part I lived mainly on the real line. Moving from $\mathbb{R}$ to $\mathbb{R}^n$ requires three structural languages that will work together rather than sit as an unrelated prerequisite appendix:

$$\begin{gathered}\text{topology: nearness, continuity, compactness, and local structure,}\\ \text{linear algebra: derivatives as linear maps,}\\ \text{determinants and orientation: volume distortion and sign.}\end{gathered}$$

The route through Part II is

$$\begin{gathered}\mathbb{R} \longrightarrow \mathbb{R}^n \longrightarrow \text{metric/topological structure}\\ \longrightarrow \text{linear approximation} \longrightarrow \text{differentiation, integration, forms, and Stokes.}\end{gathered}$$

Nearness on $\mathbb{R}$ is measured by $|x-y|$; norms and metrics provide its analogue on $\mathbb{R}^n$ and beyond. Topology retains the open-set information needed for continuity when a particular numerical distance is not central. This matters later because derivatives are defined on open sets, and the manifolds of Chapter 13 are locally modeled on Euclidean open sets.

Definition 10.1 (Metric space)**.** A **metric** on a set X is a function

$$d : X \times X \longrightarrow [0,\infty)$$

such that, for all $x, y, z \in X$,

$$d(x,y)=0 \Longleftrightarrow x=y, \qquad d(x,y)=d(y,x), \qquad d(x,z)\le d(x,y)+d(y,z).$$

The pair (X,d) is a **metric space**. For $r>0$, write

$$B_d(x,r)=\{y\in X : d(x,y)<r\}.$$

Theorem 10.2 (Cauchy–Schwarz inequality)**.** For $x, y \in \mathbb{R}^n$,

$$\left|\sum_{j=1}^n x_j y_j\right| \le \left(\sum_{j=1}^n x_j^2\right)^{1/2} \left(\sum_{j=1}^n y_j^2\right)^{1/2}.$$

Consequently,

$$d_2(x,y) = \left(\sum_{j=1}^{n}(x_j - y_j)^2\right)^{1/2}$$

is a metric on $\mathbb{R}^n$.

Proof. For $y \neq 0$, the nonnegativity, for every $t \in \mathbb{R}$, of

$$\sum_{j=1}^{n}(x_j - ty_j)^2$$

says that the discriminant of this quadratic in t is nonpositive. This gives the displayed inequality after taking square roots. Expanding

$$d_2(x,z)^2 = \sum_{j=1}^{n}\big((x_j - y_j) + (y_j - z_j)\big)^2$$

and applying Cauchy–Schwarz proves the triangle inequality. The other metric axioms are immediate. $\square$

For later estimates, if

$$\|x\|_\infty = \max_{1 \le j \le n} |x_j|,$$

then

$$|x_i| \le \|x\|_2 \le \sqrt{n}\,\|x\|_\infty.$$

The first inequality drops nonnegative terms from the sum of squares, and the second bounds each of its n terms by $\|x\|_\infty^2$.

Definition 10.3 (Open and closed sets)**.** Let (X,d) be a metric space. A set $U \subseteq X$ is **open** if for every $x \in U$ there is $r > 0$ such that $B_d(x,r) \subseteq U$. A set $F \subseteq X$ is **closed** if its complement

$$X \setminus F$$

is open.

In $\mathbb{R}$, an open interval has no endpoint while $[a,b]$ is closed. In $\mathbb{R}^2$, an open disk consists of points that have a little disk around them still inside it; the closed disk includes its boundary circle. These are the same familiar inside-versus-edge ideas, now phrased so that they work in any metric space.

Proposition 10.4 (Metric open sets)**.** The empty set and the whole metric space are open. Arbitrary unions of open sets are open, and finite intersections of open sets are open.

Proof. The first statement is immediate. If x belongs to a union of open sets, it belongs to one member of the union and hence has a ball contained in that member. If x belongs to each of

$$U_1, \ldots, U_m,$$

choose balls of radii $r_1, \ldots, r_m$ about x contained in the respective sets. The ball of radius

$$\min\{r_1, \ldots, r_m\}$$

lies in their intersection. $\square$

Definition 10.5 (Topological space)**.** A **topology** on a set X is a family of subsets containing $\varnothing$ and X, closed under arbitrary unions and finite intersections. Its members are **open**, and $(X, \mathcal{T})$ is a **topological space**.

Remark 10.6. Proposition 10.4 says that every metric determines a topology. The definitions below make sense in every topological space; a metric is needed only for statements explicitly involving distance, sequences, or epsilon–delta estimates.

Definition 10.7 (Subspaces, interior, closure, and boundary)**.** Let $Y \subseteq X$, where X is a topological space. The **subspace topology** on Y has as its open sets all sets

$$Y \cap U$$

with U open in X. For $E \subseteq X$, define

$$E^\circ = \bigcup \{U : U \text{ is open and } U \subseteq E\},$$

$$\overline{E} = \bigcap \{F : F \text{ is closed and } E \subseteq F\}, \qquad \partial E = \overline{E} \setminus E^\circ.$$

These are respectively the **interior**, **closure**, and **boundary** of E.

The interior E° consists of points definitely inside E; the closure $\overline{E}$ adds every point that can be approached from E; and the boundary consists of points approachable both from E and from its complement. For example, the interior, closure, and boundary of $(0, 1)$ in $\mathbb{R}$ are $(0, 1)$, $[0, 1]$, and $\{0, 1\}$; for an open disk in $\mathbb{R}^2$, the boundary is its surrounding circle.

Proposition 10.8 (Closure characterizations)**.** Let $E \subseteq X$, where X is a topological space. Then E° is the largest open subset of E, and $\overline{E}$ is the smallest closed superset of E. Moreover,

$$x \in \overline{E}$$

if and only if every open neighborhood of x meets E. In particular,

$$E \text{ is closed} \iff \overline{E} = E.$$

Proof. The axioms show that the defining union is open and the defining intersection is closed; their extremal properties are immediate. If

$$x \notin \overline{E},$$

then the open set $X \setminus \overline{E}$ contains x and misses E. Conversely, if an open neighborhood U of x misses E, then $X \setminus U$ is a closed superset of E, so $x \notin \overline{E}$. The last assertion follows from the extremal property. □

Theorem 10.9 (Sequential closure criterion)**.** Let (X, d) be a metric space and $E \subseteq X$. Then

$$x \in \overline{E}$$

if and only if there is a sequence (x_n) in E satisfying

$$\lim_{n \to \infty} x_n = x.$$

Thus E is closed if and only if it contains every limit of a convergent sequence in E.

Proof. If $x_n \to x$, every ball about x contains some x_n, so the neighborhood characterization applies. Conversely, choose

$$x_n \in E \cap B_d\left(x, \frac{1}{n}\right),$$

which is possible when $x \in \overline{E}$. Then $d(x_n, x) < 1/n$. The final claim follows from Proposition 10.8. $\square$

Definition 10.10 (Convergence and continuity)**.** For metric spaces (X, d_X) and (Y, d_Y), a sequence (x_n) converges to x if, for every $\varepsilon > 0$,

$$n \geq N \Longrightarrow d_X(x_n, x) < \varepsilon$$

for some N. A map $f : X \to Y$ is **continuous at** x if for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$d_X(x, y) < \delta \Longrightarrow d_Y(f(x), f(y)) < \varepsilon.$$

Definition 10.11 (Limit of a map)**.** Let $D \subseteq X$, let a be a limit point of D, and let $f : D \to Y$, where X, Y are metric spaces. We write

$$\lim_{x \to a} f(x) = L$$

if for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$x \in D, \quad 0 < d_X(x, a) < \delta \quad \Longrightarrow \quad d_Y(f(x), L) < \varepsilon.$$

For Euclidean spaces this single condition controls approach from every direction and along every path.

Example 10.12 (Paths disprove, estimates prove)**.** For

$$g(x, y) = \frac{xy}{x^2 + y^2},$$

approach $(0, 0)$ along $y = x$ and $y = -x$. The resulting values are $1/2$ and $-1/2$, so the limit does not exist. By contrast, if

$$h(x, y) = \frac{x^2 y}{x^2 + y^2} \qquad ((x, y) \neq (0, 0)),$$

then

$$|h(x, y)| \leq |y| \leq \|(x, y)\|_2,$$

which proves $h(x, y) \to 0$ as $(x, y) \to (0, 0)$. This estimate is genuinely multivariable: it controls every approach at once.

Remark 10.13 (The path test is one-sided)**.** Finding two paths with different limiting values proves nonexistence. Finding several paths with the same limiting value does not prove existence, because untested paths remain. To prove existence, seek a norm estimate that is independent of direction, or use a theorem whose hypotheses provide such uniform control.

Theorem 10.14 (Topological characterization of continuity)**.** A map $f : X \to Y$ of topological spaces is continuous if and only if

$$f^{-1}(V)$$

is open in X for every open $V \subseteq Y$. For metric spaces, this is equivalent to epsilon–delta continuity at every point.

Proof. If all inverse images of open sets are open and $f(x) \in V$, take

$$U = f^{-1}(V).$$

Conversely, if each point of $f^{-1}(V)$ has an open neighborhood contained in $f^{-1}(V)$, that inverse image is a union of open sets. In metric spaces, an epsilon-ball about $f(x)$ is contained in V precisely when the usual epsilon–delta argument supplies the needed ball about x. □

Definition 10.15 (Homeomorphisms)**.** Let X and Y be topological spaces. A map

$$h : X \longrightarrow Y$$

is a **homeomorphism** if it is bijective, continuous, and its inverse

$$h^{-1} : Y \longrightarrow X$$

is continuous. In that case, X and Y are said to be **homeomorphic**. A homeomorphism identifies spaces that have the same open-set structure, even when their elements or their ambient descriptions differ.

A homeomorphism preserves the topological shape of a space while ignoring metric quantities such as length, angle, area, and curvature. For example,

$$(0, 1) \longrightarrow \mathbb{R}, \qquad x \longmapsto \tan\bigl(\pi(x - 1/2)\bigr),$$

is a homeomorphism. Thus an open interval and the whole line are topologically equivalent despite their very different Euclidean sizes. Connectedness and compactness are preserved by homeomorphisms; Chapter 13 will use the same idea in coordinate charts on manifolds.

Proposition 10.16 (Sequential continuity)**.** A map $f : X \to Y$ between metric spaces is continuous at x if and only if

$$x_n \to x \Longrightarrow f(x_n) \to f(x).$$

Proof. The forward implication is the epsilon–delta definition. If continuity fails, some $\varepsilon_0 > 0$ permits, for every n, a choice

$$d_X(x_n, x) < \frac{1}{n}, \qquad d_Y(f(x_n), f(x)) \geq \varepsilon_0.$$

This contradicts the stated sequential property. □

10.2 Products, Compactness, and Connectedness

Because $\mathbb{R}^n = \mathbb{R} \times \cdots \times \mathbb{R}$, multivariable analysis must first understand products. The product topology is designed so that a map

$$f = (f_1, \ldots, f_n)$$

is continuous exactly when its coordinate functions are continuous, just as one checks a familiar vector-valued formula coordinate by coordinate.

Open intervals and compact intervals in Part I are the first examples of the next two notions. Compactness is the principle that turns local information into finite, hence global, control. An open cover records an arbitrary collection of local neighborhoods, and compactness says finitely many already suffice. In metric spaces, the Bolzano–Weierstrass viewpoint of Part I reappears as sequential compactness.

Definition 10.17 (Product topology)**.** For topological spaces X and Y, the **product topology** on

$$X \times Y$$

is the topology whose basic open sets are

$$U \times V,$$

where U is open in X and V is open in Y. The coordinate projections are

$$\pi_X(x, y) = x, \qquad \pi_Y(x, y) = y.$$

Proposition 10.18 (Maps into products)**.** A map

$$f : Z \longrightarrow X \times Y$$

is continuous if and only if both

$$\pi_X \circ f \qquad \text{and} \qquad \pi_Y \circ f$$

are continuous.

Proof. The projections are continuous since

$$\pi_X^{-1}(U) = U \times Y, \qquad \pi_Y^{-1}(V) = X \times V.$$

Conversely,

$$f^{-1}(U \times V) = (\pi_X \circ f)^{-1}(U) \cap (\pi_Y \circ f)^{-1}(V),$$

which is open. Basic open sets generate the product topology. □

Definition 10.19 (Compactness)**.** Let X be a topological space. An **open cover** of X is a family of open subsets

$$\{U_\alpha : \alpha \in I\}$$

such that

$$X \subseteq \bigcup_{\alpha \in I} U_\alpha.$$

A **subcover** is a subfamily with the same covering property; it is a **finite subcover** if it has the form

$$\{U_{\alpha_1}, \ldots, U_{\alpha_m}\}$$

for finitely many indices. The space X is **compact** if every open cover of X has a finite subcover. A subset is compact when it is compact in its subspace topology.

Example 10.20 (A finite subcover in action)**.** View $[0, 1]$ as a subspace of $\mathbb{R}$ and, for each $t \in [0, 1]$, let

$$U_t = [0, 1] \cap (t - 1/4, t + 1/4).$$

These relatively open intervals form an open cover: each point is surrounded by one of the displayed local neighborhoods. The five members

$$U_0, \quad U_{1/4}, \quad U_{1/2}, \quad U_{3/4}, \quad U_1$$

already cover $[0, 1]$. Thus this finite subcover keeps the same coverage while replacing a continuum of local intervals by finitely many of them. For an arbitrary open cover, compactness guarantees that some finite choice is possible even when no such choice is apparent.

The set $[0,1]$ is compact, while $(0,1)$ and $\mathbb{R}$ are not; in $\mathbb{R}^n$, the closed balls are compact by Heine–Borel, whereas open balls are not. The later payoff is substantial: compactness permits convergent subsequences, forces continuous functions to attain extrema, and upgrades local continuity estimates to uniform ones on a compact region.

Hausdorff spaces. A topological space X is **Hausdorff** if, for every two distinct points $x, y \in X$, there are disjoint open sets U, V such that

$$x \in U, \qquad y \in V.$$

Thus a Hausdorff space has enough open sets to separate distinct points. In particular, every metric space is Hausdorff: if $x \neq y$, balls about x and y of radius less than $d(x,y)/2$ are disjoint.

Theorem 10.21 (Basic compactness theorems)**.** Let X and Y be topological spaces.

(a) A closed subset of a compact space is compact.

(b) The continuous image of a compact space is compact.

(c) A compact subset of a Hausdorff space is closed. Hence compact subsets of metric spaces are closed.

Proof. For (a), adjoin $X \setminus F$ to an open cover of the closed subspace F and use compactness of X. For (b), pull an open cover of $f(X)$ back along f and choose a finite subcover of X. For (c), if $y \notin K$, separate y from each $x \in K$ by disjoint neighborhoods. Finitely many neighborhoods of the x cover K; the intersection of the corresponding neighborhoods of y misses K. Thus $X \setminus K$ is open. Distinct points in a metric space have disjoint small balls. □

Theorem 10.22 (Metric compactness)**.** For a metric space X, the following are equivalent:

(a) X is compact;

(b) every sequence in X has a convergent subsequence with limit in X.

Proof. Assume (a) and let (x_n) be a sequence. Put

$$F_N = \overline{\{x_n : n \geq N\}}.$$

Every F_N is nonempty, since the corresponding tail contains x_N, and is closed by Proposition 10.8. Moreover,

$$F_{N+1} \subseteq F_N,$$

because the $(N+1)$st tail is contained in the Nth tail. Thus the family (F_N) has the finite-intersection property: a finite intersection is its member with the largest index. Compactness implies

$$\bigcap_{N=1}^{\infty} F_N \neq \varnothing.$$

Indeed, if this intersection were empty, the open complements of the F_N would cover X; a finite subcover would force the corresponding finite intersection, hence one F_N, to be empty. Choose

$$x \in \bigcap_{N=1}^{\infty} F_N,$$

so the closure characterization in Proposition 10.8 shows that every ball about x meets every tail. Choose n_1 with

$$d(x_{n_1}, x) < 1.$$

Having chosen n_k, apply this fact to the tail beginning at $n_k + 1$ and the ball of radius $1/(k+1)$ to choose

$$n_{k+1} > n_k, \qquad d(x_{n_{k+1}}, x) < \frac{1}{k+1}.$$

Recursively, this gives increasing indices n_k satisfying

$$d(x_{n_k}, x) < \frac{1}{k}.$$

This proves (b).

Conversely, first we show directly that, for every $\varepsilon > 0$, finitely many balls of radius ε cover X. If this failed for some $\varepsilon > 0$, choose $x_1 \in X$, and, having chosen $x_1, \ldots, x_{n-1}$, choose x_n outside their balls of radius ε. Then

$$d(x_j, x_k) \geq \varepsilon \qquad (j \neq k),$$

so no subsequence converges. We next show directly that every open cover has a number $\lambda > 0$ such that each ball of radius λ is contained in a member of the cover. If no such λ existed, choose x_n such that

$$B_d\left(x_n, \frac{1}{n}\right)$$

lies in no member of the cover. Let (x_{n_j}) be a convergent subsequence with limit x. A member U of the cover containing x contains a ball $B_d(x, r)$. For all sufficiently large j,

$$B_d\left(x_{n_j}, \frac{1}{n_j}\right) \subseteq B_d(x, r) \subseteq U,$$

a contradiction. The preceding finite-ball property now yields the desired finite subcover: choose finitely many balls of radius $\lambda/2$ covering X. Each is contained in the ball of radius λ with the same center, and that larger ball is contained in a member of the original cover. □

Lemma 10.23 (Lebesgue number for a compact metric space)**.** Let K be a compact metric space and let

$$\{U_1, \ldots, U_N\}$$

be a finite open cover of K. There is $\lambda > 0$ such that every subset $E \subseteq K$ satisfying

$$\operatorname{diam}(E) < \lambda$$

is contained in one of the sets U_i.

Proof. Suppose that no such number exists. For every $k \in \mathbb{N}$, choose a nonempty set $E_k \subseteq K$ with

$$\operatorname{diam}(E_k) < \frac{1}{k}$$

which is contained in no U_i, and choose $x_k \in E_k$. By Theorem 10.22, a subsequence x_{k_j} converges to some $x \in K$. Choose U_i containing x. Since U_i is open in K, there is $r > 0$ such that

$$B(x, r) \cap K \subseteq U_i.$$

For all sufficiently large j,

$$d(x_{k_j}, x) < \frac{r}{2}, \qquad \operatorname{diam}(E_{k_j}) < \frac{r}{2}.$$

If $z \in E_{k_j}$, then

$$d(z, x) \leq d(z, x_{k_j}) + d(x_{k_j}, x) < r,$$

so $E_{k_j} \subseteq U_i$, a contradiction. □

Theorem 10.24 (Uniform continuity on compact metric spaces)**.** Let X be a compact metric space, let Y be a metric space, and let

$$f : X \longrightarrow Y$$

be continuous. Then f is uniformly continuous.

Proof. Suppose that f is not uniformly continuous. Then there are

$$\varepsilon_0 > 0$$

and points $x_n, y_n \in X$ satisfying

$$d_X(x_n, y_n) < \frac{1}{n}, \qquad d_Y(f(x_n), f(y_n)) \geq \varepsilon_0.$$

By Theorem 10.22, a subsequence of (x_n) converges to some $x \in X$. Along the corresponding indices, the triangle inequality shows that $y_n \to x$ as well. Continuity of f at x gives

$$f(x_n) \longrightarrow f(x), \qquad f(y_n) \longrightarrow f(x),$$

contradicting the displayed lower bound. □

Theorem 10.25 (Heine–Borel theorem)**.** A subset of $\mathbb{R}^n$ with its Euclidean metric is compact if and only if it is closed and bounded.

Proof. A compact set is closed by Theorem 10.21, and the balls

$$B_{d_2}(0, m) \qquad (m \in \mathbb{N})$$

give an open cover that proves boundedness. Conversely, let K be closed and bounded. Every coordinate sequence of a sequence in K is bounded. Repeated use of Theorem 4.20 produces a subsequence for which all coordinates converge. Its coordinatewise limit is its Euclidean limit, since

$$d_2(x^{(r)}, x)^2 = \sum_{j=1}^{n} |x_j^{(r)} - x_j|^2.$$

Closedness puts that limit in K, and Theorem 10.22 completes the proof. □

Definition 10.26 (Connectedness and paths)**.** A topological space is **connected** if it cannot be expressed as the union of two disjoint nonempty open sets. A **path** from x to y is a continuous map

$$\gamma : [0, 1] \longrightarrow X$$

with $\gamma(0) = x$ and $\gamma(1) = y$. The space is **path connected** if every two points are joined by a path.

Example 10.27 (Paths and separation)**.** If x, y lie in an interval $I \subseteq \mathbb{R}$, then

$$\gamma(t) = (1-t)x + ty$$

stays in I, so I is path connected (and hence connected). In contrast, $[0,1] \cup [2,3]$ is disconnected: its two displayed intervals are disjoint nonempty relatively open subsets whose union is the whole set. More generally, the same straight-line formula shows that every convex subset of $\mathbb{R}^n$ is path connected.

Later, a continuous nonzero determinant on a connected domain cannot change sign without vanishing somewhere, which is why connectedness enters orientation and change of variables.

Proposition 10.28 (Connectedness)**.** Continuous images of connected spaces are connected, every path-connected space is connected, and a subset of $\mathbb{R}$ is connected if and only if it is an interval.

Proof. The inverse images of a separation of $f(X)$ would separate X. A separation of a path-connected space pulls back along a path joining points in its two pieces, contradicting connectedness of $[0,1]$, which follows from Theorem 5.21. Finally, a noninterval subset of $\mathbb{R}$ is separated by a missing point between two of its points. Conversely, if an interval were separated, the supremum of one piece after restricting to two points in different pieces would contradict relative openness, exactly as in the usual proof of the Intermediate Value Theorem. □

10.3 Finite-Dimensional Vector Spaces and Linear Maps

The geometric objects of multivariable calculus have both an intrinsic and a coordinate description. Vectors describe directions and linear maps describe the first-order approximations that will become derivatives in Chapter 11. Choosing a basis turns those maps into arrays of real numbers—matrices—so that composition can be computed explicitly. The point of this section is to move fluently between these two descriptions.

Definition 10.29 (Real vector space)**.** A **real vector space** is a set V equipped with an addition

$$V \times V \longrightarrow V, \qquad (v, w) \longmapsto v + w,$$

and scalar multiplication

$$\mathbb{R} \times V \longrightarrow V, \qquad (c, v) \longmapsto cv,$$

such that, for all $u, v, w \in V$ and $a, b \in \mathbb{R}$,

$$(u+v)+w = u+(v+w), \qquad u+v = v+u,$$

there is a vector $0 \in V$ with $v + 0 = v$, and every $v \in V$ has an additive inverse $-v$. In addition,

$$a(u+v) = au + av, \qquad (a+b)v = av + bv,$$

$$(ab)v = a(bv), \qquad 1v = v.$$

Definition 10.30 (Bases and dimension)**.** A finite set

$$\{v_1, \dots, v_n\}$$

in a real vector space V **spans** V if every $v \in V$ is a linear combination of its members. It is **linearly independent** if

$$a_1 v_1 + \cdots + a_n v_n = 0$$

implies that all a_j vanish. A **basis** is an independent spanning set. A space having a finite spanning set is **finite dimensional**.

Lemma 10.31 (Basis reduction)**.** Every finite spanning set contains a basis.

Proof. Remove a vector whenever it is a linear combination of the others. Its removal does not alter the span. The finite process ends with an independent spanning set. □

Theorem 10.32 (Exchange theorem)**.** If an independent list of m vectors lies in a space spanned by n vectors, then

$$m \le n.$$

Consequently, all bases of a finite-dimensional vector space have the same number of elements.

Proof. Inductively replace m members of the given spanning list by the independent vectors while preserving span. At the kth step, the kth independent vector has an expression involving an unreplaced member of the old list; otherwise it would lie in the span of the preceding independent vectors. Solve for that old member and replace it. Thus $m \le n$. Applying the inequality in both directions to two bases proves the consequence. □

Definition 10.33 (Finite direct sums)**.** Let

$$V_1, \dots, V_r$$

be real vector spaces. Their **(external) direct sum** is the Cartesian product

$$V_1 \oplus \cdots \oplus V_r = V_1 \times \cdots \times V_r$$

with componentwise operations:

$$(v_1, \dots, v_r) + (w_1, \dots, w_r) = (v_1 + w_1, \dots, v_r + w_r),$$

$$c(v_1, \dots, v_r) = (cv_1, \dots, cv_r).$$

For each j, the **canonical injection** and **canonical projection** are the maps

$$\iota_j : V_j \longrightarrow V_1 \oplus \cdots \oplus V_r, \qquad \pi_j : V_1 \oplus \cdots \oplus V_r \longrightarrow V_j,$$

defined by inserting a vector in the jth component and zeros elsewhere, and by selecting the jth component, respectively.

Proposition 10.34 (Basic properties of finite direct sums)**.** The direct sum in Definition 10.33 is a real vector space, and every element has the unique decomposition

$$(v_1, \dots, v_r) = \iota_1(v_1) + \cdots + \iota_r(v_r).$$

Each ι_j and π_j is linear, and

$$\pi_i \circ \iota_j = \begin{cases} \mathrm{id}_{V_j}, & i = j, \\ 0, & i \ne j. \end{cases}$$

If every V_j is finite dimensional, then

$$\dim(V_1 \oplus \cdots \oplus V_r) = \sum_{j=1}^{r} \dim V_j.$$

Proof. The vector-space axioms hold componentwise. The displayed decomposition is obtained by adding the vectors with one nonzero component, and applying π_j to any such decomposition proves uniqueness. The formulas for the injections and projections show directly that they are linear and give the stated composite.

For each j, choose a basis

$$(v_{j,1}, \dots, v_{j,n_j})$$

of V_j. The vectors

$$\iota_j(v_{j,\ell}) \qquad (1 \le j \le r,\ 1 \le \ell \le n_j)$$

span the direct sum by the unique decomposition. A linear relation among them becomes, after applying each π_j, a linear relation in the basis of V_j; all its coefficients therefore vanish. Thus these vectors form a basis, and counting them gives the dimension formula. □

Definition 10.35 (Coordinates, linear maps, and matrices)**.** The common size of the bases of a finite-dimensional space is its **dimension**. Relative to an ordered basis

$$\mathcal{B} = (v_1, \dots, v_n),$$

the unique coefficients in

$$v = \sum_{j=1}^{n} a_j v_j$$

form the coordinate column $[v]_{\mathcal{B}}$. A map $T : V \to W$ is **linear** if

$$T(av + bw) = aT(v) + bT(w).$$

For positive integers m, n, an m-by-n **matrix** is a rectangular array

$$A = (a_{ij}) \in M_{m,n}(\mathbb{R}), \qquad 1 \le i \le m, \quad 1 \le j \le n.$$

It acts on a column $x = (x_1, \dots, x_n)^{\mathsf{T}}$ by

$$(Ax)_i = \sum_{j=1}^{n} a_{ij} x_j \qquad (1 \le i \le m).$$

If $T : V \to W$, where V has dimension n, W has dimension m, and

$$\mathcal{B} = (v_1, \dots, v_n), \qquad \mathcal{C} = (w_1, \dots, w_m)$$

are ordered bases, its **matrix** relative to these bases is the m-by-n matrix whose jth column is

$$[T(v_j)]_{\mathcal{C}}.$$

A linear map $T : V \to V$ is an **endomorphism**, also called a **linear operator** on V.

Proposition 10.36 (Coordinate rules and matrix multiplication)**.** If A is the matrix of a linear map T, then

$$[T(v)]_{\mathcal{C}} = A[v]_{\mathcal{B}}.$$

Thus T is determined by its values on a basis. If

$$A \in M_{m,n}(\mathbb{R}), \qquad B \in M_{\ell,m}(\mathbb{R}),$$

their product is the ℓ-by-n matrix

$$(BA)_{ij} = \sum_{k=1}^{m} b_{ik} a_{kj}.$$

When A and B represent composable linear maps, BA represents the composite in the corresponding order.

Proof. Apply linearity to the coordinate expansion of v. Applying the resulting formula successively to two maps gives

$$[S(T(v))]_{\mathcal{D}} = B(A[v]_{\mathcal{B}}).$$

Expanding the ith coordinate gives the displayed multiplication rule. $\square$

Example 10.37 (Composition explains matrix multiplication)**.** Let $T(x, y) = (x + y, y)$ and $S(u, v) = (2u - v, u)$. Their standard matrices are

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}.$$

Applying T and then S gives

$$S(T(x, y)) = (x + 2y, x),$$

whose matrix is

$$BA = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

Thus the matrix-product formula is not arbitrary bookkeeping: it is the coordinate rule for composition of linear maps.

Lemma 10.38 (Subspaces and extensions)**.** Every subspace of a finite-dimensional vector space has a finite basis, and every linearly independent set can be extended to a basis.

Proof. Starting with a linearly independent set, adjoin a vector outside its span whenever possible. The resulting lists remain independent, so the exchange theorem prevents more than $\dim V$ adjunctions. When the procedure stops, the list spans V and is a basis. Applying the same argument within a subspace proves its first assertion. $\square$

Theorem 10.39 (Rank–nullity)**.** For a linear map $T : V \to W$ with V finite dimensional,

$$\dim V = \dim \ker T + \dim T(V).$$

Proof. Choose a basis

$$(u_1, \ldots, u_k)$$

of $\ker T$ and extend it to a basis

$$(u_1, \ldots, u_k, v_1, \ldots, v_r)$$

of V. The vectors $T(v_j)$ span $T(V)$. A relation between them puts the corresponding combination of the v_j in $\ker T$, and independence of the full basis forces all coefficients to vanish. Hence the images form a basis of the image, proving the formula. $\square$

Corollary 10.40 (Linear bijectivity)**.** If $T : V \to W$ is linear and

$$\dim V = \dim W < \infty,$$

then injectivity, surjectivity, and bijectivity of T are equivalent.

Proof. By Theorem 10.39, injectivity is equivalent to

$$\dim T(V) = \dim V,$$

which, under the hypothesis, is equivalent to $T(V) = W$. $\square$

10.4 Norms, Permutations, Determinants, and Orientation

The determinant has three roles that will recur throughout the rest of the notes: $\det A \neq 0$ detects invertibility, $|\det A|$ measures volume scaling, and the sign of $\det A$ records orientation. Thus it will govern local inverses in Chapter 11, Jacobian factors in Chapter 12, and oriented integration and Stokes' theorem in Chapter 13. We first build the small amount of permutation language needed for its formula.

Definition 10.41 (Permutations and their sign)**.** A **permutation** of

$$\{1, \ldots, n\}$$

is a bijection of this set to itself. The set of all such permutations is denoted by S_n. Composition is its operation, the identity permutation is its identity, and every permutation has an inverse; thus S_n is a group under composition. A **transposition** exchanges two elements and fixes all others. An **inversion** of σ is a pair of positions $i < j$ whose values appear in the opposite order, $\sigma(i) > \sigma(j)$.

For $\sigma \in S_n$, its number of **inversions** is

$$\operatorname{inv}(\sigma) = \#\left\{(i, j) : 1 \leq i < j \leq n,\ \sigma(i) > \sigma(j)\right\}.$$

Define its **sign** by

$$\operatorname{sgn}(\sigma) = (-1)^{\operatorname{inv}(\sigma)}.$$

For example, in S_3, the permutation written as the list

$$\sigma = (2, 3, 1)$$

has inversions $(1, 3)$ and $(2, 3)$, so $\operatorname{inv}(\sigma) = 2$ and $\operatorname{sgn}(\sigma) = 1$. The list $(2, 1, 3)$ has the single inversion $(1, 2)$ and therefore has sign -1. Permutations of sign 1 are **even**; those of sign -1 are **odd**.

Proposition 10.42 (Sign of a permutation)**.** For $\sigma, \tau \in S_n$,

$$\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma) \operatorname{sgn}(\tau).$$

Moreover, a transposition has sign -1.

Proof. Swapping an adjacent inverted pair decreases the inversion number by exactly one. Repeatedly swap an adjacent inverted pair in the list

$$\sigma(1), \ldots, \sigma(n)$$

Each such swap lowers the nonnegative integer $\operatorname{inv}(\sigma)$, so after exactly $\operatorname{inv}(\sigma)$ swaps the list is sorted. Reversing the steps writes σ as a product of exactly that many adjacent transpositions.

More generally, every expression of a permutation as adjacent transpositions has length congruent modulo 2 to its inversion number, because each factor reverses inversion parity. Concatenating expressions for σ and τ gives one for $\sigma\tau$, so

$$\operatorname{inv}(\sigma\tau) \equiv \operatorname{inv}(\sigma) + \operatorname{inv}(\tau) \pmod 2.$$

Taking (-1) to these exponents proves multiplicativity. Finally, the transposition exchanging positions $i < j$ is a product of $2(j-i)-1$ adjacent transpositions, hence has odd sign. $\square$

Definition 10.43 (Norm)**.** A **norm** on a real vector space V is a function

$$\|\cdot\| : V \longrightarrow [0, \infty)$$

such that

$$\|v\| = 0 \iff v = 0, \qquad \|av\| = |a|\|v\|, \qquad \|v + w\| \le \|v\| + \|w\|.$$

The associated metric is

$$d(v, w) = \|v - w\|.$$

Norms measure vector size and therefore quantify errors in linear approximations. On $\mathbb{R}^2$, the unit balls of $\|x\|_1 = |x_1| + |x_2|$, $\|x\|_2$, and $\|x\|_\infty$ are, respectively, a diamond, a circle, and a square. They look different, but in finite dimensions they lead to the same convergence and continuity theory; the next theorem explains why this is important before derivatives are defined.

For a map r with values in a normed space, the notation

$$r(h) = o(\|h\|) \qquad (h \to 0)$$

means precisely

$$\frac{\|r(h)\|}{\|h\|} \longrightarrow 0.$$

Likewise, $r(h) = O(\|h\|)$ means that $\|r(h)\| \le C\|h\|$ for all sufficiently small h. This is the norm-valued version of the asymptotic notation introduced in Chapter 5; it will describe approximation errors in Chapter 11.

Proposition 10.44 (Reverse triangle inequality)**.** Every norm satisfies

$$\big|\|v\| - \|w\|\big| \le \|v - w\|.$$

Hence its norm map is continuous.

Proof. The triangle inequality gives

$$\|v\| \le \|v - w\| + \|w\|.$$

Interchange v, w and combine the two inequalities. $\square$

Theorem 10.45 (Equivalence of norms)**.** If V is finite dimensional and

$$\|\cdot\|_a, \qquad \|\cdot\|_b$$

are norms on V, then there are $c, C > 0$ such that

$$c\|v\|_a \le \|v\|_b \le C\|v\|_a$$

for every $v \in V$. Thus the two norms have the same open sets, convergent and Cauchy sequences, continuous maps, and compact subsets.

Proof. Choose a basis and identify V with $\mathbb{R}^n$. For any norm on $\mathbb{R}^n$,

$$\|x\|_b \le \sum_{j=1}^{n} |x_j| \|e_j\|_b \le C\|x\|_2$$

by Cauchy–Schwarz. Together with Proposition 10.44, this gives

$$\big|\|x\|_b - \|y\|_b\big| \le C\|x - y\|_2,$$

so $x \mapsto \|x\|_b$ is Euclidean-continuous. It has a positive minimum c on the Euclidean unit sphere, which is compact by Theorem 10.25; homogeneity gives

$$c\|x\|_2 \le \|x\|_b.$$

Comparing each of the two given norms with the Euclidean norm yields the claim. The final statements follow directly from the two-sided estimate. □

Corollary 10.46 (Completeness in finite dimensions)**.** Every finite-dimensional normed vector space is complete, and its compact subsets are exactly its closed bounded subsets.

Proof. Coordinates reduce the assertion to Euclidean space, where completeness follows coordinatewise from Theorem 3.31 and compactness from Theorem 10.25; use Theorem 10.45. □

Completeness is not merely formal here. It ensures that Cauchy approximations have limits, which is the key step in the contraction theorem and in the nonlinear existence arguments of Chapter 11.

Proposition 10.47 (Continuity of linear maps)**.** Every linear map between finite-dimensional normed spaces is continuous.

Proof. In coordinates, matrix multiplication and Cauchy–Schwarz give

$$\|T(v)\|_2 \le M\|v\|_2$$

for some M. Norm equivalence transfers this estimate to arbitrary norms, which proves continuity at zero and hence everywhere by linearity. □

The estimate $\|T(v)\| \le C\|v\|$ says quantitatively that a linear map cannot amplify a small input error without bound. Every finite-dimensional linear map has such a bound; this is the prototype for the operator-norm estimates on derivatives used later.

The determinant formula sums one contribution from each possible choice of one entry in every row and column. Its sign is essential: swapping two columns must reverse the determinant, just as swapping two ordered coordinate directions reverses orientation. The permutation sign is exactly what forces this alternating behavior. Its absolute value will measure volume, while its sign distinguishes orientation preservation from orientation reversal.

Definition 10.48 (Determinant)**.** For $A = (a_{ij}) \in M_n(\mathbb{R})$, define

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}.$$

Theorem 10.49 (Determinants)**.** The determinant is multilinear and alternating in its columns,

$$\det I_n = 1, \qquad \det(AB) = (\det A)(\det B),$$

and

$$\det A \neq 0$$

if and only if A represents an invertible linear map.

Proof. The permutation formula is linear in each column, and exchanging two columns reverses the sign of every term; it also gives $\det I_n = 1$. For fixed A, the map

$$B \longmapsto \det(AB)$$

is alternating multilinear in the columns of B. Expanding those columns in the standard basis shows that it equals its value at I_n times $\det B$, which proves multiplicativity. If one column is a linear combination of the others, multilinearity expands the determinant into a sum of determinants having repeated columns, and alternation makes each summand zero. Thus dependent columns make the determinant zero. Independent columns form a basis, so the represented map has an inverse B; then

$$1 = \det(AB) = (\det A)(\det B).$$

This proves all assertions. □

In the plane, an ordered coordinate pair can be counterclockwise or clockwise; in three dimensions this is the familiar distinction between right-handed and left-handed coordinates. In arbitrary finite dimension, the sign of a change-of-basis determinant is the algebraic version of this handedness. Orientation matters because scalar volume forgets the sign via $|\det|$, while oriented integration and Stokes' theorem retain it.

Definition 10.50 (Orientation)**.** Two ordered bases of an n-dimensional real vector space are **equivalently oriented** if the change-of-coordinate matrix between them has positive determinant. An **orientation** is one of the two equivalence classes of ordered bases. The class of the standard basis is the standard orientation of $\mathbb{R}^n$.

In $\mathbb{R}^2$, the ordered basis

$$(e_1, e_2)$$

has the standard orientation, whereas

$$(e_2, e_1)$$

has the opposite orientation because the change-of-basis determinant is -1. The basis $(e_1 + e_2, e_2)$ has the standard orientation because its change-of-basis determinant is positive.

Proposition 10.51 (Orientation)**.** Equivalent orientation is an equivalence relation. The sign of the determinant of an invertible map between oriented spaces does not depend on the positively oriented bases chosen.

Proof. The identity has determinant 1, inverses of positive-determinant matrices have positive determinant, and products of such matrices have positive determinant. A change to another positively oriented basis multiplies the matrix of a map on the left or right by a positive-determinant matrix, so multiplicativity preserves its sign. □

Exercises

Exercise 10.1. Prove that every open ball in a metric space is open and every singleton is closed.

Exercise 10.2. Prove that, for $E \subseteq X$,

$$\partial E = \overline{E} \cap \overline{X \setminus E}.$$

Exercise 10.3. Which of the sets

$$[0, 1/2), \qquad (0, 1/2), \qquad [0, 1/2]$$

are open in the subspace $[0, 1] \subseteq \mathbb{R}$?

Exercise 10.4. Prove directly from epsilon–delta definitions that a composition of continuous maps between metric spaces is continuous.

Exercise 10.5. Prove that the product topology on a finite product of metric spaces agrees with the topology from

$$d((x_1, \ldots, x_n), (y_1, \ldots, y_n)) = \max_{1 \le j \le n} d_j(x_j, y_j).$$

Exercise 10.6. Use Theorem 10.22 to prove that a compact metric space is complete.

Exercise 10.7. Prove that every convex subset of $\mathbb{R}^n$ is path connected.

Exercise 10.8. Show that the open unit ball in $\mathbb{R}^n$ is not compact.

Exercise 10.9. Prove that every independent subset of a finite-dimensional vector space can be extended to a basis.

Exercise 10.10. For

$$T(x, y, z) = (x - y, y - z),$$

find bases of the kernel and image, and verify rank–nullity.

Exercise 10.11. Let $T : V \to V$ be a linear endomorphism of a finite-dimensional real vector space and suppose that

$$T^2 = T$$

Show that

$$\ker T \cap T(V) = \{0\},$$

and that every $v \in V$ can be written uniquely in the form

$$v = u + w, \qquad u \in \ker T, \quad w \in T(V).$$

Exercise 10.12. Give maps between finite-dimensional spaces that are injective but not surjective, and surjective but not injective.

Exercise 10.13. For

$$\|x\|_1 = \sum_{j=1}^{n} |x_j|, \qquad \|x\|_\infty = \max_{1 \le j \le n} |x_j|,$$

prove

$$\|x\|_\infty \le \|x\|_2 \le \|x\|_1 \le \sqrt{n}\|x\|_2.$$

Exercise 10.14. Give an incomplete normed vector space, for example by considering polynomials on $[0,1]$ with the supremum norm.

Exercise 10.15. Use the determinant properties to prove

$$\det(A^{-1}) = \frac{1}{\det A}$$

for an invertible matrix A.

Exercise 10.16. Show that

$$(x_1, \ldots, x_n) \longmapsto (-x_1, x_2, \ldots, x_n)$$

reverses the standard orientation of $\mathbb{R}^n$.

10.5 Perspective and Transition

The derivative in Chapter 11 will be a linear map between finite-dimensional normed spaces. The equivalence of norms makes differentiability independent of a coordinate norm. Compactness supplies finite local control, while determinants and orientation distinguish the two local senses of direction needed in change of variables and Stokes' theorem.

Chapter 11

Differentiation in Finite-Dimensional Spaces

11.1 Differentiability and the Derivative

With the little-o notation introduced in Chapter 5 and extended to norm-valued remainders in Chapter 10, one-variable differentiability at a may be written as

$$f(a+h) = f(a) + f'(a)h + o(|h|).$$

The remainder is negligible relative to the increment: after subtracting the best linear approximation, its size divided by $|h|$ tends to zero. Multiplication by the scalar $f'(a)$ is already a linear map of the increment h. For a map between higher-dimensional spaces, the correct replacement is a linear map $Df(a)$, and the derivative remains the best linear approximation to the change in f.

Definition 11.1 (Fréchet derivative)**.** Let V, W be finite-dimensional normed real vector spaces, let $U \subseteq V$ be open, and let

$$f : U \longrightarrow W.$$

The function f is **differentiable at** $a \in U$ if there is a linear map

$$L : V \longrightarrow W$$

such that

$$\lim_{h \to 0} \frac{\|f(a+h) - f(a) - Lh\|_W}{\|h\|_V} = 0.$$

The map L, if it exists, is denoted by

$$Df(a)$$

and is called the **derivative** of f at a.

Proposition 11.2 (The one-dimensional case)**.** For a function $f : U \subseteq \mathbb{R} \to \mathbb{R}$, Fréchet differentiability at $a \in U$ is equivalent to differentiability in the sense of Chapter 8. Under this equivalence,

$$Df(a)h = f'(a)h.$$

Proof. Every linear map $L : \mathbb{R} \to \mathbb{R}$ has the form

$$Lh = ch$$

with $c = L(1)$. The Fréchet remainder condition is therefore exactly

$$\lim_{h\to 0} \frac{f(a+h)-f(a)}{h} = c,$$

after separating the harmless case $h = 0$. This is the ordinary derivative definition, and the asserted formula follows. ☐

Remark 11.3. Equivalently,

$$f(a+h) = f(a) + Df(a)h + r(h), \qquad \frac{\|r(h)\|_W}{\|h\|_V} \longrightarrow 0.$$

By Theorem 10.45, this definition is independent of the norms chosen on finite-dimensional spaces.

Proposition 11.4 (Uniqueness and continuity)**.** If f is differentiable at a, then $Df(a)$ is unique and f is continuous at a.

Proof. If L, M both have the required remainder property, then, for $h \neq 0$,

$$\frac{\|(L-M)h\|_W}{\|h\|_V} \le \frac{\|f(a+h)-f(a)-Lh\|_W}{\|h\|_V} + \frac{\|f(a+h)-f(a)-Mh\|_W}{\|h\|_V}.$$

Put

$$h = tv$$

and let $t \to 0$ for each fixed nonzero v. This gives

$$(L-M)v = 0,$$

so $L = M$. The remainder formula, together with the continuity of a linear map from Proposition 10.47, proves continuity of f. ☐

Proposition 11.5 (Algebra and the chain rule)**.** Let $f, g : U \to W$ be differentiable at a and let $c \in \mathbb{R}$. Then

$$D(f+g)(a) = Df(a) + Dg(a), \qquad D(cf)(a) = cDf(a).$$

For real-valued f, g,

$$D(fg)(a) = f(a)Dg(a) + g(a)Df(a).$$

If $g(a) \neq 0$, then

$$D\left(\frac{f}{g}\right)(a) = \frac{g(a)Df(a) - f(a)Dg(a)}{g(a)^2}.$$

If $f : U \to W$ is differentiable at a and $g : Z \to X$ is differentiable at $f(a)$, with $f(a) \in Z$ and Z open, then

$$D(g \circ f)(a) = Dg(f(a)) \circ Df(a).$$

Proof. Adding and scaling the remainder formulas proves the first two identities. For products, expand

$$f(a+h)g(a+h) - f(a)g(a)$$

by adding and subtracting

$$f(a)g(a+h).$$

The product of the two increments is

$$o(\|h\|_V)$$

because differentiability implies continuity. The quotient formula follows by differentiating

$$g \cdot \frac{1}{g} = 1.$$

For the chain rule, write

$$f(a+h) = f(a) + Df(a)h + r(h)$$

and apply the remainder formula for g to

$$k = f(a+h) - f(a).$$

Here

$$\|k\|_W = O(\|h\|_V),$$

so both the image of $r(h)$ under the bounded linear map $Dg(f(a))$ and the remainder for g are

$$o(\|h\|_V).$$

The surviving linear part is the asserted composite. □

Remark 11.6. The chain rule says that local linear approximations compose:

$$f(a+h) \approx f(a) + Df(a)h$$

and then

$$g(f(a)+k) \approx g(f(a)) + Dg(f(a))k.$$

Thus the linear part of $g \circ f$ is the composite

$$Dg(f(a))Df(a).$$

After coordinates are chosen, composition of linear maps is matrix multiplication, which is why that operation appears in the Jacobian formula.

Definition 11.7 (Directional derivatives and Jacobian)**.** Let $U \subseteq \mathbb{R}^n$ be open and let $f : U \to \mathbb{R}^m$. The **directional derivative** at a in direction v is

$$D_v f(a) = \lim_{t \to 0} \frac{f(a+tv) - f(a)}{t},$$

when this limit exists. The partial derivative with respect to x_j is

$$\frac{\partial f}{\partial x_j}(a) = D_{e_j} f(a).$$

The **Jacobian matrix** of a differentiable f is the standard matrix of $Df(a)$:

$$J_f(a) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(a) & \cdots & \frac{\partial f_1}{\partial x_n}(a) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(a) & \cdots & \frac{\partial f_m}{\partial x_n}(a) \end{pmatrix}.$$

Proposition 11.8 (Derivative and directional derivatives)**.** If f is differentiable at a, then

$$D_v f(a) = Df(a)v$$

for every $v \in \mathbb{R}^n$. In particular, its Jacobian is the matrix displayed above.

Proof. Put

$$h = tv$$

in the defining remainder formula and divide by t. The remainder divided by t tends to zero. Taking $v = e_j$ identifies the columns of the matrix. □

Example 11.9 (Partial derivatives need not give differentiability)**.** Define

$$f(x,y) = \begin{cases} \dfrac{xy}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Both partial derivatives at the origin exist and equal zero, since f vanishes on both coordinate axes. But along the line $y = x$,

$$f(x,x) = \frac{1}{2}$$

for $x \neq 0$; hence f is not even continuous at the origin and cannot be differentiable there. Continuous partial derivatives, as in the next theorem, are sufficient because they control all nearby coordinate increments.

Theorem 11.10 (Continuous partial derivatives)**.** Let $U \subseteq \mathbb{R}^n$ be open. If all partial derivatives of

$$f : U \longrightarrow \mathbb{R}^m$$

exist on a neighborhood of a and are continuous at a, then f is differentiable at a and its derivative is $J_f(a)$.

Proof. It is enough to consider a real-valued component. For small h, telescope the increment from a to $a + h$ along the coordinate polygonal path. Put

$$p_0 = a, \qquad p_j = a + h_1 e_1 + \cdots + h_j e_j \qquad (1 \le j \le n).$$

For sufficiently small h, every segment from p_{j-1} to p_j lies in the neighborhood on which the partial derivatives exist. Applying the one-variable Mean Value Theorem to the function that varies only its jth coordinate gives a point

$$a_j = p_{j-1} + t_j h_j e_j \qquad (0 < t_j < 1)$$

such that

$$f(a+h) - f(a) = \sum_{j=1}^{n} \frac{\partial f}{\partial x_j}(a_j) h_j.$$

Each a_j tends to a with h. Subtract the same sum with a_j replaced by a. Given $\varepsilon > 0$, continuity of the partial derivatives makes the absolute value of the difference at most

$$\varepsilon \sum_{j=1}^{n} |h_j| \le \varepsilon \sqrt{n} \|h\|_2.$$

This is the required remainder estimate. Apply it componentwise. □

Definition 11.11 (Class C^1)**.** A differentiable map on an open set is of class

$$C^1$$

when its derivative is continuous as a map into the finite-dimensional space of linear maps.

Remark 11.12 (A practical differentiability workflow)**.** First compute the only possible derivative: if differentiability holds, its matrix must have the partial derivatives as columns. If the partial derivatives exist in a neighborhood and are continuous at the point, apply Theorem 11.10. Otherwise return to the definition, subtract the candidate linear map, and estimate the remainder divided by $\|h\|$. Partial or directional derivatives inspect individual lines; total differentiability requires one linear approximation that controls all small increments simultaneously.

Example 11.13 (A direct remainder estimate)**.** Define $F(0,0) = 0$ and, away from the origin,

$$F(x, y) = \frac{x^2 y^2}{x^2 + y^2}.$$

The candidate derivative at the origin is zero. Since $4x^2y^2 \leq (x^2 + y^2)^2$,

$$\frac{|F(x, y)|}{\|(x, y)\|_2} \leq \frac{1}{4}\|(x, y)\|_2 \longrightarrow 0.$$

Hence F is differentiable at the origin by the definition, even without first proving continuity of its nearby partial derivatives.

Example 11.14 (Directional derivatives are still not enough)**.** Set $G(0,0) = 0$ and

$$G(x, y) = \frac{x^3}{x^2 + y^2}$$

elsewhere. For every $v \neq 0$,

$$D_v G(0) = \frac{v_1^3}{v_1^2 + v_2^2}.$$

All directional derivatives therefore exist, but their dependence on v is not linear. They cannot equal $DG(0)v$ for a linear map, so G is not differentiable at the origin. Existence of all partial derivatives is an even weaker condition.

Definition 11.15 (Operator norm)**.** For a linear map $L : V \to W$ between normed spaces, define

$$\|L\|_{\text{op}} = \sup \left\{ \|Lv\|_W : \|v\|_V \leq 1 \right\}.$$

This number is finite in finite dimensions, and

$$\|Lv\|_W \leq \|L\|_{\text{op}} \|v\|_V.$$

11.2 Higher Derivatives and Local Estimates

The one-variable Mean Value and Taylor theorems describe a function by its successive local derivatives. The same mechanism works along a line segment in $\mathbb{R}^n$: restrict the function to that segment, apply the Part I theorem, and interpret the resulting coefficients as multilinear maps.

Theorem 11.16 (Mean-value estimate)**.** Let $U \subseteq \mathbb{R}^n$ be open, let $f : U \to \mathbb{R}^m$ be differentiable, and suppose the segment from x to y lies in U. If

$$\|Df((1-t)x+ty)\|_{\mathrm{op}} \leq M \qquad (0 \leq t \leq 1),$$

then

$$\|f(y)-f(x)\|_2 \leq M\|y-x\|_2.$$

Proof. If $f(y) \neq f(x)$, put

$$u = \frac{f(y)-f(x)}{\|f(y)-f(x)\|_2}$$

and apply the one-variable Mean Value Theorem to

$$t \longmapsto u \cdot f((1-t)x+ty).$$

For some $c \in (0,1)$,

$$\|f(y)-f(x)\|_2 = u \cdot Df((1-c)x+cy)(y-x).$$

Cauchy–Schwarz and the operator-norm bound prove the assertion. The zero case is immediate. □

Definition 11.17 (Higher derivatives)**.** For a differentiable map on an open subset of a finite-dimensional space, define inductively

$$D^0 f = f, \qquad D^{k+1}f(a) = D(D^k f)(a),$$

when the right side exists. Under the standard identification,

$$D^k f(a)$$

is a k-linear map. The map is of class

$$C^k$$

when all derivatives through order k exist and are continuous.

Theorem 11.18 (Equality of mixed partials)**.** Let $U \subseteq \mathbb{R}^n$ be open and let $f : U \to \mathbb{R}$ have continuous second partial derivatives. Then

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(a) = \frac{\partial^2 f}{\partial x_j \partial x_i}(a).$$

Proof. For small nonzero s, t, form

$$\Delta = f(a+se_i+te_j) - f(a+se_i) - f(a+te_j) + f(a).$$

Applying the one-variable Mean Value Theorem first in one coordinate and then the other expresses

$$\frac{\Delta}{st}$$

as either mixed partial at a point tending to a. Letting

$$(s,t) \to (0,0)$$

and using continuity gives the result. □

Definition 11.19 (Gradient, Hessian, and critical points)**.** Let $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$ be differentiable at a. Its **gradient** is the coordinate vector

$$\nabla f(a) = \left(\frac{\partial f}{\partial x_1}(a), \ldots, \frac{\partial f}{\partial x_n}(a)\right).$$

Thus the invariant derivative has the coordinate form

$$Df(a)h = \nabla f(a) \cdot h.$$

If f is of class C^2, its **Hessian** at a is the matrix

$$H_f(a) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(a)\right)_{i,j=1}^{n}.$$

Under the standard coordinate identification,

$$D^2 f(a)[h, k] = h^\mathsf{T} H_f(a) k.$$

By Theorem 11.18, the Hessian is symmetric. A point $a \in U$ is a **critical point** of a differentiable scalar-valued function when

$$Df(a) = 0,$$

or equivalently when $\nabla f(a) = 0$.

Theorem 11.20 (Taylor's theorem along a segment)**.** Let $U \subseteq \mathbb{R}^n$ be open, let $f : U \to \mathbb{R}^m$ be of class

$$C^{k+1},$$

and suppose the segment from a to $a + h$ is in U. If $m = 1$, there is

$$\theta \in (0, 1)$$

such that

$$f(a + h) = \sum_{j=0}^{k} \frac{1}{j!} D^j f(a)[h, \ldots, h] + \frac{1}{(k+1)!} D^{k+1} f(a + \theta h)[h, \ldots, h].$$

For vector-valued f, this formula holds componentwise, possibly with a different value of θ for each component.

Proof. For a scalar component define

$$g(t) = f(a + th).$$

Repeated application of the chain rule gives

$$g^{(j)}(t) = D^j f(a + th)[h, \ldots, h].$$

Apply the one-variable Taylor theorem Theorem 8.21 to g at zero and evaluate at one. Do this for each component. □

Corollary 11.21 (Second-order test)**.** Let $f : U \to \mathbb{R}$ be of class

$$C^2$$

and satisfy

$$Df(a) = 0.$$

If

$$D^2 f(a)[h,h] > 0 \qquad (h \neq 0),$$

then a is a strict local minimum. If the displayed inequality is reversed, then a is a strict local maximum.

Proof. On the Euclidean unit sphere, the continuous quadratic form has a positive minimum by Theorem 10.25. Hence

$$D^2 f(a)[h,h] \geq c\|h\|_2^2$$

for some $c > 0$. Taylor's theorem with $k = 1$ and continuity of the second derivative make

$$f(a+h) - f(a)$$

positive for all sufficiently small nonzero h. Apply this to $-f$ for the maximum assertion. □

11.3 The Contraction Principle

Definition 11.22 (Contraction)**.** A map

$$T : X \longrightarrow X$$

on a metric space is a **contraction** if

$$d(Tx, Ty) \leq q d(x,y)$$

for all $x, y \in X$ and some

$$0 \leq q < 1.$$

Theorem 11.23 (Contraction mapping theorem)**.** A contraction of a complete metric space has a unique fixed point. For every x_0, the sequence

$$x_{n+1} = T(x_n)$$

converges to it.

Proof. The contraction inequality gives

$$d(x_{n+1}, x_n) \leq q^n d(x_1, x_0).$$

Thus, for $m > n$,

$$d(x_m, x_n) \leq \frac{q^n}{1-q} d(x_1, x_0).$$

The sequence is Cauchy and has a limit x. Continuity of T implies

$$T(x) = x.$$

If x, y are fixed, then

$$d(x,y) \leq q d(x,y),$$

so $x = y$. □

Example 11.24 (Solving $x = \cos x$ by iteration)**.** On $X = [1/2, 1]$ with the usual metric, $T(x) = \cos x$ maps X into itself. Indeed, the elementary-function theory gives $1/2 < \cos 1 \leq \cos x < 1$ there. The Mean Value Theorem and $|\sin x| \leq \sin 1 < 1$ give

$$|T(x) - T(y)| \leq (\sin 1)|x - y|.$$

The closed interval is complete, so the contraction theorem gives a unique solution in X, and the numerical iteration $x_{n+1} = \cos x_n$ converges to it from every starting point in X. Completeness is used precisely when the Cauchy iterates are asserted to have a limit. The same mechanism later underlies existence arguments for differential equations, integral equations, and problems in functional analysis.

Proposition 11.25 (Perturbation of an invertible map)**.** Let $A, B : \mathbb{R}^n \to \mathbb{R}^n$ be linear and let A be invertible. If

$$\|I - A^{-1}B\|_{\text{op}} < 1,$$

then B is invertible.

Proof. Put

$$T = I - A^{-1}B.$$

The geometric-series argument gives

$$(I - T)^{-1} = \sum_{k=0}^{\infty} T^k,$$

because the series converges in the finite-dimensional matrix space. Since

$$B = A(I - T),$$

the result follows. □

11.4 Inverse and Implicit Functions

For an invertible linear map $T : \mathbb{R}^n \to \mathbb{R}^n$, invertibility is equivalent to $\det T \neq 0$. In one variable, the corresponding hypothesis $f'(a) \neq 0$ says that the linear approximation $h \mapsto f'(a)h$ can be inverted. For a map

$$\mathbb{R}^n \longrightarrow \mathbb{R}^n,$$

the exact replacement is that $Df(a)$ is an invertible linear map. Since

$$f(a + h) = f(a) + Df(a)h + o(\|h\|),$$

the natural question is whether an invertible best linear approximation forces f itself to be invertible nearby. The inverse function theorem answers yes, locally: close to a, the map neither folds space nor collapses a direction, and nearby points can be recovered uniquely from their images.

Example 11.26 (Local is not global)**.** The map

$$P(u, v) = (e^u \cos v, e^u \sin v)$$

has

$$\det DP(u, v) = e^{2u} > 0$$

at every point. It is therefore locally invertible everywhere, but it is not globally one-to-one because $P(u, v) = P(u, v + 2\pi)$. A nonzero determinant at a point, or even at every point, is a local conclusion unless additional global information is supplied.

Theorem 11.27 (Inverse function theorem)**.** Let $U \subseteq \mathbb{R}^n$ be open and let

$$f : U \longrightarrow \mathbb{R}^n$$

be of class

$$C^1.$$

If

$$Df(a)$$

is invertible, then there are open neighborhoods

$$U_0 \subseteq U, \qquad V_0 \subseteq \mathbb{R}^n$$

of a and $f(a)$ such that

$$f : U_0 \longrightarrow V_0$$

is bijective. Its inverse g is of class

$$C^1$$

and

$$Dg(y) = \big(Df(g(y))\big)^{-1}.$$

Proof. Put

$$A = Df(a)^{-1}.$$

Choose q with

$$0 < q < 1.$$

Because U is open and Df is continuous at a, there is $r > 0$ such that the closed ball

$$K = \{x \in \mathbb{R}^n : \|x - a\|_2 \le r\}$$

is contained in U and

$$\|I - ADf(x)\|_{\mathrm{op}} \le q \qquad (x \in K).$$

The set K is complete because it is closed in the complete space $\mathbb{R}^n$. Define

$$G(x) = x - Af(x).$$

Since

$$DG(x) = I - ADf(x),$$

the segment between any two points $x, z \in K$ lies in the convex set K, and Theorem 11.16 gives

$$\|G(x) - G(z)\|_2 \le q\|x - z\|_2.$$

For $y \in \mathbb{R}^n$, define

$$T_y(x) = x - A(f(x) - y).$$

Then

$$T_y(x) - T_y(z) = G(x) - G(z),$$

so T_y is a q-contraction on K. Moreover, for $x \in K$,

$$\begin{aligned}\|T_y(x) - a\|_2 &= \|[x - Af(x)] - [a - Af(a)] + A(y - f(a))\|_2 \\ &\le q\|x - a\|_2 + \|A\|_{\mathrm{op}}\|y - f(a)\|_2.\end{aligned}$$

Set
$$\rho = \frac{(1-q)r}{\|A\|_{\mathrm{op}}}.$$
If $\|y - f(a)\|_2 < \rho$, the last expression is strictly smaller than r. Thus $T_y(K) \subseteq K$, and Theorem 11.23 gives a unique fixed point $g(y) \in K$. The fixed-point equation is equivalent to
$$f(g(y)) = y.$$
If $x, z \in K$ and $f(x) = f(z)$, then
$$\|x - z\|_2 = \|[x - Af(x)] - [z - Af(z)]\|_2 \leq q\|x - z\|_2.$$
Since $q < 1$, this forces $x = z$, so f is injective on K.

Take
$$V_0 = B_2\left(f(a), \frac{\rho}{2}\right)$$
and set
$$U_0 = B_2(a, r) \cap f^{-1}(V_0).$$
The set V_0 is open, contains $f(a)$, and its closure lies in the range of y's for which the contraction argument applies. Since f is continuous, U_0 is open and contains a. For every $y \in V_0$, the self-map estimate with $\|y - f(a)\|_2 < \rho/2$ puts the fixed point $g(y)$ in the open ball $B_2(a, r)$, so $g(y) \in U_0$. Thus $f(U_0) = V_0$, and injectivity on K shows that
$$f : U_0 \longrightarrow V_0$$
is bijective with inverse g.

Subtracting the fixed-point equations
$$g(y) = G(g(y)) + Ay, \qquad g(z) = G(g(z)) + Az$$
and using the Lipschitz estimate for G gives
$$(1 - q)\|g(y) - g(z)\|_2 \leq \|A\|_{\mathrm{op}}\|y - z\|_2,$$
and hence
$$\|g(y) - g(z)\|_2 \leq \frac{\|A\|_{\mathrm{op}}}{1 - q}\|y - z\|_2.$$
In particular, g is continuous. For $x \in K$,
$$\|I - Df(a)^{-1}Df(x)\|_{\mathrm{op}} = \|I - ADf(x)\|_{\mathrm{op}} \leq q < 1.$$
Thus Proposition 11.25, with its invertible map equal to $Df(a)$ and its perturbed map equal to $Df(x)$, shows that every $Df(x)$ on K is invertible.

Fix $y \in V_0$, put $x = g(y)$, and, for h small enough that $y + h \in V_0$, set
$$\Delta x = g(y + h) - g(y).$$
Differentiability of f at x gives
$$h = f(x + \Delta x) - f(x) = Df(x)\Delta x + r(\Delta x),$$

where
$$\frac{\|r(\Delta x)\|_2}{\|\Delta x\|_2} \longrightarrow 0 \qquad (\Delta x \to 0).$$
The Lipschitz estimate for g gives
$$\|\Delta x\|_2 \le \frac{\|A\|_{\text{op}}}{1-q}\|h\|_2.$$
Consequently,
$$\frac{\|r(\Delta x)\|_2}{\|h\|_2} \le \frac{\|A\|_{\text{op}}}{1-q}\frac{\|r(\Delta x)\|_2}{\|\Delta x\|_2} \longrightarrow 0,$$
where for $h \neq 0$ one has $\Delta x \neq 0$ because g is injective. Therefore $r(\Delta x) = o(\|h\|_2)$. Multiplying by $Df(x)^{-1}$ yields
$$\Delta x = Df(x)^{-1}h + o(\|h\|_2),$$
and therefore
$$Dg(y) = Df(g(y))^{-1}.$$

It remains to justify continuity of this derivative. Fix an invertible matrix C. If
$$\|C^{-1}(B-C)\|_{\text{op}} < \frac{1}{2},$$
then the geometric-series proof of Proposition 11.25 gives
$$\|B^{-1}\|_{\text{op}} \le 2\|C^{-1}\|_{\text{op}}.$$
Together with
$$B^{-1} - C^{-1} = B^{-1}(C-B)C^{-1},$$
this proves that inversion is continuous at C. Since g and Df are continuous and $Df(g(y))$ is invertible throughout V_0, the formula for Dg is continuous. Thus g is of class C^1. □

Example 11.28 (Using the inverse function theorem)**.** Let
$$f(u,v) = (u+v^2, u^2+v).$$
Its Jacobian is
$$Df(u,v) = \begin{pmatrix} 1 & 2v \\ 2u & 1 \end{pmatrix}, \qquad \det Df(u,v) = 1 - 4uv.$$
At $(0,0)$ the determinant is 1, so the theorem supplies neighborhoods of $(0,0)$ on which f has a unique C^1 inverse. In ordinary language, every output sufficiently close to $(0,0)$ comes from exactly one nearby input, and that input varies differentiably with the output. The theorem does not make a global claim; on the curve $uv = 1/4$ the derivative is singular, so the Inverse Function Theorem gives no conclusion about local invertibility there.

The derivative formula is exactly what formal differentiation predicts: from $f^{-1} \circ f = \text{id}$, the chain rule suggests
$$D(f^{-1})(f(a))Df(a) = I,$$
and hence $D(f^{-1})(f(a)) = Df(a)^{-1}$. The theorem proves that this formal calculation applies to an actual local inverse.

Definition 11.29 (Diffeomorphisms)**.** Let $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ be open. A bijection

$$f : U \longrightarrow V$$

is a C^r **diffeomorphism** if both f and f^{-1} are of class C^r. When $r = \infty$, it is a **smooth diffeomorphism**. Thus the inverse function theorem says, in this language, that a C^1 map with invertible derivative at a point is a C^1 diffeomorphism after restriction to sufficiently small open neighborhoods.

Remark 11.30 (From an equation to a local graph)**.** The circle

$$x^2 + y^2 = 1$$

is not the graph of one function $y = g(x)$ globally: most x values have two points on the circle. Yet near $(0, 1)$, the equation

$$F(x, y) = x^2 + y^2 - 1 = 0$$

can be solved for y because $F_y(0, 1) = 2 \neq 0$. Near $(1, 0)$ it can instead be solved for x, since $F_x(1, 0) = 2 \neq 0$. The Implicit Function Theorem is the general existence-and-local-parametrization principle behind this familiar calculus picture: it solves selected variables locally even when there is no useful explicit algebraic formula.

Conceptually it is another form of local invertibility. For $F : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^m$, the auxiliary map

$$H(x, y) = (x, F(x, y))$$

keeps the prospective independent variables and records the equations as the remaining output coordinates. Invertibility of D_yF makes DH invertible, so a local inverse for H recovers the desired dependent variables.

Theorem 11.31 (Implicit function theorem)**.** Let $U \subseteq \mathbb{R}^n \times \mathbb{R}^m$ be open and let

$$F : U \longrightarrow \mathbb{R}^m$$

be of class

$$C^1.$$

Suppose

$$F(a, b) = 0$$

and

$$D_yF(a, b)$$

is invertible. Then there are open neighborhoods

$$A \subseteq \mathbb{R}^n, \qquad B \subseteq \mathbb{R}^m$$

of a and b, respectively, and a unique C^1 map

$$\varphi : A \longrightarrow B$$

such that

$$F(x, \varphi(x)) = 0.$$

Moreover,

$$D\varphi(x) = -\big(D_yF(x, \varphi(x))\big)^{-1} D_xF(x, \varphi(x)).$$

Proof. Geometrically, the theorem solves the equation

$$F(x, y) = 0$$

near (a, b) for the y-variables as functions of the x-variables. To convert this into an inverse-function problem, retain x as an output coordinate and define

$$H(x, y) = (x, F(x, y)).$$

Its derivative at (a, b) is

$$DH(a, b)(u, v) = \big(u, D_xF(a, b)u + D_yF(a, b)v\big).$$

It is invertible because, given (p, q), its inverse is

$$(p, q) \longmapsto \Big(p, \big(D_yF(a, b)\big)^{-1}(q - D_xF(a, b)p)\Big).$$

Continuity of D_yF and Proposition 11.25 give an open product neighborhood $P \times R$ of (a, b), contained in U, on which every $D_yF(x, y)$ is invertible: shrink the product until

$$\|I - D_yF(a, b)^{-1}D_yF(x, y)\|_{\text{op}} < 1.$$

Apply Theorem 11.27 to H. There are open neighborhoods W of (a, b) and Z of $(a, 0)$ such that

$$H : W \longrightarrow Z$$

is a C^1 diffeomorphism. Intersect W with $P \times R$ and replace Z by the image of this intersection; because $H : W \to Z$ is a homeomorphism, these are still open neighborhoods and the restricted map is still a C^1 diffeomorphism. Thus assume

$$W \subseteq P \times R.$$

Because W is open, choose open neighborhoods A_1 of a and B of b such that

$$A_1 \times B \subseteq W.$$

Because Z is open at $(a, 0)$, choose an open neighborhood A_2 of a such that

$$A_2 \times \{0\} \subseteq Z.$$

Continuity of H^{-1} at $(a, 0)$ permits A_2 to be shrunk so that

$$H^{-1}(A_2 \times \{0\}) \subseteq \mathbb{R}^n \times B.$$

Set

$$A = A_1 \cap A_2.$$

Then $A \times B \subseteq W$ and $A \times \{0\} \subseteq Z$.

For $x \in A$, define

$$\varphi(x) = \pi_2\big(H^{-1}(x, 0)\big).$$

The preceding inclusion shows that $\varphi(x) \in B$. Since H^{-1}, the map $x \mapsto (x, 0)$, and the projection π_2 are C^1, so is φ. If

$$H^{-1}(x, 0) = (x', y'),$$

then

$$H(x', y') = (x, 0).$$

The first component forces $x' = x$, and the second gives $F(x, y') = 0$. Therefore

$$F(x, \varphi(x)) = 0.$$

If $x \in A$, $y \in B$, and $F(x, y) = 0$, then both (x, y) and $(x, \varphi(x))$ lie in W, and

$$H(x, y) = (x, 0) = H(x, \varphi(x)).$$

Injectivity of H on W gives $y = \varphi(x)$, proving the stated local uniqueness. Moreover, $D_y F(x, \varphi(x))$ is invertible for every $x \in A$, because the graph lies in $W \subseteq P \times R$.

Differentiate

$$F(x, \varphi(x)) = 0$$

using the chain rule. As linear maps from $\mathbb{R}^n$ to $\mathbb{R}^m$, this gives

$$D_x F(x, \varphi(x)) + D_y F(x, \varphi(x)) D\varphi(x) = 0.$$

Here $D_x F$ is an m-by-n map, $D_y F$ is an invertible m-by-m map, and $D\varphi$ is an m-by-n map. Multiplication on the left by $D_y F(x, \varphi(x))^{-1}$ proves

$$D\varphi(x) = -D_y F(x, \varphi(x))^{-1} D_x F(x, \varphi(x)).$$

□

Example 11.32 (A local curve)**.** Near $(1, 0)$, the equation

$$x^2 + y^2 = 1$$

defines x as a C^1 function of y, since the derivative with respect to x is

$$2x \neq 0.$$

Its derivative is

$$\frac{\mathrm{d}x}{\mathrm{d}y} = -\frac{y}{x}.$$

This is a local, rather than a global, parametrization of the circle.

Example 11.33 (Implicit differentiation justified)**.** For $F(x, y) = x^2 + y^2 - 1$, take $(a, b) = (0, 1)$. Then

$$F(0, 1) = 0, \qquad F_y(0, 1) = 2 \neq 0.$$

The Implicit Function Theorem therefore gives a unique differentiable function $y = g(x)$ near 0 with $g(0) = 1$ and $x^2 + g(x)^2 = 1$. Its derivative is

$$g'(x) = -\frac{F_x(x, g(x))}{F_y(x, g(x))} = -\frac{x}{g(x)}.$$

This is the familiar implicit-differentiation rule from calculus, now backed by a theorem that first supplies the local function being differentiated. At $(0, -1)$ one obtains the lower local graph; at $(\pm 1, 0)$ one must solve locally for x as a function of y instead.

Theorem 11.34 (Lagrange multipliers for one constraint)**.** Let $f, g : U \subseteq \mathbb{R}^n \to \mathbb{R}$ be of class C^1. Suppose that a is a local extremum of f subject to

$$g(x) = c,$$

with $g(a) = c$, and that $\nabla g(a) \neq 0$. Then there is $\lambda \in \mathbb{R}$ such that

$$\nabla f(a) = \lambda \nabla g(a).$$

Proof. The Implicit Function Theorem makes the level set $g = c$ near a a smooth hypersurface. Its tangent directions form

$$T = \ker Dg(a).$$

Indeed, after relabeling coordinates so that a partial derivative of g is nonzero, the theorem writes the level set as a local graph, whose derivative has precisely this tangent range. Restricting f to that graph gives a local unconstrained extremum, so

$$Df(a)v = 0 \qquad (v \in T).$$

Thus $Df(a)$ vanishes on $\ker Dg(a)$. Since $Dg(a)$ is a nonzero linear functional, any linear functional vanishing on this kernel is a scalar multiple of $Dg(a)$. Hence $Df(a) = \lambda Dg(a)$, which is the displayed gradient equation. □

Example 11.35 (Extrema of xy on the unit circle)**.** To maximize or minimize

$$f(x, y) = xy$$

subject to $g(x, y) = x^2 + y^2 = 1$, observe that

$$\nabla f = (y, x), \qquad \nabla g = (2x, 2y).$$

The Lagrange equations are

$$(y, x) = \lambda(2x, 2y), \qquad x^2 + y^2 = 1.$$

They give $\lambda = 1/2$ and $y = x$, or $\lambda = -1/2$ and $y = -x$. Thus $f = 1/2$ at

$$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \qquad \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right),$$

and $f = -1/2$ at the two points with $y = -x$. These are respectively the maximum and minimum, as also follows from $2|xy| \le x^2 + y^2 = 1$.

Remark 11.36 (How to recognize which theorem to use)**.** For a map $f : \mathbb{R}^n \to \mathbb{R}^n$, a question about a local inverse calls for computing $Df(a)$ and checking that it is invertible (equivalently, that its determinant is nonzero). For equations $F(x, y) = 0$ and a request to solve for the y variables, compute the block $D_y F(a, b)$ instead. In either case, the result concerns neighborhoods of the specified point, not a global inverse or a global parametrization.

Exercises

Exercise 11.1. For

$$f(x, y) = (x^2 y, \sin(x + y)),$$

compute $Df(x, y)$ and $Df(x, y)(u, v)$.

Exercise 11.2. Give a function from $\mathbb{R}^2$ to $\mathbb{R}$ with both partial derivatives at the origin but which is not differentiable there.

Exercise 11.3. Prove that a linear map between finite-dimensional normed spaces is differentiable everywhere and is its own derivative.

Exercise 11.4. Prove directly from the definition that differentiability at a implies continuity at a.

Exercise 11.5. Use Theorem 11.16 to prove that a C^1 map with zero derivative on a convex open set is constant.

Exercise 11.6. Compute the Hessian of

$$x^3 + xy^2 - 3x$$

and classify its critical points.

Exercise 11.7. Apply Taylor's theorem to find the quadratic approximation of

$$e^x \cos y$$

at the origin.

Exercise 11.8. Give a function with unequal mixed partial derivatives at the origin and explain why Theorem 11.18 does not apply.

Exercise 11.9. Use the contraction mapping theorem to prove that

$$x = \cos x$$

has a unique solution in

$$[0, 1].$$

Exercise 11.10. Let A be a matrix with

$$\|I - A\|_{\text{op}} < 1.$$

Show that

$$x_{k+1} = (I - A)x_k + b$$

converges to the unique solution of

$$Ax = b.$$

Exercise 11.11. Apply the inverse function theorem to

$$f(x, y) = (e^x \cos y, e^x \sin y)$$

at $(0, 0)$ and compute the derivative of its local inverse at $(1, 0)$.

Exercise 11.12. Near $(0, 1)$, use the implicit function theorem on

$$x^2 + y^2 + xy = 1$$

to solve for y as a C^1 function of x and find its derivative at zero.

Exercise 11.13. Explain why

$$y^2 = x^3$$

does not define y as a C^1 function of x near the origin.

11.5 Perspective and Transition

Chapter 12 uses differentiability, determinants, and local inverses to study transformations of domains. Chapter 13 reinterprets derivatives as pullbacks of alternating multilinear forms.

Chapter 12

Riemann Integration in Several Variables

12.1 Rectangles, Partitions, and Darboux Sums

This is the direct extension of Chapter 9: intervals become rectangles, length becomes volume, and one partitions each coordinate interval at once. The Darboux definition is unchanged in spirit. Chapter 10 supplies the compactness and uniform-continuity facts that make continuous functions integrable on these higher-dimensional compact rectangles.

Definition 12.1 (Rectangles and volume)**.** A **rectangle** in $\mathbb{R}^n$ is a product

$$Q = [a_1, b_1] \times \cdots \times [a_n, b_n],$$

where

$$a_j < b_j \qquad (1 \le j \le n).$$

Its **volume** is

$$\operatorname{vol}(Q) = \prod_{j=1}^{n} (b_j - a_j).$$

Definition 12.2 (Grid partitions and Darboux sums)**.** Let Q be a rectangle. A **grid partition** of Q is obtained by choosing a finite partition of each interval $[a_j, b_j]$ and taking all products of the resulting subintervals. Its members are **subrectangles**. For bounded

$$f : Q \longrightarrow \mathbb{R}$$

and a grid partition $\mathcal{P}$, put

$$U(f, \mathcal{P}) = \sum_{R \in \mathcal{P}} \sup_R f \ \operatorname{vol}(R), \qquad L(f, \mathcal{P}) = \sum_{R \in \mathcal{P}} \inf_R f \ \operatorname{vol}(R).$$

The function f is **Riemann integrable** on Q if

$$\inf_{\mathcal{P}} U(f, \mathcal{P}) = \sup_{\mathcal{P}} L(f, \mathcal{P}).$$

The common value is denoted by

$$\int_Q f(x)\ \mathrm{d}x.$$

Proposition 12.3 (Darboux criterion)**.** A bounded function $f : Q \to \mathbb{R}$ is Riemann integrable if and only if, for every

$$\varepsilon > 0,$$

there is a grid partition $\mathcal{P}$ such that

$$U(f, \mathcal{P}) - L(f, \mathcal{P}) < \varepsilon.$$

Proof. Refining a grid partition decreases upper sums and increases lower sums. Thus every lower sum is at most every upper sum. If the two extremal values are equal, choose partitions whose upper and lower sums are within

$$\varepsilon/2$$

of that common value and take their common refinement. Conversely, the displayed condition forces the difference of the two extremal values to be smaller than every positive ε, hence zero by Lemma 4.2. □

Theorem 12.4 (Continuous functions are integrable)**.** Every continuous function

$$f : Q \longrightarrow \mathbb{R}$$

is Riemann integrable.

Proof. The rectangle Q is compact by Theorem 10.25, so Theorem 10.24 makes f uniformly continuous. Choose a grid whose subrectangles have Euclidean diameter less than δ. On each subrectangle R,

$$\sup_R f - \inf_R f < \frac{\varepsilon}{\mathrm{vol}(Q)}.$$

Therefore

$$U(f, \mathcal{P}) - L(f, \mathcal{P}) < \varepsilon,$$

and Proposition 12.3 applies. □

Example 12.5 (Continuity succeeds; boundedness alone fails)**.** On $Q = [0, 1]^2$, the function $f(x, y) = x^2 + y^2$ is integrable by Theorem 12.4. In contrast, define

$$g(x, y) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Every nondegenerate subrectangle contains points of both kinds, so its oscillation is 1. Consequently every grid partition satisfies

$$U(g, \mathcal{P}) - L(g, \mathcal{P}) = \mathrm{vol}(Q) = 1,$$

and g is not Riemann integrable. The example separates boundedness from integrability in several variables just as the Dirichlet function does in one variable.

Theorem 12.6 (Basic properties)**.** If $f, g : Q \to \mathbb{R}$ are integrable and $c \in \mathbb{R}$, then

$$\int_Q (f + g)\ \mathrm{d}x = \int_Q f\ \mathrm{d}x + \int_Q g\ \mathrm{d}x, \qquad \int_Q cf\ \mathrm{d}x = c \int_Q f\ \mathrm{d}x.$$

If

$$f \le g,$$

then their integrals have the same order. Also,

$$\left|\int_Q f \, \mathrm{d}x\right| \le \int_Q |f| \, \mathrm{d}x,$$

and

$$\left|\int_Q f \, \mathrm{d}x\right| \le M \operatorname{vol}(Q)$$

whenever

$$|f(x)| \le M \qquad (x \in Q).$$

Proof. The upper and lower sum inequalities used in the proof of Theorem 9.6 hold verbatim for subrectangles:

$$U(f+g,\mathcal{P}) \le U(f,\mathcal{P}) + U(g,\mathcal{P}),$$
$$L(f+g,\mathcal{P}) \ge L(f,\mathcal{P}) + L(g,\mathcal{P}).$$

Taking a common refinement of partitions that make the two individual gaps small proves integrability of the sum. The same inequalities, now compared with the integrals, prove equality of its integral with the sum of the integrals. Scalar multiplication and order preservation are identical to the one-variable arguments.

The inequality

$$||u| - |v|| \le |u - v|$$

shows from a common partition that $|f|$ is integrable. Since

$$-|f| \le f \le |f|,$$

order preservation gives the first estimate. The second follows by integrating the constant functions $-M$ and M. □

12.2 Iterated Integrals and Jordan-Measurable Sets

The equality of an integral over a rectangle with repeated one-variable integrals is not a new definition. It is a theorem saying that the same finite rectangular approximations can be grouped in either order. The continuity hypothesis keeps the oscillation on every sufficiently small product rectangle under control.

Theorem 12.7 (Riemann–Fubini theorem for continuous functions)**.** Let

$$Q = A \times B,$$

where $A \subseteq \mathbb{R}^p$ and $B \subseteq \mathbb{R}^q$ are rectangles, and let

$$f : Q \longrightarrow \mathbb{R}$$

be continuous. Then the functions

$$x \longmapsto \int_B f(x,y) \, \mathrm{d}y, \qquad y \longmapsto \int_A f(x,y) \, \mathrm{d}x$$

are continuous, and

$$\int_{A\times B} f(x,y) \, \mathrm{d}(x,y) = \int_A \left(\int_B f(x,y) \, \mathrm{d}y\right) \mathrm{d}x = \int_B \left(\int_A f(x,y) \, \mathrm{d}x\right) \mathrm{d}y.$$

Proof. Uniform continuity of f on the compact rectangle implies that, when

$$x' \to x,$$

the functions

$$y \longmapsto f(x', y)$$

converge uniformly to

$$y \longmapsto f(x, y).$$

The bounded-function estimate from Chapter 9 therefore proves continuity of the first displayed function; the other is similar.

Let $\eta > 0$. Choose product grids in A and B so fine that the oscillation of f on every product subrectangle is less than

$$\frac{\eta}{\operatorname{vol}(A)\operatorname{vol}(B)}.$$

Choose one point (x_R, y_S) in each product cell $R \times S$, and form the common finite sum

$$\Sigma = \sum_{R,S} f(x_R, y_S)\operatorname{vol}(R)\operatorname{vol}(S).$$

The upper and lower Darboux sums on the product grid differ by less than η, so Σ differs from

$$\int_{A\times B} f$$

by less than η. For each fixed R, the same oscillation estimate shows that

$$\left|\int_B f(x_R, y)\ \mathrm{d}y - \sum_S f(x_R, y_S)\operatorname{vol}(S)\right| < \frac{\eta}{\operatorname{vol}(A)}.$$

Multiplying by $\operatorname{vol}(R)$, summing over R, and then using the analogous one-variable-in-x estimate shows that Σ differs from

$$\int_A \left(\int_B f(x, y)\ \mathrm{d}y\right) \mathrm{d}x$$

by at most 2η. Since η is arbitrary, the first equality follows. Interchanging A and B proves the second. $\square$

Example 12.8 (A double integral on a rectangle)**.** For the continuous function $f(x, y) = x + 2y$ on $[0, 1] \times [0, 2]$, Riemann–Fubini gives

$$\begin{aligned}\int_{[0,1]\times[0,2]} (x + 2y)\ \mathrm{d}(x, y) &= \int_0^1 \left(\int_0^2 (x + 2y)\ \mathrm{d}y\right) \mathrm{d}x \\ &= \int_0^1 (2x + 4)\ \mathrm{d}x = 5.\end{aligned}$$

Here the rectangle makes iterated integration natural; no coordinate change is needed.

Theorem 12.9 (Coordinatewise integration by parts)**.** Let

$$Q = [a_1, b_1] \times \cdots \times [a_n, b_n],$$

let u, v be of class C^1 on an open neighborhood of Q, and fix

$$j \in \{1, \ldots, n\}.$$

Write

$$Q_{\widehat{j}} = \prod_{\substack{1 \le i \le n \\ i \ne j}} [a_i, b_i], \qquad \mathrm{d}x_{\widehat{j}} = \mathrm{d}x_1 \cdots \mathrm{d}x_{j-1}\, \mathrm{d}x_{j+1} \cdots \mathrm{d}x_n.$$

Then

$$\int_Q u(x) \frac{\partial v}{\partial x_j}(x)\ \mathrm{d}x = \int_{Q_{\widehat{j}}} [u(x)v(x)]_{x_j=a_j}^{x_j=b_j}\ \mathrm{d}x_{\widehat{j}} - \int_Q v(x) \frac{\partial u}{\partial x_j}(x)\ \mathrm{d}x.$$

In the middle integral, the coordinates other than x_j are held fixed while the bracket is evaluated at the two endpoints.

Proof. For fixed $x_{\widehat{j}} \in Q_{\widehat{j}}$, apply Theorem 9.19 to the C^1 functions of x_j obtained from u and v. The resulting identity has continuous dependence on $x_{\widehat{j}}$. Integrating it over $Q_{\widehat{j}}$ and using Theorem 12.7 gives the displayed formula. □

Remark 12.10 (Rectangular and intrinsic integration by parts)**.** The preceding formula uses only one-variable integration by parts and Fubini, so it is naturally a statement about rectangles and their coordinate faces. At this point no intrinsic integral over a general boundary has been developed. Chapter 13 replaces the coordinate-face term by an integral over the oriented boundary of a manifold, obtained from the graded Leibniz rule and Stokes' theorem.

12.3 Uniform Limits and Multiple Integration

The same principle from Chapter 9 survives unchanged on a rectangle: a uniform error in the integrand is at most that error times the volume of the domain after integration. Pointwise convergence alone is not the Riemann theory's general interchange principle.

Theorem 12.11 (Uniform interchange for multiple Riemann integrals)**.** Let $Q \subseteq \mathbb{R}^n$ be a rectangle. Suppose that every function

$$f_k : Q \longrightarrow \mathbb{R}$$

is Riemann integrable and that f_k converges uniformly to f on Q. Then f is Riemann integrable and

$$\lim_{k\to\infty} \int_Q f_k(x)\ \mathrm{d}x = \int_Q f(x)\ \mathrm{d}x.$$

Moreover,

$$\lim_{k\to\infty} \int_Q |f_k(x) - f(x)|\ \mathrm{d}x = 0.$$

Proof. The oscillation argument in the proof of Theorem 9.9 applies to grid subrectangles, using Proposition 12.3. Once integrability of f is known, Theorem 12.6 gives

$$\left| \int_Q f_k\ \mathrm{d}x - \int_Q f\ \mathrm{d}x \right| \le \int_Q |f_k - f|\ \mathrm{d}x \le \mathrm{vol}(Q) \sup_{x\in Q} |f_k(x) - f(x)|.$$

The right-hand side tends to zero, and the same estimate gives the final claim. □

Remark 12.12 (Domination in several variables)**.** If the f_k above are also dominated by a nonnegative Riemann-integrable function, that domination is compatible with the conclusion but does not replace uniform convergence in this elementary Riemann theorem. It is included only to foreshadow the multivariable Lebesgue Dominated Convergence Theorem, which replaces uniform convergence with appropriate almost-everywhere convergence in a measure-theoretic setting.

As in one variable, pointwise convergence alone does not control a Riemann integral. On a compact rectangle, continuity and monotonicity provide the extra structure that can force uniform convergence.

Theorem 12.13 (Dini's theorem)**.** Let $Q \subseteq \mathbb{R}^n$ be a rectangle, and let

$$f_k : Q \longrightarrow \mathbb{R}$$

be continuous functions satisfying

$$f_k(x) \le f_{k+1}(x) \qquad (x \in Q,\ k \in \mathbb{N}).$$

If f_k converges pointwise on Q to a continuous function f, then the convergence is uniform and

$$\lim_{k\to\infty} \int_Q f_k(x)\ \mathrm{d}x = \int_Q f(x)\ \mathrm{d}x.$$

The same conclusion holds for a pointwise decreasing sequence.

Proof. Put $h_k = f - f_k$. If h_k did not converge uniformly to zero, there would be $\varepsilon > 0$ and $x_k \in Q$ with

$$h_k(x_k) \ge \varepsilon.$$

By compactness of Q and Theorem 10.22, some subsequence x_{k_j} tends to a point $x \in Q$. For every fixed m, monotonicity gives

$$h_m(x_{k_j}) \ge h_{k_j}(x_{k_j}) \ge \varepsilon$$

for all sufficiently large j. Continuity yields $h_m(x) \ge \varepsilon$ for every m, contradicting $h_m(x) \to 0$. Thus the convergence is uniform, and Theorem 12.11 gives the formula. The decreasing case follows by negation. □

Remark 12.14 (Dini versus Lebesgue monotone convergence)**.** This is Dini's theorem on a compact rectangle: continuity, compactness, and monotonicity upgrade pointwise convergence to uniform convergence, after which Theorem 12.11 gives the integral formula. The full multivariable Lebesgue Monotone Convergence Theorem instead assumes nonnegative measurable functions and does not require continuity or uniform convergence; it lies beyond the present Riemann framework.

Rectangles already have a natural volume. To assign a volume to a more general bounded set E, view the set through its indicator function $\mathbf{1}_E$: integrating this function records the amount of space occupied by E. Riemann integration succeeds when rectangular approximations can resolve the boundary well enough; this leads directly to the following definition.

Definition 12.15 (Jordan measurability)**.** Let E be a bounded subset of $\mathbb{R}^n$ and choose a rectangle Q containing E. The set E is **Jordan measurable** if its indicator

$$\mathbf{1}_E(x) = \begin{cases} 1, & x \in E, \\ 0, & x \notin E \end{cases}$$

is Riemann integrable on Q. Its **Jordan volume** is

$$\mathrm{vol}(E) = \int_Q \mathbf{1}_E(x)\ \mathrm{d}x.$$

Definition 12.16 (Jordan-null set)**.** A bounded set $N \subseteq \mathbb{R}^n$ is **Jordan-null** if, for every

$$\varepsilon > 0,$$

it can be covered by finitely many rectangles whose total volume is less than ε.

Proposition 12.17 (Boundary criterion for Jordan measurability)**.** A bounded set $E \subseteq \mathbb{R}^n$ is Jordan measurable if and only if its boundary is Jordan-null.

Proof. Let Q be a rectangle containing E. If ∂E is covered by rectangles of total volume less than ε, refine their faces and the faces of Q to a common grid. Every grid cell not meeting the cover lies either in the interior of E or in the interior of its complement, so the upper and lower sums of $\mathbf{1}_E$ differ only on the covering cells. Their gap is at most the total covering volume. Thus $\mathbf{1}_E$ is integrable.

Conversely, apply the Darboux criterion to $\mathbf{1}_E$. On every subrectangle which meets both E and its complement, its oscillation is one. The union of these finitely many cells covers ∂E, apart from grid faces; enlarge those faces by arbitrarily thin rectangles. The Darboux gap and the enlargement may be made smaller than any prescribed positive number, proving that ∂E is Jordan-null. □

Proposition 12.18 (Continuous functions on Jordan sets)**.** Let $E \subseteq \mathbb{R}^n$ be bounded with Jordan-null boundary, and let

$$f : E \longrightarrow \mathbb{R}$$

be bounded and continuous for the relative topology on E. Then its zero extension to a containing rectangle is Riemann integrable.

Proof. Let M bound $|f|$. Cover ∂E by finitely many open rectangles of arbitrarily small total volume. Every point of E outside that cover belongs to E°, so relative continuity supplies a neighborhood on which the oscillation of f is as small as prescribed. The compact part of E outside the cover has a finite such subcover. A grid refining the boundary-cover rectangles and these finitely many neighborhoods has arbitrarily small oscillation on every cell not meeting the boundary cover. The covering cells contribute at most $2M$ times their total volume. The Darboux criterion proves integrability. □

Proposition 12.19 (Independence and elementary Jordan sets)**.** The Jordan volume is independent of the containing rectangle. Every rectangle, and every finite union of rectangles with pairwise disjoint interiors, is Jordan measurable; its volume is the sum of the volumes of its rectangles.

Proof. Extending an indicator by zero from one containing rectangle to a larger one changes no upper or lower sum except on a finite family of adjoining rectangles, where the function is identically zero. Common refinements therefore give the same integral.

The indicator of a rectangle is constant except on its finitely many faces. Given

$$\varepsilon > 0,$$

cover these faces by finitely many thin rectangles of total volume less than

$$\varepsilon.$$

Outside that cover, use a grid refining the rectangle's faces; there the upper and lower sums agree. The Darboux criterion proves integrability. The same argument works for a finite union, and additivity gives the volume formula. □

Proposition 12.20 (Null modifications and finite additivity)**.** Let $N \subseteq \mathbb{R}^n$ be bounded and Jordan-null.

(a) The set N is Jordan measurable and has volume zero.

(b) A bounded function supported on N, extended by zero to a containing rectangle, is Riemann integrable with integral zero. Consequently, changing a bounded Riemann-integrable function on N does not change its integral.

(c) Let $E_1, \dots, E_m$ be Jordan measurable and suppose every $E_i \cap E_j$, $i \neq j$, is Jordan-null. Then their union is Jordan measurable and

$$\operatorname{vol}\left(\bigcup_{i=1}^{m} E_i\right) = \sum_{i=1}^{m} \operatorname{vol}(E_i).$$

More generally, if h is bounded on the union and the zero extension of $h|_{E_i}$ is integrable for every i, then the zero extension of h from the union is integrable and

$$\int_{\bigcup_i E_i} h = \sum_{i=1}^{m} \int_{E_i} h.$$

Proof. Given $\varepsilon > 0$, cover N by finitely many rectangles of total volume less than ε, and refine their faces to a grid in a containing rectangle. The indicator of N vanishes on every grid cell outside the cover, so its Darboux gap is at most the covering volume. Hence N is Jordan measurable and has volume zero.

If $|h| \leq M$ and h is supported on N, then

$$|h| \leq M\mathbf{1}_N.$$

The same grid argument proves integrability, and order preservation gives zero integral. The difference of two functions which agree off N is of this form, proving the modification assertion.

Put

$$N_0 = \bigcup_{i<j} (E_i \cap E_j).$$

If $m \leq 1$, there is nothing to prove. If $m \geq 2$, this finite union is Jordan-null: combine finite covers whose total volumes are each less than $\varepsilon/\binom{m}{2}$. Away from N_0,

$$\mathbf{1}_{\cup_i E_i} = \sum_i \mathbf{1}_{E_i}.$$

The right side is integrable, so the null-modification result proves integrability of the left side and the volume formula. The same argument applied to the zero extensions of $h|_{E_i}$ proves the final assertion. $\square$

Definition 12.21 (Integration over a Jordan set)**.** Let $E \subseteq \mathbb{R}^n$ be Jordan measurable and let $f : E \to \mathbb{R}$ be bounded. If the extension

$$\widetilde{f}(x) = \begin{cases} f(x), & x \in E, \\ 0, & x \notin E \end{cases}$$

is integrable on a containing rectangle Q, define

$$\int_E f(x)\, \mathrm{d}x = \int_Q \widetilde{f}(x)\, \mathrm{d}x.$$

Remark 12.22. This definition is intentionally more restrictive than later Lebesgue integration. A bounded function on a Jordan-measurable set need not have an integrable zero extension. The qualification in the definition is therefore mathematical content, not merely a technicality.

12.4 Linear Changes of Variables

The one-variable substitution rule says that $|\phi'(t)|$ measures the local stretching of length. In several variables, the corresponding linear fact comes first: an invertible linear map L sends a small box to a parallelepiped and scales its volume by $|\det L|$. A coordinate scaling by c has factor $|c|$, a diagonal map has the product of its coordinate scalings, a shear has factor 1, and an orthogonal map preserves volume. Thus the determinant is the natural volume-scaling number, not a formal correction inserted after the fact.

Example 12.23 (A rectangle becomes a parallelogram)**.** Let

$$L(s,t) = (2s+t, s+3t).$$

The unit square is sent to the parallelogram spanned by $(2,1)$ and $(1,3)$. Since

$$\det L = \det \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} = 5,$$

its area is 5. More generally, the image of a rectangle of area A has area $5A$. The linear change-of-variables theorem below turns this geometric calculation into an integral formula.

Lemma 12.24 (Volume under an invertible linear map)**.** Let

$$L : \mathbb{R}^n \longrightarrow \mathbb{R}^n$$

be invertible and let Q be a rectangle. Then $L(Q)$ is Jordan measurable and

$$\operatorname{vol}(L(Q)) = |\det L| \operatorname{vol}(Q).$$

More generally, if $E \subseteq \mathbb{R}^n$ is Jordan measurable, then $L(E)$ is Jordan measurable and

$$\operatorname{vol}(L(E)) = |\det L| \operatorname{vol}(E).$$

Proof. We first record the effect of the three elementary invertible linear maps on Jordan volume. A coordinate permutation merely permutes the side intervals of a rectangle and preserves its volume. Multiplication of one coordinate by $c \neq 0$ takes a rectangle to a rectangle and multiplies its volume by $|c|$.

For the remaining operation, fix $i \neq j$ and consider the shear

$$S(x)_i = x_i + cx_j, \qquad S(x)_k = x_k \quad (k \neq i).$$

Let $R = \prod_k [a_k, b_k]$, put $d_i = b_i - a_i$, and write z for the coordinates other than x_i. Above $z \in R_{\widehat{i}}$, the x_i-fiber of $S(R)$ is

$$[\,\ell(z), \ell(z) + d_i\,], \qquad \ell(z) = a_i + cz_j.$$

Partition $R_{\widehat{i}}$ into finitely many rectangles P on each of which the oscillation $\omega_P(\ell)$ is less than a prescribed γ, where $0 < \gamma < d_i$. Over each P, the fibered set is contained in the rectangular prism

$$P \times [\,\inf_P \ell,\ d_i + \sup_P \ell\,]$$

and contains the prism
$$P \times [\sup_P \ell,\ d_i + \inf_P \ell].$$

The outer and inner prisms have respective fiber lengths

$$d_i + \omega_P(\ell) \qquad \text{and} \qquad d_i - \omega_P(\ell).$$

After summing over P, their total volume difference is at most

$$2\gamma \operatorname{vol}(R_{\widehat{i}}).$$

The upper and lower Jordan integrals of $\mathbf{1}_{S(R)}$ are therefore squeezed to $\operatorname{vol}(R)$ as $\gamma \to 0$. Hence $S(R)$ is Jordan measurable and
$$\operatorname{vol}(S(R)) = \operatorname{vol}(R).$$

These conclusions extend from rectangles to every Jordan-measurable set. Indeed, for such a set E, the Darboux criterion applied to $\mathbf{1}_E$ gives finite unions I and O of grid rectangles with pairwise disjoint interiors such that

$$I \subseteq E \subseteq O, \qquad \operatorname{vol}(O) - \operatorname{vol}(I) < \varepsilon.$$

Images of distinct grid rectangles under an elementary map overlap only in images of their common faces. If M is the operator norm of the elementary map, a fixed face can be covered by at most $C_0 h^{-(n-1)}$ face-cubes of side at most h. Each image face-cube lies in an axis-parallel box of side $2M\sqrt{n}\,h$, so these boxes have total volume at most

$$C_0 (2M\sqrt{n})^n h,$$

which tends to zero with h. Thus these overlaps are Jordan-null, and Proposition 12.20 shows that the volumes of the images of I and O are the sums of the volumes of their pieces. The inclusions

$$S(I) \subseteq S(E) \subseteq S(O)$$

then make the upper and lower Jordan integrals of $\mathbf{1}_{S(E)}$ differ by less than ε. The same argument applies to coordinate permutations and coordinate scalings. Consequently, every elementary map sends Jordan-measurable sets to Jordan-measurable sets and multiplies their volumes by the absolute value of its determinant.

Let $K = [0,1]^n$. Gaussian elimination supplies elementary matrices $E_1, \dots, E_k$ such that

$$E_k \cdots E_1 L = I.$$

Applying the preceding volume rule successively to the intermediate Jordan sets gives

$$1 = \operatorname{vol}(K) = \left(\prod_{\nu=1}^{k} |\det E_\nu| \right) \operatorname{vol}(L(K)).$$

Taking determinants in the elimination identity gives

$$\left(\prod_{\nu=1}^{k} |\det E_\nu| \right) |\det L| = 1.$$

Combining the two equalities yields

$$\operatorname{vol}(L(K)) = |\det L|.$$

The same calculation applies to an arbitrary Jordan-measurable set E. The elementary-map result already proved ensures that every intermediate image is Jordan measurable, and successive application of that result gives

$$\operatorname{vol}(E) = \left(\prod_{\nu=1}^{k} |\det E_\nu|\right) \operatorname{vol}(L(E)).$$

The determinant identity therefore gives

$$\operatorname{vol}(L(E)) = |\det L| \operatorname{vol}(E).$$

For a general rectangle $Q = t + DK$, where D is diagonal with positive diagonal entries equal to the side lengths of Q, translation preserves Jordan volume and

$$L(Q) = Lt + (LD)(K).$$

Thus

$$\operatorname{vol}(L(Q)) = |\det(LD)| = |\det L| \operatorname{vol}(Q).$$

For completeness, measurability can also be read directly from the boundary. Since L is a homeomorphism,

$$\partial L(Q) = L(\partial Q).$$

Each face of Q can be covered by at most $C_0 h^{-(n-1)}$ face-cubes of side at most h. If $M = \|L\|_{\mathrm{op}}$, the image of each such piece has diameter at most $M\sqrt{n}\,h$, and hence lies in an axis-parallel n-box of side $2M\sqrt{n}\,h$. The total volume of these boxes is at most

$$C_0(2M\sqrt{n})^n h,$$

which tends to zero. A finite union over the faces proves that $\partial L(Q)$ is Jordan-null, in agreement with Proposition 12.17. ☐

Theorem 12.25 (Linear change of variables)**.** Let $L : \mathbb{R}^n \to \mathbb{R}^n$ be invertible, let E be Jordan measurable, and let

$$f : L(E) \longrightarrow \mathbb{R}$$

have an integrable zero extension. Then $L(E)$ is Jordan measurable and

$$\int_{L(E)} f(y)\, \mathrm{d}y = \int_E f(Lx)|\det L|\, \mathrm{d}x.$$

Proof. Let R be a rectangle and let

$$s = \sum_{\nu=1}^{N} c_\nu \mathbf{1}_{R_\nu}$$

be a rectangular step function obtained from a grid partition of R. The sets $L(R_\nu)$ are Jordan measurable by Lemma 12.24. Two of them intersect only in the image of a common grid face, which is Jordan-null by the quantitative face estimate in that lemma. Hence Proposition 12.20 gives

$$\int_{L(R)} s(L^{-1}y)\, \mathrm{d}y = \sum_{\nu=1}^{N} c_\nu \operatorname{vol}(L(R_\nu)) = |\det L| \int_R s(x)\, \mathrm{d}x.$$

Thus the formula is established first for rectangular step functions; their transforms are finite sums of indicators of the Jordan sets $L(R_\nu)$, not rectangular step functions in the y-coordinates.

Now let $h : R \to \mathbb{R}$ be Riemann integrable. For every $\varepsilon > 0$, the Darboux criterion supplies lower and upper rectangular step functions ℓ and u satisfying

$$\ell \leq h \leq u, \qquad \int_R (u - \ell) < \varepsilon.$$

Their transforms are integrable on $L(R)$, and

$$\ell(L^{-1}y) \leq h(L^{-1}y) \leq u(L^{-1}y).$$

The gap between the transformed upper and lower integrals is

$$|\det L| \int_R (u - \ell) < |\det L|\varepsilon.$$

To make the integrability conclusion explicit, choose Darboux partitions for the two transformed integrable step functions whose upper–lower gaps have sum less than an arbitrary $\delta > 0$, and pass to a common refinement. On that refinement, the upper sum of $h \circ L^{-1}$ is at most the upper sum of $u \circ L^{-1}$, while its lower sum is at least the lower sum of $\ell \circ L^{-1}$. Its Darboux gap is consequently less than

$$|\det L|\varepsilon + \delta.$$

Since ε and δ are arbitrary, $h \circ L^{-1}$ is integrable on $L(R)$. Squeezing its integral between the two transformed step-function integrals proves

$$\int_{L(R)} h(L^{-1}y)\ \mathrm{d}y = |\det L| \int_R h(x)\ \mathrm{d}x.$$

All integrations here may be read through zero extensions to containing rectangles, as in the definition of integration over a Jordan set.

Apply the same lower–upper construction to $\mathbf{1}_E$ on a rectangle containing E. The transformed indicators squeeze $\mathbf{1}_{L(E)}$, with integral gap multiplied by $|\det L|$. Hence $L(E)$ is Jordan measurable before any integral over it is used.

Finally, choose a rectangle P containing $L(E)$, and let $\widetilde{f}$ be the integrable zero extension of f to P. Apply the preceding argument to L^{-1} and $\widetilde{f}$. It proves that

$$x \longmapsto \widetilde{f}(Lx)$$

has an integrable zero extension from $L^{-1}(P)$, and

$$\int_P \widetilde{f}(y)\ \mathrm{d}y = |\det L| \int_{L^{-1}(P)} \widetilde{f}(Lx)\ \mathrm{d}x.$$

The integrands vanish, respectively, outside $L(E)$ and outside E. Using the definition of integration over Jordan sets therefore turns this identity into the asserted formula. □

12.5 Nonlinear Changes of Variables

For a differentiable nonlinear map,

$$\Phi(a + h) = \Phi(a) + D\Phi(a)h + o(\|h\|).$$

Consequently, a sufficiently small box near a behaves approximately like the parallelepiped produced by the linear map $D\Phi(a)$. The progression is therefore

$$\text{linear volume scaling} \longrightarrow \text{local linearization} \longrightarrow |\det D\Phi(a)| \text{ as local volume distortion.}$$

The following lemma makes this geometric principle quantitative; the theorem then adds the Riemann-sum bookkeeping needed to integrate over a whole region.

Lemma 12.26 (Local volume comparison on cubes)**.** Let $U \subseteq \mathbb{R}^n$ be open, let

$$\Phi : U \longrightarrow \mathbb{R}^n$$

be of class C^1, and suppose that $D\Phi(a)$ is invertible. For every

$$\varepsilon > 0,$$

there is a neighborhood N of a such that, for every closed cube

$$C \subseteq N$$

on which Φ is injective,

$$|\mathrm{vol}(\Phi(C)) - |\det D\Phi(a)|\,\mathrm{vol}(C)| \leq \varepsilon\,\mathrm{vol}(C).$$

Moreover, $\Phi(C)$ is Jordan measurable.

Proof. Put

$$A = D\Phi(a).$$

Choose $\delta > 0$ so small that

$$0 < \delta < \min\left\{1, \frac{1}{\sqrt{n}}\right\}$$

and

$$|\det A| \max\left\{(1+\delta\sqrt{n})^n - 1,\, 1 - (1-\delta\sqrt{n})^n\right\} < \varepsilon.$$

Continuity of $D\Phi$ permits us to choose an open convex ball N about a, with closure contained in U, such that

$$\|A^{-1}D\Phi(x) - I\|_{\mathrm{op}} < \delta \qquad (x \in N).$$

Let $C = c + sK$, where

$$K = [-1/2, 1/2]^n.$$

Define

$$\Psi(u) = s^{-1}A^{-1}\big(\Phi(c+su) - \Phi(c)\big) \qquad (u \in K).$$

Because $C \subseteq N$, every segment used below lies in C, hence in N. The chain rule gives

$$D\Psi(u) = A^{-1}D\Phi(c+su).$$

Put $g(u) = \Psi(u) - u$. Then

$$\Psi(u) = u + g(u), \qquad \|Dg(u)\|_{\mathrm{op}} < \delta.$$

The mean-value estimate in the Euclidean norm gives

$$\|g(u) - g(v)\|_2 \leq \delta\|u - v\|_2.$$

Since

$$\|w\|_\infty \le \|w\|_2 \le \sqrt{n}\,\|w\|_\infty,$$

we also have

$$\|g(u) - g(v)\|_\infty \le \delta\sqrt{n}\,\|u - v\|_\infty.$$

Because $\Psi(0) = 0$, one has $g(0) = 0$, and hence

$$\|g(u)\|_\infty \le \frac{\delta\sqrt{n}}{2} \qquad (u \in K).$$

It follows that

$$\Psi(K) \subseteq \left\{ y : \|y\|_\infty \le \frac{1 + \delta\sqrt{n}}{2} \right\}.$$

Conversely, suppose

$$\|y\|_\infty \le \frac{1 - \delta\sqrt{n}}{2}$$

and define $T_y(u) = y - g(u)$. For $u \in K$,

$$\|T_y(u)\|_\infty \le \|y\|_\infty + \|g(u)\|_\infty \le \frac{1}{2},$$

so $T_y(K) \subseteq K$. Moreover,

$$\|T_y(u) - T_y(v)\|_2 \le \delta\|u - v\|_2.$$

Thus T_y is a contraction in the Euclidean norm. The closed cube K is complete, so Theorem 11.23 gives a fixed point $u \in K$, and the fixed-point equation is $\Psi(u) = y$. Therefore

$$\left\{ y : \|y\|_\infty \le \frac{1 - \delta\sqrt{n}}{2} \right\} \subseteq \Psi(K).$$

The estimate

$$\|\Psi(u) - \Psi(v)\|_2 \ge (1 - \delta)\|u - v\|_2$$

also proves directly that Ψ, and hence Φ, is injective on C.

We next justify Jordan measurability. For $x \in N$,

$$\|I - A^{-1}D\Phi(x)\|_{\mathrm{op}} < 1,$$

so Proposition 11.25, with invertible map A and perturbed map $D\Phi(x)$, shows that $D\Phi(x)$ is invertible. The continuous bijection $\Phi|_C : C \to \Phi(C)$ is a homeomorphism: a closed subset of the compact set C is compact, its image is compact, and compact subsets of the Hausdorff space $\mathbb{R}^n$ are closed. If $x \in C^\circ$, the inverse function theorem gives a neighborhood $O \subseteq C^\circ$ for which $\Phi(O)$ is open. Thus $\Phi(x)$ is interior to $\Phi(C)$. Since $\Phi(C)$ is compact and therefore closed, this proves

$$\partial\Phi(C) \subseteq \Phi(\partial C).$$

Because $\overline{N}$ is compact and contained in U, continuity of $D\Phi$ gives

$$L = \max_{x \in \overline{N}} \|D\Phi(x)\|_{\mathrm{op}} < \infty.$$

The mean-value estimate makes Φ L-Lipschitz on N. Each face of C can be covered by at most

$$C_0(s/h)^{n-1}$$

face-cubes of side at most h. The image of each lies in an axis-parallel box of side $2L\sqrt{n}\,h$, so the total volume of these boxes is at most

$$C_0(2L\sqrt{n})^n s^{n-1}h.$$

Letting $h \to 0$ proves that $\Phi(\partial C)$ is Jordan-null, and Proposition 12.17 shows that $\Phi(C)$ is Jordan measurable.

Finally,

$$\Phi(C) = \Phi(c) + sA\Psi(K).$$

Translation invariance and Lemma 12.24 give

$$\operatorname{vol}(\Phi(C)) = |\det A| s^n \operatorname{vol}(\Psi(K)).$$

The inner and outer cube containments, together with $\operatorname{vol}(C) = s^n$, yield

$$|\det A|(1-\delta\sqrt{n})^n \operatorname{vol}(C) \le \operatorname{vol}(\Phi(C)) \le |\det A|(1+\delta\sqrt{n})^n \operatorname{vol}(C).$$

The choice of δ proves the estimate. □

Theorem 12.27 (Change of variables)**.** Let $Q \subseteq \mathbb{R}^n$ be a rectangle, let U be an open neighborhood of Q, and let

$$\Phi : U \longrightarrow \mathbb{R}^n$$

be of class C^1. Assume that Φ is injective on Q and that

$$\det D\Phi(x) \neq 0 \qquad (x \in Q).$$

Then $\Phi(Q)$ is Jordan measurable. If

$$f : \Phi(Q) \longrightarrow \mathbb{R}$$

is continuous, then

$$\int_{\Phi(Q)} f(y)\ \mathrm{d}y = \int_Q f(\Phi(x))|\det D\Phi(x)|\ \mathrm{d}x.$$

Proof. Write

$$J(x) = |\det D\Phi(x)|.$$

The restriction $\Phi|_Q : Q \to \Phi(Q)$ is a homeomorphism. Indeed, it is a continuous bijection, Q is compact, and $\Phi(Q) \subseteq \mathbb{R}^n$ is Hausdorff. More explicitly, a closed subset of Q is compact, its image is compact, and compact subsets of $\mathbb{R}^n$ are closed; hence the inverse is continuous. If $x \in Q^\circ$, the inverse function theorem gives an open neighborhood $O \subseteq Q^\circ$ such that $\Phi(O)$ is open. Therefore $\Phi(x)$ is interior to $\Phi(Q)$, and

$$\partial\Phi(Q) \subseteq \Phi(\partial Q).$$

We shall use one uniform Lipschitz constant. Since Q is compact and U is open, there is $\tau > 0$ such that the closed τ-neighborhood of Q lies in U. Put

$$K_\tau = \{x \in \mathbb{R}^n : \operatorname{dist}(x, Q) \le \tau\}.$$

This compact set is contained in U. The open τ-neighborhood W is convex because Q is a rectangle, and $W \subseteq K_\tau$. Put

$$L = \max_{x \in K_\tau} \|D\Phi(x)\|_{\mathrm{op}}.$$

Then Theorem 11.16 shows that Φ is L-Lipschitz on W. Every face of Q can be covered by at most $C_0 h^{-(n-1)}$ face-cubes of side at most h, where C_0 is independent of h. Their images lie in n-boxes of side $2L\sqrt{n}\,h$, whose total volume is at most

$$C_0(2L\sqrt{n})^n h.$$

After summing over the finitely many faces and letting $h \to 0$, $\Phi(\partial Q)$ is Jordan-null. By the boundary inclusion just proved and Proposition 12.17, $\Phi(Q)$ is Jordan measurable before any integral over it is used.

We also need a uniform estimate for images of thin rectangles. If $R \subseteq Q$ is a nondegenerate rectangle, cover it, for arbitrarily small $d > 0$, by grid cubes of side d. Taking d smaller than every side length of R and smaller than $\tau/\sqrt{n}$, the covering cubes lie in W, and the sum of their volumes is at most

$$2^n \operatorname{vol}(R).$$

Each image cube lies in an axis-parallel box of side $2L\sqrt{n}\,d$. The same boundary argument as above shows that $\Phi(R)$ is Jordan measurable, and the covering estimate gives

$$\operatorname{vol}(\Phi(R)) \le C_L \operatorname{vol}(R), \qquad C_L = 2^n(2L\sqrt{n})^n.$$

Degenerate rectangles have Jordan-null images by the face estimate.

The functions J and

$$H(x) = f(\Phi(x))J(x)$$

are continuous on Q. Let

$$B = \max_{y\in\Phi(Q)} |f(y)|, \qquad M_H = \max_{x\in Q}|H(x)|.$$

For $t \ge 0$, let $\omega_f(t)$ and $\omega_H(t)$ be the suprema of $|f(y) - f(z)|$ and $|H(x) - H(w)|$, respectively, over pairs at distance at most t in their compact domains. Uniform continuity gives

$$\lim_{t\to 0} \omega_f(t) = 0, \qquad \lim_{t\to 0} \omega_H(t) = 0.$$

Fix $\varepsilon > 0$. Choose $\eta > 0$ so small that

$$B\eta \operatorname{vol}(Q) < \frac{\varepsilon}{4}.$$

If $B = 0$, any positive η satisfies this requirement. For each $a \in Q$, apply Lemma 12.26 with error $\eta/2$, and then shrink its neighborhood $N_a \subseteq W$ so that

$$|J(x) - J(a)| < \frac{\eta}{2} \qquad (x \in N_a \cap Q).$$

Choose finitely many of the sets $N_a \cap Q$ which cover Q. By Lemma 10.23, there is $\lambda > 0$ such that every subset of Q of diameter less than λ lies in one member of this finite cover. Hence, if $C \subseteq Q$ is a cube with $\operatorname{diam}(C) < \lambda$, and $x_C \in C$, then for a suitable center a,

$$\begin{aligned} |\operatorname{vol}(\Phi(C)) - J(x_C)\operatorname{vol}(C)| &\le |\operatorname{vol}(\Phi(C)) - J(a)\operatorname{vol}(C)| \\ &\quad + |J(a) - J(x_C)|\operatorname{vol}(C) \\ &\le \eta \operatorname{vol}(C). \end{aligned}$$

Here Φ is injective on C because it is injective on Q.

Write

$$Q = \prod_{i=1}^{n} [a_i, b_i], \qquad d_i = b_i - a_i.$$

Choose $h > 0$, smaller than every d_i, and put

$$m_i = \left\lfloor \frac{d_i}{h} \right\rfloor, \qquad Q_h = \prod_{i=1}^{n} [a_i, a_i + m_i h].$$

The rectangle Q_h is the union of finitely many closed cubes of side h, called the retained cubes. For $1 \le i \le n$, set

$$R_{h,i} = \left(\prod_{k<i} [a_k, a_k + m_k h] \right) \times [a_i + m_i h, b_i] \times \left(\prod_{k>i} [a_k, b_k] \right),$$

and put

$$R_h = \bigcup_{i=1}^{n} R_{h,i}.$$

Thus R_h is the closure of $Q \setminus Q_h$. The rectangles $R_{h,i}$ have pairwise Jordan-null overlaps: two distinct pieces can meet only where a coordinate equals an endpoint of a shortened interval. A telescoping product estimate gives

$$\operatorname{vol}(R_h) = \operatorname{vol}(Q) - \operatorname{vol}(Q_h) \le C_Q h, \qquad C_Q = \sum_{i=1}^{n} \prod_{k \ne i} d_k.$$

By the rectangle-image bound, finite additivity, and the nullity of the overlaps,

$$\operatorname{vol}(\Phi(R_h)) \le C_L \operatorname{vol}(R_h).$$

Choose h small enough that

$$\sqrt{n}\, h < \lambda,$$
$$\omega_f(L\sqrt{n}\, h) \operatorname{vol}(\Phi(Q)) < \frac{\varepsilon}{4}, \qquad \omega_H(\sqrt{n}\, h) \operatorname{vol}(Q) < \frac{\varepsilon}{4},$$

and

$$(BC_L + M_H) C_Q h < \frac{\varepsilon}{4}.$$

For each retained cube C, choose a tag $x_C \in C$ and put $J_C = J(x_C)$. Then

$$\begin{aligned}
&\int_{\Phi(C)} f(y)\, \mathrm{d}y - f(\Phi(x_C)) J_C \operatorname{vol}(C) \\
&\quad = \int_{\Phi(C)} \big(f(y) - f(\Phi(x_C))\big)\, \mathrm{d}y \\
&\qquad + f(\Phi(x_C)) \big(\operatorname{vol}(\Phi(C)) - J_C \operatorname{vol}(C)\big).
\end{aligned}$$

Since

$$\operatorname{diam}(\Phi(C)) \le L \operatorname{diam}(C) \le L\sqrt{n}\, h,$$

the bounded-function estimate and the local tag estimate above imply

$$\begin{aligned}
&\left| \int_{\Phi(C)} f(y)\, \mathrm{d}y - f(\Phi(x_C)) J_C \operatorname{vol}(C) \right| \\
&\qquad \le \omega_f(L\sqrt{n}\, h) \operatorname{vol}(\Phi(C)) + B\eta \operatorname{vol}(C).
\end{aligned}$$

Distinct retained cubes meet only on faces. Injectivity gives

$$\Phi(C) \cap \Phi(C') = \Phi(C \cap C'),$$

and the Lipschitz face estimate makes this intersection Jordan-null. Therefore Proposition 12.20 gives

$$\int_{\Phi(Q_h)} f = \sum_C \int_{\Phi(C)} f, \qquad \operatorname{vol}(\Phi(Q_h)) = \sum_C \operatorname{vol}(\Phi(C)).$$

Summing this local estimate yields

$$\begin{aligned}\left| \int_{\Phi(Q_h)} f - \sum_C f(\Phi(x_C)) J_C \operatorname{vol}(C) \right| \\ \le \omega_f(L\sqrt{n}\, h) \operatorname{vol}(\Phi(Q)) + B\eta \operatorname{vol}(Q).\end{aligned}$$

Similarly, continuity of H on each retained cube gives

$$\left| \sum_C H(x_C) \operatorname{vol}(C) - \int_{Q_h} H(x) \, \mathrm{d}x \right| \le \omega_H(\sqrt{n}\, h) \operatorname{vol}(Q).$$

Finally, $Q = Q_h \cup R_h$, and their overlap is Jordan-null. Their images also have Jordan-null overlap, by injectivity and the Lipschitz face estimate. Finite additivity and the bounded-function estimate give

$$\left| \int_{\Phi(R_h)} f \right| \le B \operatorname{vol}(\Phi(R_h)) \le BC_L \operatorname{vol}(R_h)$$

and

$$\left| \int_{R_h} H \right| \le M_H \operatorname{vol}(R_h).$$

Combining these two estimates with the summed local estimate and the Riemann-sum estimate, and observing that $H(x_C) = f(\Phi(x_C))J_C$, gives

$$\begin{aligned}\left| \int_{\Phi(Q)} f - \int_Q H \right| &\le \omega_f(L\sqrt{n}\, h) \operatorname{vol}(\Phi(Q)) + B\eta \operatorname{vol}(Q) \\ &\quad + \omega_H(\sqrt{n}\, h) \operatorname{vol}(Q) + (BC_L + M_H) \operatorname{vol}(R_h) \\ &< \varepsilon\end{aligned}$$

by the choices of η and h, together with the strip-volume bound. Since $\varepsilon > 0$ was arbitrary, the change-of-variables formula follows. □

Remark 12.28 (A working strategy for multiple integrals)**.** First identify the domain. On a rectangle, iterated integration is often the simplest computational tool. For a nonrectangular region, ask whether it decomposes into simple pieces or suggests coordinates adapted to its geometry. For a proposed change of variables, compute $D\Phi$ and $|\det D\Phi|$, transform both the integrand and the domain, and verify the injectivity and nonvanishing-Jacobian hypotheses (allowing only carefully controlled boundary exceptions). Keep two tasks distinct: continuity, Darboux estimates, and Jordan measurability address existence of an integral; iterated integration and change of variables usually address its calculation.

12.5.1 Alternative Viewpoint: Pullbacks and Manifold Integration

The proof of Theorem 12.27 is necessarily detailed because it establishes an analytic fact in the Riemann/Jordan setting. Its geometric core is simple. Differentiability gives

$$\Phi(x+h) = \Phi(x) + D\Phi(x)h + o(\|h\|),$$

so $D\Phi(x)$ is the infinitesimal linear approximation to Φ. For a linear map, $\det L$ records oriented volume scaling and $|\det L|$ records unsigned volume scaling; see the determinant properties in Theorem 10.49 and the linear volume result Lemma 12.24. Thus

$$|\det D\Phi(x)|$$

is not an arbitrary correction factor: it is the local unsigned volume distortion of the derivative. The proof above makes this statement quantitative by comparing small cubes with their linearized images and then passing to Riemann sums.

Accordingly, the scalar formula just proved has the schematic form

$$\int_V f(y)\ \mathrm{d}y = \int_U f(\Phi(x))|\det D\Phi(x)|\ \mathrm{d}x.$$

The absolute value occurs because ordinary scalar volume is unsigned. Chapter 13 gives a complementary interpretation. For a top-degree form on V,

$$\omega = f(y)\ \mathrm{d}y_1 \wedge \cdots \wedge \mathrm{d}y_n,$$

the pullback by a smooth map $\Phi : U \to V$ is

$$\Phi^*\omega = f(\Phi(x)) \det D\Phi(x)\ \mathrm{d}x_1 \wedge \cdots \wedge \mathrm{d}x_n.$$

Indeed,

$$\Phi^*(\mathrm{d}y_j) = \sum_{i=1}^{n} \frac{\partial \Phi^j}{\partial x_i}\ \mathrm{d}x_i,$$

and the alternating multilinearity of the wedge product produces the determinant. In this language, the determinant appears automatically when a top-degree form is transported to the parameter domain.

The sign is now meaningful. For an orientation-preserving diffeomorphism, the oriented formula is

$$\int_V \omega = \int_U \Phi^*\omega;$$

an orientation-reversing diffeomorphism changes the sign. This distinction between $\det D\Phi$ and $|\det D\Phi|$ is why orientation is part of the natural language of integration on manifolds. One can formulate unsigned integration intrinsically using densities, but that theory is beyond the scope of these notes.

This pullback identity is the form of change of variables that survives on a smooth manifold, where no single global coordinate system need exist. Chapter 13 develops pullbacks and top-degree forms in Section 13.3, and its change-of-parametrization result Proposition 13.19 uses the present theorem to establish coordinate independence. The logical order is essential:

$$\begin{aligned} &\text{Euclidean change of variables} \\ \Longrightarrow\ &\text{coordinate invariance of form integration} \\ \Longrightarrow\ &\text{well-defined integration on manifolds.} \end{aligned}$$

Thus the differential-form discussion is a conceptual, coordinate-invariant reformulation, not an independent proof of the Euclidean analytic theorem.

Finally, Lebesgue integration later provides a substantially more flexible change-of-variables theory for measurable sets and functions under weaker hypotheses. It replaces much of the present Jordan-boundary and Riemann-sum bookkeeping by measurable sets, null sets, Lebesgue integration, and countable decompositions. Measure theory is not introduced here; see Section 13.7 for its place among the natural continuations of these notes.

Example 12.29 (Polar coordinates and the unit disk)**.** Consider

$$\Phi(\theta, r) = (r\cos\theta, r\sin\theta).$$

Its derivative and Jacobian are

$$D\Phi(\theta, r) = \begin{pmatrix} -r\sin\theta & \cos\theta \\ r\cos\theta & \sin\theta \end{pmatrix}, \qquad |\det D\Phi(\theta, r)| = r.$$

The factor r is geometric: an angular change $\mathrm{d}\theta$ at radius r sweeps an arc of length approximately $r\ \mathrm{d}\theta$, while the radial change has length approximately $\mathrm{d}r$.

To use the theorem without hiding its hypotheses, first take

$$Q_{\varepsilon,\eta} = [\eta, 2\pi - \eta] \times [\varepsilon, 1]$$

with $\varepsilon, \eta > 0$. On this rectangle the map is injective and its Jacobian never vanishes. Its image is an annular sector. For example, since $x^2 + y^2 = r^2$,

$$\begin{aligned} \int_{\Phi(Q_{\varepsilon,\eta})} (x^2 + y^2)\ \mathrm{d}(x, y) &= \int_{\eta}^{2\pi-\eta} \int_{\varepsilon}^{1} r^2\, r\ \mathrm{d}r\ \mathrm{d}\theta \\ &= \frac{(2\pi - 2\eta)(1 - \varepsilon^4)}{4}. \end{aligned}$$

Letting $\varepsilon, \eta \to 0^+$ gives

$$\int_{x^2+y^2\le 1} (x^2 + y^2)\ \mathrm{d}(x, y) = \frac{\pi}{2}.$$

The omitted central disk and angular wedge have areas tending to zero, and the integrand is bounded by 1 on the unit disk, so their contributions tend to zero by the bounded-integral estimate. Equivalently, the origin and the two angular boundary rays are Jordan-null exceptional sets. The same calculation with integrand 1 gives the area of the unit disk:

$$\int_{x^2+y^2\le 1} 1\ \mathrm{d}(x, y) = \pi.$$

Exercises

Exercise 12.1. Prove that replacing a grid partition by a refinement decreases its upper sum and increases its lower sum.

Exercise 12.2. Compute

$$\int_{[0,1]\times[0,2]} (x + 2y)\ \mathrm{d}(x, y)$$

directly from iterated one-variable integrals.

Exercise 12.3. Use Theorem 12.9 with $j = 1$ to evaluate

$$\int_{[0,1]^2} x \frac{\partial}{\partial x}(x^2 y) \; \mathrm{d}(x, y).$$

Exercise 12.4. Prove that a constant function c on Q has integral

$$c \operatorname{vol}(Q).$$

Exercise 12.5. Show that a finite set in $\mathbb{R}^n$ has Jordan volume zero.

Exercise 12.6. Prove that the open unit ball in $\mathbb{R}^n$ is Jordan measurable, assuming that its boundary can be covered by finitely many boxes of arbitrarily small total volume.

Exercise 12.7. Give a bounded subset of $[0, 1]$ whose indicator is not Riemann integrable.

Exercise 12.8. Use Theorem 12.25 to compute the area of the parallelogram spanned by two vectors in $\mathbb{R}^2$.

Exercise 12.9. Verify the determinant computation in the polar-coordinate example.

Exercise 12.10. Use polar coordinates to compute

$$\int_{\{(x,y):x^2+y^2\leq 1\}} (x^2 + y^2) \; \mathrm{d}(x, y).$$

Do not apply the preceding theorem directly to the full disk. First apply it on

$$[0, 2\pi - \delta] \times [\varepsilon, 1],$$

where it is injective, and then let

$$\delta \to 0, \qquad \varepsilon \to 0.$$

Justify the omitted sector and inner disk by the bounded-function estimate and their Jordan volumes.

12.6 Limits of the Riemann Theory

The Riemann integral handles continuous functions and geometrically regular domains very effectively. Its limitations become sharper in several variables: indicators of sets with complicated boundaries may fail to be integrable, iterated integrals need not be available for arbitrary bounded functions, and the present change-of-variables theorem requires continuity and a globally injective parametrization. Lebesgue measure and integration replace Jordan volume by a much more flexible notion of size and later permit theorems under substantially weaker hypotheses. Those extensions are not developed in these notes.

Chapter 13

Differential Forms, Stokes' Theorem, and Calculus on Manifolds

13.1 Why Manifolds? A Preview

Many geometric spaces are locally indistinguishable from Euclidean space while being globally very different. A circle S^1 is locally like a line, a sphere S^2 is locally like a plane, and a torus T^2 is locally like a plane even though none of these spaces is globally $\mathbb{R}^n$. A smooth n-manifold is, informally, a space covered by coordinate neighborhoods

$$\varphi : U \longrightarrow V \subseteq \mathbb{R}^n$$

whose changes of coordinates are smooth. The guiding principle is to do calculus locally in Euclidean coordinates and then prove that the resulting objects do not depend on the chosen coordinates.

At a point p of such a space, the tangent directions form an n-dimensional vector space

$$T_pM.$$

Eventually, a differential k-form at p will be an alternating k-linear map

$$\omega_p : (T_pM)^k \longrightarrow \mathbb{R}.$$

This explains why the chapter begins with alternating multilinear algebra rather than with the formal manifold definition.

Chapters 10–12 supplied precisely the Euclidean ingredients that this local-to-global viewpoint needs: topology and finite-dimensional linear algebra in Chapter 10, differentiation on open subsets of $\mathbb{R}^n$ in Chapter 11, and integration, Jacobians, and change of variables in Chapter 12. Chapter 13 asks how these operations can be transported consistently between coordinate charts. Pullbacks and differential forms are designed to encode that coordinate-change law, so they are the natural objects for integration on oriented manifolds. The rigorous manifold definitions appear later in the chapter.

13.2 Alternating Multilinear Algebra

Chapter 11 identified the derivative of a scalar-valued function with a linear functional. Differential forms enlarge that observation: they are the multilinear objects that can be integrated over oriented domains. The alternating condition records the fact that an oriented area or volume vanishes when two directions coincide.

Definition 13.1 (Dual spaces and coordinate covectors)**.** Let V be a finite-dimensional real vector space. Its **dual space** is the vector space

$$V^* = \{\ell : V \longrightarrow \mathbb{R} : \ell \text{ is linear}\}.$$

Its elements are **covectors**. If

$$(e_1, \ldots, e_n)$$

is a basis of V, the **dual basis**

$$(e^1, \ldots, e^n)$$

is determined by

$$e^i(e_j) = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

For $\mathbb{R}^n$ with its standard basis, write

$$\mathrm{d}x_i = e^i.$$

Thus

$$\mathrm{d}x_i(v) = v_i$$

for $v = (v_1, \ldots, v_n) \in \mathbb{R}^n$. On a target copy of $\mathbb{R}^m$, the analogous coordinate covectors are denoted by

$$\mathrm{d}y_1, \ldots, \mathrm{d}y_m.$$

Definition 13.2 (Multilinear and alternating maps)**.** Let V be a finite-dimensional real vector space. A map

$$A : V^k \longrightarrow \mathbb{R}$$

is k**-multilinear** if it is linear in each argument separately. A 2-multilinear map is, equivalently, called a **bilinear** map. It is **alternating** if

$$A(v_1, \ldots, v_k) = 0$$

whenever two of its arguments are equal. The vector space of alternating k-multilinear maps is denoted by

$$\Lambda^k(V^*).$$

Its elements are called k**-covectors**.

Proposition 13.3 (Alternation)**.** If A is alternating, then interchanging two arguments changes its sign:

$$A(v_1, \ldots, v_i, \ldots, v_j, \ldots, v_k) = -A(v_1, \ldots, v_j, \ldots, v_i, \ldots, v_k).$$

In particular, an alternating map vanishes on every linearly dependent list.

Proof. Apply alternation to the list with $v_i + v_j$ in both the ith and jth positions and expand by multilinearity. The two diagonal terms vanish, and the remaining two terms give the first assertion. If a list is linearly dependent, express one vector as a linear combination of the others and use multilinearity; every resulting term has two equal arguments. □

Definition 13.4 (Wedge product)**.** For

$$\alpha \in \Lambda^p(V^*), \qquad \beta \in \Lambda^q(V^*),$$

their **wedge product** is the alternating

$$(p+q)\text{-multilinear map}$$

defined by

$$(\alpha \wedge \beta)(v_1, \dots, v_{p+q}) = \frac{1}{p!q!} \sum_{\sigma \in S_{p+q}} \operatorname{sgn}(\sigma)\alpha(v_{\sigma(1)}, \dots, v_{\sigma(p)})\beta(v_{\sigma(p+1)}, \dots, v_{\sigma(p+q)}).$$

The direct sum

$$\Lambda^*(V^*) = \bigoplus_{k=0}^{\dim V} \Lambda^k(V^*)$$

with this product is the **exterior algebra** of V^*. The finite direct-sum construction is the one defined in Definition 10.33; it records forms of all degrees as tuples, with addition and scalar multiplication performed degree by degree.

A form in $\Lambda^p(V^*)$ is called **homogeneous of degree** p. A product of homogeneous forms is **graded commutative** if interchanging forms of degrees p and q multiplies the product by

$$(-1)^{pq}.$$

Proposition 13.5 (Exterior-algebra rules)**.** The wedge product is bilinear, associative, and graded commutative:

$$\alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha.$$

If

$$(e^1, \dots, e^n)$$

is the dual basis of a basis of V, then the forms

$$e^{i_1} \wedge \cdots \wedge e^{i_k}, \qquad 1 \le i_1 < \cdots < i_k \le n,$$

form a basis of

$$\Lambda^k(V^*).$$

Proof. The permutation definition gives bilinearity and graded commutativity by reindexing. Associativity follows by grouping the same signed permutations of

$$p+q+r$$

arguments. Expanding an alternating form on basis vectors shows that its values on strictly increasing lists determine it; the displayed forms take value one on their own list and zero on the other increasing lists. Thus they are a basis. □

13.3 Differential Forms, Pullbacks, and Exterior Derivatives

For a smooth function $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$, Chapter 11 gives

$$Df(x) : \mathbb{R}^n \longrightarrow \mathbb{R}.$$

Thus $Df(x)$ is a 1-covector. The notation $df_x = Df(x)$ turns the familiar one-variable identity

$$df = f'(x)\,\mathrm{d}x$$

into a coordinate-free statement that continues to make sense in every dimension. In coordinates it becomes

$$df = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}\,\mathrm{d}x_i.$$

This is the direct chain from the Part I derivative, through the Fréchet derivative of Chapter 11, to a differential 1-form.

Definition 13.6 (Differential forms)**.** Let $U \subseteq \mathbb{R}^n$ be open and let

$$0 \leq k \leq n.$$

A **smooth k-form** on U is a function that assigns to each

$$x \in U$$

an element of

$$\Lambda^k((\mathbb{R}^n)^*)$$

whose coordinate coefficients are smooth. Thus it has a unique expression

$$\omega = \sum_{1\leq i_1<\cdots<i_k\leq n} a_{i_1\cdots i_k}(x)\,\mathrm{d}x_{i_1} \wedge \cdots \wedge \mathrm{d}x_{i_k}.$$

Here the sum is over the increasing multi-indices

$$I = (i_1, \ldots, i_k), \qquad 1 \leq i_1 < \cdots < i_k \leq n,$$

and we abbreviate

$$\mathrm{d}x_I = \mathrm{d}x_{i_1} \wedge \cdots \wedge \mathrm{d}x_{i_k}.$$

By convention,

$$\Lambda^0((\mathbb{R}^n)^*) = \mathbb{R},$$

so a smooth 0-form is simply a smooth real-valued function. The integer

$$k$$

is the **degree** of a k-form.

Proposition 13.7 (Differential of a function)**.** For a smooth function $f : U \to \mathbb{R}$,

$$df = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}\,\mathrm{d}x_i.$$

At each $x \in U$, the form df_x is the derivative $Df(x)$.

Proof. Both sides are linear functionals. Evaluating the displayed sum on the standard basis vector e_i gives
$$\frac{\partial f}{\partial x_i}(x) = Df(x)e_i.$$
The coordinate formula for a linear functional proves equality. □

Definition 13.8 (Pullback)**.** Let
$$\Phi : U \longrightarrow V$$
be a smooth map, where
$$U \subseteq \mathbb{R}^n, \qquad V \subseteq \mathbb{R}^m$$
are open. For a k-form ω on V, its **pullback** is the smooth k-form
$$\Phi^*\omega$$
on U defined by
$$(\Phi^*\omega)_x(v_1, \dots, v_k) = \omega_{\Phi(x)}(D\Phi(x)v_1, \dots, D\Phi(x)v_k).$$
Its smoothness follows from the smoothness of the coefficients of ω and of Φ.

Remark 13.9. Pullback is the form-theoretic version of substitution. For the standard volume form on $\mathbb{R}^n$,
$$\Phi^*(\mathrm{d}y_1 \wedge \cdots \wedge \mathrm{d}y_n) = \det D\Phi(x)\ \mathrm{d}x_1 \wedge \cdots \wedge \mathrm{d}x_n.$$
Thus the determinant in Chapter 12 is exactly the coefficient produced by pulling a volume form back to the parameter domain.

Proposition 13.10 (Pullback rules)**.** For smooth maps
$$U \xrightarrow{\Phi} V \xrightarrow{\Psi} W,$$
and forms of compatible degree,
$$(\Psi \circ \Phi)^* = \Phi^* \circ \Psi^*, \qquad \Phi^*(\alpha \wedge \beta) = \Phi^*\alpha \wedge \Phi^*\beta.$$

Proof. The chain rule from Proposition 11.5 gives
$$D(\Psi \circ \Phi)(x) = D\Psi(\Phi(x))D\Phi(x).$$
Substitution into the definition proves the first identity. Substitution into the alternating sum defining the wedge product proves the second. □

Definition 13.11 (Exterior derivative)**.** For a smooth k-form
$$\omega = \sum_I a_I\ \mathrm{d}x_I$$
on an open subset of $\mathbb{R}^n$, define its **exterior derivative** by
$$\mathrm{d}\omega = \sum_I \sum_{j=1}^n \frac{\partial a_I}{\partial x_j}\,\mathrm{d}x_j \wedge \mathrm{d}x_I.$$

Remark 13.12. The exterior derivative raises degree by one:

$$\text{functions} \longrightarrow \text{1-forms} \longrightarrow \text{2-forms} \longrightarrow \cdots .$$

It generalizes ordinary differentiation while retaining the product rule. The identities $d^2 = 0$ and pullback-commutation below are the two algebraic facts that allow a local integral identity to pass through coordinates and then to global manifolds.

Theorem 13.13 (Fundamental exterior-derivative rules)**.** For smooth forms α, β,

$$\mathrm{d}(\alpha \wedge \beta) = \mathrm{d}\alpha \wedge \beta + (-1)^{\deg \alpha} \alpha \wedge \mathrm{d}\beta,$$

$$\mathrm{d}(\mathrm{d}\alpha) = 0, \qquad \Phi^*(\mathrm{d}\omega) = \mathrm{d}(\Phi^*\omega).$$

Proof. The first identity follows from the product rule for the coefficient functions. Write

$$\alpha = \sum_I a_I \, \mathrm{d}x_I.$$

Applying the definition twice gives the double sum

$$\mathrm{d}(\mathrm{d}\alpha) = \sum_I \sum_{j,k} \frac{\partial^2 a_I}{\partial x_k \partial x_j} \, \mathrm{d}x_k \wedge \mathrm{d}x_j \wedge \mathrm{d}x_I.$$

If a differential is repeated, the corresponding term vanishes because

$$\mathrm{d}x_j \wedge \mathrm{d}x_j = 0.$$

For distinct indices, the terms indexed by (j, k) and (k, j) pair: their coefficients agree by Theorem 11.18, while

$$\mathrm{d}x_k \wedge \mathrm{d}x_j = -\,\mathrm{d}x_j \wedge \mathrm{d}x_k.$$

They therefore cancel, proving $\mathrm{d}^2 = 0$.

For a smooth function f, the multivariable chain rule gives

$$\Phi^*(\mathrm{d}f) = \mathrm{d}(f \circ \Phi).$$

In particular, pullback commutes with d on coordinate functions. Linearity and compatibility of pullback with wedge products then extend this identity to arbitrary forms, proving the final assertion. □

Example 13.14. In $\mathbb{R}^3$, write

$$\omega = P \, \mathrm{d}x + Q \, \mathrm{d}y + R \, \mathrm{d}z.$$

Then

$$\mathrm{d}\omega = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) \mathrm{d}x \wedge \mathrm{d}y + \left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right) \mathrm{d}x \wedge \mathrm{d}z + \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right) \mathrm{d}y \wedge \mathrm{d}z.$$

The three coefficients encode the usual curl.

13.4 Integration and the Local Stokes Theorem

Before the higher-dimensional statement, recall its one-dimensional model. For a smooth function on an oriented interval,

$$\int_{[a,b]} df = f(b) - f(a) = \int_{\partial[a,b]} f.$$

The local theorem is the same assertion on a rectangle: its proof applies the Part I Fundamental Theorem of Calculus one coordinate at a time.

Definition 13.15 (Integration of top-degree forms)**.** Let $Q \subseteq \mathbb{R}^n$ be a rectangle with the standard orientation. For a continuous n-form

$$\omega = f \; \mathrm{d}x_1 \wedge \cdots \wedge \mathrm{d}x_n,$$

define

$$\int_Q \omega = \int_Q f(x) \; \mathrm{d}x.$$

Reversing the orientation changes the sign of the integral.

Integration over an open set. Let $V \subseteq \mathbb{R}^n$ be open, and let

$$\omega = f \; \mathrm{d}x_1 \wedge \cdots \wedge \mathrm{d}x_n$$

be a continuous top-degree form whose support

$$K = \operatorname{supp}(\omega) = \overline{\{x \in V : f(x) \neq 0\}}$$

is compact and contained in V, where the closure is taken in V. Because a compact subset of the open set V is separated from $\mathbb{R}^n \setminus V$ (by a positive distance when $K \neq \varnothing$), there is a neighborhood of $\mathbb{R}^n \setminus V$ disjoint from K. The function f vanishes on the part of that neighborhood lying in V. Consequently, extending f by zero outside V gives a continuous function $\widetilde{f}$. Choose any rectangle Q whose interior contains K, and define

$$\int_V \omega = \int_Q \widetilde{f}(x) \; \mathrm{d}x.$$

This is independent of Q. Indeed, after enlarging two such rectangles to a common containing rectangle, finite additivity of the Riemann integral and the vanishing of $\widetilde{f}$ off K show that both integrals equal the integral over the larger rectangle. If ω is smooth, then the extended coefficients are identically zero on a neighborhood of the artificial boundary of V, so the same zero extension is smooth.

Definition 13.16 (Boundary orientation)**.** The boundary of the standard rectangle is oriented by the **outward-normal-first rule**: a basis of a boundary face is positive when adjoining an outward normal before it gives the orientation of the rectangle. The integral over the boundary is the signed sum over its faces.

Theorem 13.17 (Stokes theorem on a rectangle)**.** Let $Q \subseteq \mathbb{R}^n$ be a rectangle and let ω be a smooth

$$(n-1)\text{-form}$$

on an open neighborhood of Q. Then

$$\int_Q \mathrm{d}\omega = \int_{\partial Q} \omega.$$

Proof. Write

$$\omega = \sum_{j=1}^{n} (-1)^{j-1} f_j \; \mathrm{d}x_1 \wedge \cdots \wedge \widehat{\mathrm{d}x_j} \wedge \cdots \wedge \mathrm{d}x_n.$$

Only

$$\frac{\partial f_j}{\partial x_j}$$

contributes to the coefficient of

$$\mathrm{d}x_1 \wedge \cdots \wedge \mathrm{d}x_n$$

in $\mathrm{d}\omega$. Apply the one-variable Fundamental Theorem of Calculus in the x_j variable and then Theorem 12.7 to integrate the remaining variables. The two endpoint terms are exactly the integrals over the opposite oriented faces. Summing over j proves the formula. □

Definition 13.18 (Parametrized integration)**.** Let $D \subseteq \mathbb{R}^n$ be a compact rectangle and let

$$\Phi : D \longrightarrow \mathbb{R}^m$$

be a smooth injective parametrization with injective derivative on the interior. For a continuous n-form ω on a neighborhood of

$$\Phi(D),$$

define

$$\int_{\Phi(D)} \omega = \int_D \Phi^* \omega,$$

where the image carries the orientation induced by Φ.

Proposition 13.19 (Change of parametrization)**.** Let $D, D' \subseteq \mathbb{R}^n$ be compact rectangles, and suppose that $\Psi : D' \to D$ is the restriction of a smooth diffeomorphism between open neighborhoods of these rectangles, with $\Psi(D') = D$. Replacing Φ by $\Phi \circ \Psi$ leaves the integral unchanged when $\det D\Psi > 0$ on D', and changes its sign when $\det D\Psi < 0$ on D'.

Proof. Pullback functoriality gives

$$(\Phi \circ \Psi)^* \omega = \Psi^*(\Phi^* \omega).$$

Write $\Phi^*\omega = g(x) \; \mathrm{d}x_1 \wedge \cdots \wedge \mathrm{d}x_n$. Then

$$\Psi^*(\Phi^*\omega) = g(\Psi(u)) \det D\Psi(u) \; \mathrm{d}u_1 \wedge \cdots \wedge \mathrm{d}u_n.$$

Because D' is connected and the determinant is continuous and nowhere zero, its sign is constant. The scalar change-of-variables theorem Theorem 12.27 therefore gives

$$\int_{D'} g(\Psi(u)) \det D\Psi(u) \; \mathrm{d}u = \begin{cases} \displaystyle\int_D g(x) \; \mathrm{d}x, & \det D\Psi > 0, \\ \displaystyle -\int_D g(x) \; \mathrm{d}x, & \det D\Psi < 0. \end{cases}$$

These are precisely the asserted orientation-preserving and orientation-reversing cases. □

13.5 Manifolds and the Global Stokes Theorem

The local Stokes theorem works inside one coordinate region. A curved global space generally needs many overlapping charts, so a global integral and boundary formula require coordinate invariance together with a mechanism by which artificial internal boundaries cancel. The manifold and refinement machinery introduced next supplies exactly that passage from local Euclidean calculations to one coherent global theorem.

Topology terminology. The following modest amount of general-topology language will be used only to state the refinement theorem below. A topological space is **second countable** if there is a countable collection of open sets such that every open set is a union of members of that collection. A family

$$\{U_\alpha : \alpha \in I\}$$

is **locally finite** if every point has a neighborhood meeting only finitely many of its members. A cover

$$\{V_j : j \in J\}$$

refines a cover $\{U_\alpha : \alpha \in I\}$ if every V_j is contained in some U_α. A subset E is **precompact** if its closure is compact. Finally, a topological space is **paracompact** if every open cover admits a locally finite open refinement. These definitions use the notions of closure and compactness from Chapter 10.

Supports. If f is a function on a topological space X, its **support** is

$$\operatorname{supp}(f) = \overline{\{x \in X : f(x) \neq 0\}}.$$

A function has **compact support** when this closed set is compact. This definition makes precise the condition that a function vanish outside a compact region.

Definition 13.20 (Topological and smooth manifolds)**.** Let M be a topological space. A **coordinate chart of dimension** n on M is a pair (U, φ) in which $U \subseteq M$ is open and

$$\varphi : U \longrightarrow V \subseteq \mathbb{R}^n,$$

is a homeomorphism onto an open subset V of $\mathbb{R}^n$. (The displayed indexed notation will be used for a family of charts.) An **atlas** is a family of coordinate charts

$$\{(U_\alpha, \varphi_\alpha) : \alpha \in I\}$$

whose domains cover M. Thus coordinate charts provide the local Euclidean coordinates, and an atlas makes them available at every point. Recall the notion of homeomorphism from Definition 10.15.

Two charts $(U_\alpha, \varphi_\alpha)$ and (U_β, φ_β) are **smoothly compatible** if each transition map

$$\varphi_\beta \circ \varphi_\alpha^{-1}$$

on its open domain is a C^∞ diffeomorphism. An atlas whose charts are pairwise smoothly compatible is a **smooth atlas**. Recall the notion of diffeomorphism from Definition 11.29. A maximal smooth atlas is a **smooth structure**. A **topological n-manifold** is a Hausdorff, second-countable topological space equipped with an atlas of dimension n. A topological n-manifold with a smooth structure is a **smooth n-manifold**.

A **smooth manifold with boundary** is defined in the same way with charts into relatively open subsets of

$$\mathbb{H}^n = \{(x_1, \ldots, x_n) : x_n \geq 0\}.$$

The transition maps are required to be smooth diffeomorphisms in the half-space sense (equivalently, locally extend smoothly across the boundary). The boundary ∂M consists of points represented by $x_n = 0$.

Hausdorffness gives the expected separation and uniqueness behavior for points, while second countability excludes excessively large pathological spaces and is one ingredient behind the refinement and partition-of-unity theory used later. Every smooth atlas determines a unique maximal smooth atlas: adjoin every chart smoothly compatible with it. Thus maximality adds all allowable coordinate systems without changing the underlying geometry.

Definition 13.21 (Smooth maps and tangent vectors)**.** A map $F : M \to N$ between smooth manifolds is **smooth** if every coordinate representative

$$\psi \circ F \circ \varphi^{-1}$$

is smooth wherever it is defined. A tangent vector at $p \in M$ may be represented by the velocity at zero of a smooth curve through p; two curves are equivalent when their coordinate velocities agree in one chart. The resulting vector space is denoted by T_pM. A chart identifies T_pM with $\mathbb{R}^n$, and a smooth map induces the linear map

$$dF_p : T_pM \longrightarrow T_{F(p)}N.$$

This equivalence is independent of the chart. Indeed, if φ and ψ are charts at p, the chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}t}(\psi \circ \gamma)(0) = D(\psi \circ \varphi^{-1})(\varphi(p))\frac{\mathrm{d}}{\mathrm{d}t}(\varphi \circ \gamma)(0).$$

Thus equal φ-coordinate velocities give equal ψ-coordinate velocities; applying the same argument to the inverse transition map gives the converse.

Definition 13.22 (Forms and orientation on a manifold)**.** A smooth k-form on M is a collection of smooth coordinate k-forms that agree on overlaps under pullback by transition maps. An atlas whose transition maps have positive Jacobian determinant is an **oriented atlas**. An **orientation** is an equivalence class of oriented atlases. On an oriented manifold with boundary, the boundary orientation is again the outward-normal-first orientation.

Two oriented atlases determine the same orientation if their union is an oriented atlas; equivalently, every transition map from a chart in one atlas to a chart in the other has positive Jacobian determinant wherever it is defined. This is the manifold version of the orientation of a finite-dimensional vector space from Chapter 10: it chooses the positive class of local coordinate bases consistently from chart to chart.

Support of a differential form. For a differential form ω on M, define

$$\operatorname{supp}(\omega) = \overline{\{p \in M : \omega_p \neq 0\}}.$$

The form has **compact support** when this closed set is compact.

Theorem 13.23 (External refinement theorem (used without proof))**.** Every second-countable Hausdorff smooth manifold, possibly with boundary, is paracompact. Given an open cover of M, there are countable locally finite families of open coordinate neighborhoods V_j and W_j, with the V_j covering M, such that $\overline{V_j}$ is compact and

$$\overline{V_j} \subseteq W_j \subseteq U_{\alpha(j)}$$

for a suitable index $\alpha(j) \in I$.

We use this standard topological refinement theorem without proof; see, for example, [1, Chapters 1–2]. Its proof belongs to general topology rather than to the Euclidean calculus developed here. The smooth bump-function construction and normalization required for partitions of unity are proved below.

Theorem 13.24 (Partition of unity)**.** Let M be a smooth manifold, possibly with boundary, and let

$$\{U_\alpha : \alpha \in I\}$$

be an open cover. There are smooth functions

$$\rho_\alpha : M \to [0, 1]$$

such that the family is locally finite,

$$\operatorname{supp}(\rho_\alpha) \subseteq U_\alpha,$$

and

$$\sum_{\alpha \in I} \rho_\alpha = 1.$$

In this situation, the partition of unity is said to be **subordinate** to the cover: its support condition says precisely that each

$$\operatorname{supp}(\rho_\alpha)$$

lies in the corresponding U_α.

Proof. Apply the external refinement theorem Theorem 13.23 to obtain

$$\overline{V_j} \subseteq W_j \subseteq U_{\alpha(j)}.$$

For a manifold with boundary, use boundary charts in this construction; the smooth-extension convention in the definition of a manifold with boundary makes the same Euclidean bump-function construction applicable.

For each j, choose a chart $\varphi_j : W_j \to \varphi_j(W_j) \subseteq \mathbb{R}^n$. The function

$$\theta(t) = \begin{cases} 0, & t \le 0, \\ \exp(-1/t), & t > 0 \end{cases}$$

is smooth. Since the compact set $\varphi_j(\overline{V_j})$ lies in the open set $\varphi_j(W_j)$, finitely many Euclidean balls whose closures lie in that open set cover $\varphi_j(\overline{V_j})$. For those balls, the corresponding translated and rescaled functions $\theta(1 - \|x - c\|_2^2/r^2)$, combined by

$$1 - \prod(1 - \theta),$$

give a smooth nonnegative function b_j on $\mathbb{R}^n$ which is positive on $\varphi_j(\overline{V_j})$ and whose support is contained in the finite union of the closed balls, hence in $\varphi_j(W_j)$. Define χ_j on W_j by $b_j \circ \varphi_j$ and extend it by zero to M. The compact support of b_j inside $\varphi_j(W_j)$ makes this extension smooth, and

$$\chi_j \geq 0, \qquad \chi_j > 0 \text{ on } V_j, \qquad \operatorname{supp}(\chi_j) \subseteq W_j.$$

Because the family (W_j) is locally finite and $\operatorname{supp}(\chi_j) \subseteq W_j$, the family of supports $(\operatorname{supp}(\chi_j))$ is locally finite. Therefore the sum

$$\chi = \sum_{j=1}^{\infty} \chi_j$$

is locally finite, smooth, and well defined; it is positive because the V_j cover M. Set

$$\rho_j = \frac{\chi_j}{\chi}.$$

The denominator is positive, and local finiteness makes χ smooth and each quotient smooth. The family (ρ_j) is locally finite, satisfies

$$\operatorname{supp}(\rho_j) \subseteq \operatorname{supp}(\chi_j) \subseteq W_j \subseteq U_{\alpha(j)},$$

and sums to one. For each original index α, define

$$\rho_\alpha = \sum_{\{j:\alpha(j)=\alpha\}} \rho_j.$$

These sums are locally finite, so the ρ_α are smooth and the regrouped family remains locally finite. Moreover, a locally finite union of closed sets is closed, and therefore

$$\operatorname{supp}(\rho_\alpha) \subseteq \bigcup_{\{j:\alpha(j)=\alpha\}} \operatorname{supp}(\rho_j) \subseteq U_\alpha.$$

Regrouping the locally finite sum also gives $\sum_\alpha \rho_\alpha = 1$. Thus the partition is subordinate to the original cover. $\square$

Definition 13.25 (Integration on an oriented manifold)**.** Let M be an oriented n-manifold and let ω be a smooth n-form with compact support. Choose a partition of unity subordinate to oriented coordinate charts and define

$$\int_M \omega = \sum_\alpha \int_{\varphi_\alpha(U_\alpha)} (\varphi_\alpha^{-1})^*(\rho_\alpha \omega).$$

Only finitely many supports $\operatorname{supp}(\rho_\alpha)$ meet the compact set $\operatorname{supp}(\omega)$. Furthermore,

$$\operatorname{supp}(\rho_\alpha \omega) \subseteq \operatorname{supp}(\rho_\alpha) \cap \operatorname{supp}(\omega) \subseteq U_\alpha.$$

This support is a closed subset of the compact set $\operatorname{supp}(\omega)$, so it is compact. Its image under φ_α is therefore compact and contained in the open chart image. Thus every nonzero summand is legitimately the compactly supported Euclidean open-set integral defined above, and the displayed sum is finite.

Proposition 13.26 (Well-definedness)**.** The preceding integral is independent of the oriented charts and subordinate partition of unity.

Proof. Let (ρ_α) and (σ_β) be two subordinate partitions of unity, with their respective oriented charts. On the compact support of ω, local finiteness implies that only finitely many ρ_α, only finitely many σ_β, and hence only finitely many products $\rho_\alpha\sigma_\beta$ contribute. There one has

$$\rho_\alpha\omega = \sum_\beta \rho_\alpha\sigma_\beta\omega, \qquad \sigma_\beta\omega = \sum_\alpha \rho_\alpha\sigma_\beta\omega.$$

Each product has compact support contained in the overlap of the corresponding chart domains, since its support is contained in $\operatorname{supp}(\rho_\alpha)\cap\operatorname{supp}(\sigma_\beta)\cap\operatorname{supp}(\omega)$. On that overlap, Proposition 13.19 identifies its two oriented coordinate integrals. Linearity and the finite double sum then show that both definitions equal the same sum of the integrals of $\rho_\alpha\sigma_\beta\omega$, proving independence. □

Theorem 13.27 (General Stokes theorem)**.** Let M be an oriented smooth n-manifold with boundary, and let ω be a smooth

$$(n-1)\text{-form}$$

with compact support. Then

$$\int_M \mathrm{d}\omega = \int_{\partial M} \omega.$$

Proof. Choose a partition of unity subordinate to oriented manifold-with-boundary charts that carry M locally into $\mathbb{H}^n$ and carry ∂M into $\{x_n = 0\}$. Compactness of $\operatorname{supp}(\omega)$ and local finiteness show that only finitely many forms

$$\rho_\alpha\omega$$

are nonzero. Each has compact support contained in the corresponding chart, because $\operatorname{supp}(\rho_\alpha\omega) \subseteq \operatorname{supp}(\rho_\alpha) \cap \operatorname{supp}(\omega)$. By the product rule,

$$\mathrm{d}(\rho_\alpha\omega) = \mathrm{d}\rho_\alpha \wedge \omega + \rho_\alpha\,\mathrm{d}\omega.$$

After summing, the first terms cancel because

$$\sum_\alpha \mathrm{d}\rho_\alpha = \mathrm{d}\left(\sum_\alpha \rho_\alpha\right) = 0.$$

This identity is legitimate because the partition is locally finite: near each point both sums are finite, so exterior differentiation commutes with the displayed sum. It is therefore enough to prove the formula for a form supported in one chart, where "supported in" means that its closure-based support lies in the chart domain. In an interior chart, the coordinate support is compact in the open chart image. Hence the coordinate representative vanishes on a neighborhood of the artificial boundary of that image and extends smoothly by zero to $\mathbb{R}^n$. Choose a rectangle whose interior contains the extended support. The extension vanishes near the rectangle boundary, so Theorem 13.17 gives the formula with no manifold boundary contribution. In a boundary chart, the half-space definition of smoothness gives a smooth (not necessarily zero) extension of the coordinate representative across

$$x_n = 0$$

to a neighborhood in $\mathbb{R}^n$. Compact coordinate support permits zero extension across the remaining artificial boundary of the coordinate image; no zero extension is made across the genuine face $x_n = 0$. Choose a rectangle

$$[a_1, b_1] \times \cdots \times [a_{n-1}, b_{n-1}] \times [0, b_n]$$

whose relative interior contains its coordinate support and whose artificial faces are disjoint from it. Using a cutoff which is one near that support, retain the smooth extension across $x_n = 0$ and extend by zero only across the artificial faces; the resulting form vanishes near all of those faces. Apply Theorem 13.17 to the smooth extension. Its only surviving boundary term is the face $x_n = 0$, which is the coordinate representative of the genuine manifold boundary. The outward-normal-first convention gives exactly the induced boundary sign. Pullback compatibility from Theorem 13.13 transfers each coordinate formula back to the chart. Summing the finitely many contributions proves the result. □

Corollary 13.28 (Integration by parts for differential forms)**.** Let M be an oriented smooth n-manifold with boundary, let

$$0 \le p \le n-1,$$

let α be a smooth p-form with compact support, and let β be a smooth

$$(n-p-1)\text{-form}.$$

Then

$$\int_M \mathrm{d}\alpha \wedge \beta = \int_{\partial M} \alpha \wedge \beta - (-1)^p \int_M \alpha \wedge \mathrm{d}\beta.$$

Proof. The form

$$\alpha \wedge \beta$$

has degree $n-1$ and compact support, so Theorem 13.27 applies. The graded Leibniz rule from Theorem 13.13 gives

$$\mathrm{d}(\alpha \wedge \beta) = \mathrm{d}\alpha \wedge \beta + (-1)^p \alpha \wedge \mathrm{d}\beta.$$

Integrate this identity over M, apply Stokes to its left side, and rearrange. □

Remark 13.29 (Three levels of integration by parts)**.** The same structural identity appears at three increasing levels of generality:

$$\begin{gathered}
\text{product rule plus the Fundamental Theorem of Calculus} \\
\implies \text{one-variable integration by parts,} \\
\text{one-variable integration by parts plus Fubini} \\
\implies \text{the coordinatewise rectangular formula,} \\
\text{graded Leibniz rule plus Stokes} \\
\implies \text{integration by parts for forms.}
\end{gathered}$$

The final formula is coordinate-free. Green's identities and weak formulations of partial differential equations are important later descendants, but they require additional analytic machinery.

13.6 Classical Consequences

Classical vector-calculus notation. The differential-form integrals below are primary; familiar flux notation is only a translation. For a vector field

$$F = (F_1, \dots, F_n),$$

the associated flux form is

$$\sum_{j=1}^{n}(-1)^{j-1}F_j \ \mathrm{d}x_1 \wedge \cdots \wedge \widehat{\mathrm{d}x_j} \wedge \cdots \wedge \mathrm{d}x_n.$$

Its integral over an oriented boundary is conventionally denoted by

$$\int_{\partial\Omega} F \cdot n \ \mathrm{d}S.$$

In dimension three, the exterior derivative of

$$P \ \mathrm{d}x + Q \ \mathrm{d}y + R \ \mathrm{d}z$$

corresponds, under the usual Euclidean identification, to

$$(\nabla \times F) \cdot n \ \mathrm{d}S.$$

Thus n and $\mathrm{d}S$ here are conventional vector-calculus shorthand, not separately developed surface-measure machinery.

Corollary 13.30 (Fundamental theorem of calculus)**.** For a smooth function f on

$$[a, b],$$

$$\int_a^b f'(x) \ \mathrm{d}x = f(b) - f(a).$$

Proof. Apply Theorem 13.27 to the oriented interval and the 0-form f. □

Corollary 13.31 (Green's theorem)**.** Let $D \subseteq \mathbb{R}^2$ be a compact smooth 2-manifold with boundary, equipped with the orientation induced by the standard orientation of $\mathbb{R}^2$, and let P, Q be smooth on a neighborhood of D. The induced boundary orientation is the usual positive orientation. Then

$$\int_{\partial D} P \ \mathrm{d}x + Q \ \mathrm{d}y = \int_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) \mathrm{d}x \ \mathrm{d}y.$$

Proof. Apply Theorem 13.27 to

$$\omega = P \ \mathrm{d}x + Q \ \mathrm{d}y.$$

□

Corollary 13.32 (Classical Stokes theorem)**.** Let S be an oriented smooth compact surface with boundary in $\mathbb{R}^3$ and let

$$F = (P, Q, R)$$

be smooth near S. Then

$$\int_{\partial S} P \ \mathrm{d}x + Q \ \mathrm{d}y + R \ \mathrm{d}z = \int_S (\nabla \times F) \cdot n \ \mathrm{d}S.$$

Proof. Apply Theorem 13.27 to the 1-form

$$P \ \mathrm{d}x + Q \ \mathrm{d}y + R \ \mathrm{d}z.$$

The coordinate calculation in the preceding example identifies its exterior derivative with curl flux. □

Corollary 13.33 (Divergence theorem)**.** Let $\Omega \subseteq \mathbb{R}^n$ be a compact smooth n-manifold with boundary, equipped with the orientation induced by the standard orientation of $\mathbb{R}^n$, and let

$$F = (F_1, \dots, F_n)$$

be smooth near Ω. The induced boundary orientation is the usual outward orientation. Then

$$\int_{\partial\Omega} F \cdot n \, \mathrm{d}S = \int_\Omega \left(\sum_{j=1}^n \frac{\partial F_j}{\partial x_j} \right) \mathrm{d}x.$$

Proof. Apply Theorem 13.27 to

$$\omega = \sum_{j=1}^n (-1)^{j-1} F_j \, \mathrm{d}x_1 \wedge \cdots \wedge \widehat{\mathrm{d}x_j} \wedge \cdots \wedge \mathrm{d}x_n.$$

Its exterior derivative is the divergence times the standard volume form, and its boundary integral is the outward flux. □

Exercises

Exercise 13.1. Prove that

$$\dim \Lambda^k((\mathbb{R}^n)^*) = \binom{n}{k}.$$

Exercise 13.2. Let α, β be 1-covectors. Prove

$$\alpha \wedge \beta = -\beta \wedge \alpha$$

and give an example in which

$$\alpha \wedge \beta \neq 0.$$

Exercise 13.3. For

$$\omega = x \, \mathrm{d}y + y \, \mathrm{d}z + z \, \mathrm{d}x,$$

compute

$$\mathrm{d}\omega.$$

Exercise 13.4. Let

$$\Phi(u, v) = (u^2 - v^2, 2uv).$$

Compute

$$\Phi^*(x \, \mathrm{d}y - y \, \mathrm{d}x).$$

Exercise 13.5. Use Theorem 13.17 in dimension one to recover the Fundamental Theorem of Calculus.

Exercise 13.6. Apply Corollary 13.28 to the oriented interval

$$[a, b]$$

and the forms $\alpha = u$ and $\beta = v$ to recover the one-variable integration-by-parts formula for smooth functions u and v.

Exercise 13.7. For the unit square with its standard orientation, evaluate

$$\int_{\partial Q} x \, \mathrm{d}y$$

and compare it with

$$\int_{Q} \mathrm{d}(x \, \mathrm{d}y).$$

Exercise 13.8. Use Green's theorem to compute the area enclosed by a positively oriented smooth simple closed plane curve.

Exercise 13.9. Use the divergence theorem to compute the outward flux of

$$F(x, y, z) = (x, y, z)$$

across the unit sphere.

Exercise 13.10. Explain why

$$\mathrm{d}^2 = 0$$

implies that the boundary of a boundary has no contribution in Stokes' theorem.

13.7 Perspective and Further Directions

Differential forms separate the three ingredients hidden in vector-calculus formulas: differentiation is the exterior derivative, orientation controls sign, and integration pairs top-degree forms with domains. Stokes' theorem is the single compatibility law joining them. The familiar one-, two-, and three-dimensional formulas differ only in the notation used to identify forms with scalar and vector fields.

Measure Theory and Lebesgue Integration. The limitations of the Riemann and Jordan theories noted in Chapters 9 and 12 lead first to measure theory and Lebesgue integration. There one develops sigma-algebras, measures, measurable functions, and a more flexible integral. The genuine Monotone Convergence Theorem, Dominated Convergence Theorem, Fatou's lemma, and the Tonelli–Fubini theorems then hold under hypotheses substantially weaker than the uniform-convergence conditions available here. This is the setting anticipated by the Lebesgue-convergence remarks in the integration chapters.

General Topology. General topology studies the open-set language of Chapter 10 beyond metric and finite-dimensional spaces: bases and subbases, quotient and product spaces, separation axioms, compactness, connectedness, paracompactness, and metrization questions. The external refinement theorem used above is a small but important example of this broader theory.

Functional Analysis. Functional analysis extends normed-space methods to Banach and Hilbert spaces, bounded linear operators, infinite-dimensional differentiation, contraction principles, spectral theory, and weak convergence. In infinite dimensions, the equivalence of norms and the compactness behavior familiar from Chapter 10 no longer persist automatically.

Complex Analysis. The power-series, differentiation, and integration methods also open the way to complex analysis. Complex differentiability for maps

$$f : \mathbb{C} \longrightarrow \mathbb{C}$$

is far more restrictive than real differentiability and leads to holomorphic functions, contour integrals, Cauchy's theorem and integral formula, and residues.

Differential Geometry. Chapter 13 begins differential geometry: tangent and cotangent bundles, vector fields, Riemannian metrics, geodesics, connections, covariant differentiation, curvature, and integration on manifolds elaborate the local linear structures and differential forms already present here.

De Rham Theory and Algebraic Topology. One especially direct continuation starts with the identity

$$\mathrm{d}^2 = 0.$$

A form ω is called **closed** when

$$\mathrm{d}\omega = 0,$$

and **exact** when

$$\omega = \mathrm{d}\eta.$$

Every exact form is closed. De Rham cohomology records the extent to which the converse can fail globally:

$$H^k_{\mathrm{dR}}(M) = \frac{\ker\bigl(\mathrm{d} : \Omega^k(M) \longrightarrow \Omega^{k+1}(M)\bigr)}{\mathrm{im}\bigl(\mathrm{d} : \Omega^{k-1}(M) \longrightarrow \Omega^k(M)\bigr)}.$$

Thus it measures the obstruction to solving $\omega = \mathrm{d}\eta$ on all of M. De Rham's theorem, not proved here, identifies these analytic groups with topological invariants and forms a central bridge to algebraic topology.

Partial Differential Equations. Multivariable derivatives, divergence, integration by parts, and Stokes-type identities lead naturally to partial differential equations: the Laplace and Poisson equations, the heat and wave equations, weak derivatives, Sobolev spaces, and variational methods. Modern PDE joins the ideas of multivariable analysis with measure theory and functional analysis.

Probability. Modern probability provides a related application of measure theory: probability spaces, random variables, expectation as an integral, and convergence theorems give a rigorous language for random phenomena.

References and Further Reading

[1] John M. Lee. *Introduction to Smooth Manifolds.* Graduate Texts in Mathematics 218, second edition, Springer, 2013. Cited for the external manifold-refinement theorem used in Chapter 13.

[2] Walter Rudin. *Principles of Mathematical Analysis.* International Series in Pure and Applied Mathematics, third edition, McGraw–Hill, 1976. Further reading for real analysis.

[3] James R. Munkres. *Topology.* Second edition, Prentice Hall, 2000. Further reading for the general-topology background used sparingly in Part II.

[4] Michael Spivak. *Calculus.* Fourth edition, Publish or Perish, 2008. Further reading for rigorous single-variable calculus.

Attribution note. Except for the external refinement theorem explicitly cited in Chapter 13, the principal results in these notes are proved internally. The references listed here provide verification and further reading; they are not claims that the manuscript's proofs or exposition were adapted from a particular source.